OECD

OECD ECONOMIC SURVEYS

1998

SPECIAL FEATURES

Structural Policies

The Labour Market in a Knowledge-Based Economy

CANADA

OECD ECONOMIC SURVEYS

1997-1998

CANADA

ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

© OECD, 1998.
© Software: 1987-1996, Acrobat is a trademark of ADOBE.
All rights reserved. OECD grants you the right to use one copy of this Program for your personal use only. Unauthorised reproduction, lending, hiring, transmission or distribution of any data or software is prohibited. You must treat the Program and associated materials and any elements thereof like any other copyrighted material.

All requests should be made to:

Head of Publications Service,
OECD Publications Service,
2, rue André-Pascal, 75775 Paris
Cedex 16, France.

OECD ECONOMIC SURVEYS

1997-1998

CANADA

ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT

Pursuant to Article 1 of the Convention signed in Paris on 14th December 1960, and which came into force on 30th September 1961, the Organisation for Economic Co-operation and Development (OECD) shall promote policies designed:

- to achieve the highest sustainable economic growth and employment and a rising standard of living in Member countries, while maintaining financial stability, and thus to contribute to the development of the world economy;
- to contribute to sound economic expansion in Member as well as non-member countries in the process of economic development; and
- to contribute to the expansion of world trade on a multilateral, non-discriminatory basis in accordance with international obligations.

The original Member countries of the OECD are Austria, Belgium, Canada, Denmark, France, Germany, Greece, Iceland, Ireland, Italy, Luxembourg, the Netherlands, Norway, Portugal, Spain, Sweden, Switzerland, Turkey, the United Kingdom and the United States. The following countries became Members subsequently through accession at the dates indicated hereafter: Japan (28th April 1964), Finland (28th January 1969), Australia (7th June 1971), New Zealand (29th May 1973), Mexico (18th May 1994), the Czech Republic (21st December 1995), Hungary (7th May 1996), Poland (22nd November 1996) and Korea (12th December 1996). The Commission of the European Communities takes part in the work of the OECD (Article 13 of the OECD Convention).

Publié également en français.

© OECD 1998
Permission to reproduce a portion of this work for non-commercial purposes or classroom use should be obtained through the Centre français d'exploitation du droit de copie (CFC), 20, rue des Grands-Augustins, 75006 Paris, France, Tel. (33-1) 44 07 47 70, Fax (33-1) 46 34 67 19, for every country except the United States. In the United States permission should be obtained through the Copyright Clearance Center, Customer Service, (508)750-8400, 222 Rosewood Drive, Danvers, MA 01923 USA, or CCC Online: http://www.copyright.com/. All other applications for permission to reproduce or translate all or part of this book should be made to OECD Publications, 2, rue André-Pascal, 75775 Paris Cedex 16, France.

Table of contents

Assessment and recommendations 1

I. Macroeconomic performance 11

Forces shaping economic activity 11
Recent developments 22
Short-term prospects 32
The medium-term outlook 35

II. Structural issues and policies 42

Microeconomic reform to date: an overall assessment 42
Recent progress 51
Remaining problems 99

III. The labour market in a knowledge-based economy 103

The knowledge-based economy: what it means for labour 104
Is there a shift toward a KBE in the Canadian labour market? 106
Improving workers' skills: the state of current government programmes 111
Scope for further action 125

Notes 130

Bibliography 134

Annexes

I. Indicators of monetary policy 137
II. Federal and provincial fiscal policy 141
III. Classification of industries on the basis of technology 146
IV. Estimating potential output 147

V. Some illustrative scenarios for economic growth 149
VI. The labour market in a knowledge based economy: empirical evidence 152
VII. Chronology of economic events 172
Notes 179

Statistical annex and structural indicators 181

Boxes

1. Task Force on the Future of the Canadian Financial Services Sector: Summary 54
2. Issues arising from the Dispute Settlement Procedures of the Free Trade Agreement 68
3. Report of the Technical Committee on Business Taxation: main recommendations 88
4. Social assistance reform and labour markets: the case of British Columbia 96

Tables

Text

1. Demand and output 23
2. Labour supply and demand 27
3. Wages, prices and profits 29
4. Balance of payments 31
5. Short-term projections 33
6. A medium-term scenario 36
7. Indicators of structural change in manufacturing 43
8. Indicators of structural performance in selected OECD countries 44
9. Contributions to potential output growth 47
10. Indicators of provincial economic performance 49
11. Number of financial institutions by sector 57
12. Developments in federally-regulated financial institutions 57
13. Assets of financial institutions held in Canada 58
14. Tariffs on US imports 64
15. Trade developments with the United States 66

16. Manufacturing labour productivity levels in major OECD economies 70
17. Relative levels of unit labour costs in manufacturing 71
18. The public sector procurement market 74
19. Agreement on Internal Trade: major outstanding obligations 75
20. Agreement on Internal Trade: remaining (major) barriers to trade 78
21. Developments in provincial exports 79
22. The EU, NAFTA and AIT trade frameworks: an overview of the main features 80
23. Implementing the OECD Jobs Strategy – an overview of progress 101
24. Government expenditure on labour market programmes 119

Annexes

A1. Provincial deficits and debt 142
A2. Provincial-territorial legislated fiscal rules 144
A3. Classification of industries based on technology 146
A4. Economic growth scenarios 150
A5. Labour force participation rates 157
A6. Annual average employment growth by level of knowledge intensity 159

Statistical annex and structural indicators

A. Selected background statistics 182
B. Supply and use of resources 183
C. Industrial production, employment and other business indicators 185
D. Prices, wages and finance 186
E. Balance of payments 187
F. Public sector 188
G. Financial markets 189
H. Labour market indicators 190
I. Production structure and performance indicators 191

Figures

Text

1. Conditions for growth 12
2. Monetary conditions 14

3. Interest rate differentials and the yield gap 16
4. Decomposition of the general government financial balance 19
5. Measures of personal saving 24
6. Business fixed investment 26
7. Government debt 39
8. Per capita income 40
9. Projections of GDP per head in selected OECD countries 47
10. Canada pension plan reform: contribution rates and fund/benefit ratio 90
11. Unemployment insurance indicators 94
12. Educational attainment in OECD countries 113

Annexes

A1. Monetary conditions index 138
A2. Measures of underlying inflation 139
A3. Labour market trends by international comparison 153
A4. Smoothed employment/population rates by cohort: females 154
A5. Smoothed employment/population rates by cohort: males 155
A6. Smoothed full year/full time rates by cohort: females 161
A7. Smoothed full year/full time rates by cohort: males 162
A8. Self employment and labour market slack 165
A9. Real weekly earnings by cohort: males 168
A10. Real weekly earnings by cohort: females 169

BASIC STATISTICS OF CANADA

THE LAND

Area (thousand sq. km)	9 976
Agricultural area (1990, as per cent of total area)	6.4
Population of major cities, including metropolitan areas (thousands) (1995)	
Montréal	3 337
Toronto	4 344

THE PEOPLE

Population (1997)	29 960 148
Number of inhabitants per sq. km	3.0
Population, annual net natural increase (average 1993-1997)	165 400
Natural increase rate per 1 000 inhabitants (average 1993-1997)	5.7
Civilian labour force (1997)	15 354 400
Employment agriculture (1997)	423 100
Immigration (annual average 1993-1997)	226 735
Average annual increase in civilian labour force (1993-1997), per cent	1.2

PRODUCTION

GDP in 1997 (millions of Canadian dollars)	855 103
GDP per head (Canadian dollars)	28 541
Gross fixed investment (private and public) per head (Canadian dollars)	5 293
Gross fixed investment (private and public) as per cent of GDP	18.5
Origin of gross domestic product (1997, per cent of total, 1992 prices):	
Agricultural, forestry and fishing	2.5
Mining and quarrrying	4.1
Manufacturing	17.6
Construction	5.6
Non business sector	18.2
Other	54.5

THE GOVERNMENT

Government current expenditure on goods and services (1997, as per cent of GDP)	19.7
Government gross fixed capital formation (1997, as per cent of GDP)	2.1
Federal Government current revenue (1997, as per cent of GDP)	20.3
Federal direct and guaranteed debt (1997, as per cent of current expenditure)	285.6

Composition of Parliament (October 1997)	Number of seats House of Commons	Number of seats Senate
Progressive Conservative	19	43
Liberal	156	55
New Democratic	21	–
Bloc Québécois	45	–
Independent	1	5
Reform	59	–

FOREIGN TRADE

Exports (1997)	
Exports of goods and services, as per cent of GDP	40.2
Main exports (per cent of commodity exports):	
Wheat	1.7
Natural gas	2.9
Lumber and sawmill products	5.6
Pulp and paper	6.0
Other metals and minerals	9.5
Motor vehicles and parts	23.3
Other manufactured goods	29.2
Main customers (per cent of commodity exports):	
United States	80.8
EU	5.5
Japan	4.0
Imports (1997)	
Imports of goods and services, as per cent of GDP	38.5
Main imports (per cent of commodity exports):	
Industrial materials	18.7
Motor vehicles and parts	21.9
Producers' equipment	32.9
Consumer goods	10.7
Main suppliers (per cent of commodity imports):	
United States	76.1
EU	8.7
Japan	3.1

THE CURRENCY

Monetary unit: Canadian dollar

Currency unit per US dollar, average of daily figures:	
Year 1997	1.385

Note: An international comparison of certain basic statistics is given in an annex table.

This Survey is based on the Secretariat's study prepared for the annual review of Canada by the Economic and Development Review Committee on 22 September 1998.

•

After revisions in the light of discussions during the review, final approval of the Survey for publication was given by the Committee on 30 October 1998.

•

The previous Survey of Canada was issued in November 1997.

Assessment and recommendations

Following two years of robust growth...

Although growth slowed by mid-1998 in response to a weaker external environment, Canada has enjoyed good economic performance over most of the past two years, with real GDP expanding at an annual rate of 3½ per cent since mid-1996. Previously relying primarily on external sources, growth has been increasingly driven by domestic demand, underpinned by low interest rates and rising business and consumer confidence in a context of sharp fiscal improvement. With a lag, stronger economic activity has resulted in substantial job creation in the private sector, which has brought the unemployment rate down to levels last seen at the beginning of the decade. Although the spurt in growth has reduced economic slack substantially, a margin of unused capacity still exists, exerting downward pressure on costs and prices. As a result, inflation has remained subdued, running near the lower end of the official 1 to 3 per cent target range in recent months. Since demand has been especially strong for import-intensive goods and declining prices for natural resources have dampened export earnings, the external current account has moved back into deficit.

... the recovery is projected to continue at a slower pace...

The fallout from the Asia crisis and increases in short-term interest rates over the past year are expected to dampen activity in the period ahead. However, there are a number of mitigating factors: growth in domestic demand

is projected to continue in the United States, Canada's major trading partner, albeit at a more moderate rate; exchange-rate depreciation has partially offset the impact of falling commodity prices; and long-term interest rates have declined along with lower inflation expectations and government debt reduction. Hence, economic growth is projected by the OECD Secretariat to proceed at a moderate pace, roughly in line with potential. With little further narrowing of the output gap over the 1998-99 period, inflation is expected to stay low and consistent with the official target. Despite slower export growth, the current account deficit is projected to remain broadly stable as imports slow in line with domestic demand and the deterioration of Canada's terms of trade comes to a halt.

... with external developments and financial market turbulence posing some downward risks

Given the current turbulence in the world economy, uncertainties surrounding that outlook are greater than usual, however. With Asia Pacific economies accounting for only 8 per cent of total Canadian exports, directly related downside risks would seem to be comparatively small. Although a further fall in commodity prices could adversely affect business confidence and investment, direct effects of the Asia crisis are likely to remain of limited importance, except for British Columbia, whose trade with that region represents a significant portion of activity. In contrast, indirect effects of the crisis through the US economy could prove to be more consequential as they would accentuate significantly the adverse implications for both exports and private sector confidence – the more so in the context of increased financial market volatility in Canada and on global markets. Weaker equity markets also reinforce the risk of consumer retrenchment associated with the low level of the personal saving ratio.

Careful monetary management is thus required...

Such uncertainties pose a challenge to monetary authorities. A year ago, the rapid closing of the output gap seemed to warrant some tightening of monetary conditions to prevent the emergence of inflationary pressure. Indeed, the Bank of Canada raised short-term interest rates five times since October 1997. However, only the first increase was intended to moderate the extent of monetary stimulus. Those in late 1997 and early 1998, were designed to counteract the easing of monetary conditions, caused by the decline in the Canadian dollar, that was considered excessive by the authorities. The last increase in late August 1998 – which has been partly reversed since – was a response to diminished confidence in Canadian dollar investments as signalled by intensifying downward pressure on the exchange rate and rising long-term interest rates. Despite these moves, there has been a decline in the Monetary Conditions Index (MCI), a measure of the combined effect of short-term interest and exchange rate movements that serves as an operational guide for policy. This easing in monetary conditions is appropriate given the weaker and more uncertain outlook for the world economy and commodity prices and the absence of inflation pressure. Maintaining such an easier monetary stance would be warranted until the closing of the output gap resumes.

... as well as prudent fiscal policy

Given the high level of public debt in Canada by international comparison, the uncertain external environment also heightens the importance of maintaining a balanced budget to make the economy less vulnerable to shocks. Following several years of severe retrenchment, fiscal policy is now gradually moving toward a neutral stance as both federal and provincial governments have begun to reduce taxes and raise spending. With public finances moving toward increasing surpluses, the question has indeed arisen as to how to use this ‘‘fiscal dividend’’ and what the budgetary

policy objectives should be over the medium term. Since, at the current pace of debt reduction, it will probably take another five years even to bring Canada's debt-to-GDP ratio in line with the OECD average, priority should be attached to maintaining the ratio on a clear downward trend. Given the risks to the outlook, this reinforces the need for prudent fiscal planning. Such a course would probably provide some room for automatic stabilisers to work in the event of an unanticipated economic slowdown. It also would not exclude further tax reductions over time. As discussed in last year's Survey, desirable adjustments to the Canadian tax system include, in particular, the abolition of surtaxes, re-introduction of effective indexation in the personal income tax system, cuts in unemployment insurance premia to restore balance in the Employment Insurance Account, and corporate tax changes to improve competitiveness. Limited expenditure initiatives in priority areas (such as human capital development and health) could also be considered, as, together with the above tax changes, they would serve to enhance economic efficiency.

Continued emphasis on structural reform is also necessary...

More generally, raising the potential of the economy will necessitate putting additional focus on structural reform. In retrospect, it appears that the expected positive effects on productivity and potential output of the numerous structural policy measures taken since the mid-1980s have been slow to materialise. While the reasons for this are not yet fully understood, a number of developments seem to have been working to offset the beneficial effects of such measures, including: slow adjustment in some sectors to new policy settings and initiatives; the wide margin of spare capacity that has existed over much of the 1990s and the apparent absence of other conditions deemed to be favourable to productivity and growth (such as high rates of innovation, R&D, and investment); and the fact that small and medium-sized enterprises – whose share of activity is relatively high

in Canada – have shown themselves to be less internationally oriented, innovative and productive than elsewhere. However, there are some trends more recently that give rise to optimism that the apparent pick-up of economic efficiency and potential output growth can be sustained: the macroeconomic environment has become more favourable; there has been a significant reallocation of productive resources from the public toward the private sector; the increasing openness of the economy to trade continues to enhance competitive pressures; and rigidities in some sectors where the reform process had been lagging have finally been addressed. Nonetheless, there is no room for complacency, and a number of structural policy challenges remain, in addition to those mentioned above in the field of taxation, to respond, in particular, to the evolving nature of technological and market forces.

... as unfinished business remains in the financial sector...

The financial sector is one area where further restructuring is likely to occur. A recently released official report focused on structural issues in that sector, including: the ability of foreign and domestic entities to enter into the sector; the nature of the powers available to different categories of financial institutions; the type of ownership structures available to them; and examining the regulatory framework governing their activities. The report's recommendations are currently being reviewed by the government. They should also influence the decisions to be made by the government about recently-proposed bank mergers, together with their impact on competition. In addition, policy-makers are conducting a review of the Payments System with a view to granting more financial institutions direct access to the system.

... and in the labour market

Although recent labour market reforms seem to have reduced structural unemployment, not enough has been done to deal with the problem of the underutilisation of

resources in several regions or with the longer-term issue of a declining supply of labour, which limits potential output growth. In the former case, a number of measures, outlined in previous Surveys, should be taken, such as eliminating regional differentiation of unemployment insurance benefits and further exploring the notion of experience rating on employer premia, to enhance job creation and deal with inequities across industries and regions; finishing the work set out in the Agreement on Internal Trade might also help in this regard. In the latter case, consideration needs to be given to strengthening labour force participation through, for example: strengthening laws that would prevent a mandatory retirement age; removing features of the pension system which discourage recipients from earning income; making the early retirement feature of the public pension plan less attractive; and extending the working lives of Canadians by a variety of active labour market and training measures.

With the economy becoming more knowledge based...

Further action to improve the functioning of the labour market should also alleviate transitional problems as the economy becomes more knowledge based. Although there is a shift in demand favouring skilled over unskilled workers, the Canadian labour market has not been dominated by such a trend. For instance, there are gender and generational elements to labour market trends that do not appear to be directly related to changes in returns to education and training. Hence, other forces, such as demographic supply shifts and institutional factors, have played an important role. Nonetheless, skill-related differences in labour market outcomes are substantial and, even though the pace of change has been gradual in Canada, trends in technology and trade suggest that the economy will become more and more knowledge based. Moreover, by international comparison, the knowledge-intensive sector is relatively small.

Thus, it appears reasonable to emphasise human capital development as part of a national policy agenda that attempts to accelerate the country's move toward a knowledge based economy that could turn out to be an opportunity for greater economic growth.

... it is important to address problems in the fields of education...

Although Canada has been successful in terms of higher proportions of young people entering and staying in school longer, there are still some disturbingly poor educational outcomes given the high level of expenditure by international standards. For instance, the school drop-out rate remains substantial, and literacy levels, in particular, continue to be surprisingly low compared with other countries. More emphasis on vocational courses, in which few students participate, may make the system more attractive and help lower the drop-out rate. More job-relevant curricula may also hold students' attention to a greater extent and help them to acquire higher levels of literacy. A comparison with other, better performing OECD countries would seem to indicate, however, that the lack of success in raising average literacy levels reflects failures in Canada's education system. Moreover, there are strong regional differences in literacy skills, suggesting the need for a co-operative effort across the country to ensure that education levels and curricula correspond to the needs of a knowledge based economy. At the same time, it will be important to promote mobility and portability of qualifications, providing for their recognition across provinces.

... and training...

The federal government has agreed to withdraw from direct training policies wherever the provinces request it to do so. Training closely tailored to local conditions and opportunities can be more successful, and provincial control offers the possibility to improve flexibility, accountability and integration with related services (such as welfare and edu-

cation). However, the devolution of training policy to the provinces must be designed to avoid some risks. In the absence of inter-provincial recognition of qualifications, such a move could effectively reduce labour mobility within Canada, adding to the impact of existing inter-provincial barriers. Moreover, devolution raises the possibility of some provinces acting as free riders, as seems to be the case in university education. Finally, assessing training needs and evaluating policy outcomes requires considerable research infrastructure and expertise which many provinces do not currently possess. It will be important that the handover of training policy to the provinces be accomplished in a way that ensures that potential benefits are realised and that these concerns are addressed. In particular, provinces will need to implement the objective of developing a more pervasive ‘‘evaluation culture’’ towards training programmes and build on past experience at the federal level.

... while encouraging private sector efforts in that area

One feature of the current training system is that, based on experience with past successful programmes, it tends to help those who already have some skills. It thus exacerbates the skills gap between the more and less educated, creating a virtuous circle for those initially well-endowed. While such an approach is likely to produce results and may promote the shift toward a knowledge based economy, it needs to be complemented by programmes aimed at developing more basic skills to help those not in a position to take advantage of more advanced schemes. Furthermore, firm-based training of existing employees tends to focus on people who already have good skills. In addition, although employer-sponsored training is important in terms of total training funding, it is low by international standards. This suggests that ways need to be found to encourage employers to increase their training efforts. Options include a train-

ing tax/tax credit scheme, the obligation to spend a minimum amount of the payroll on training, and government assistance to joint firm/worker trust funds. However, it has to be ensured that any such schemes lead to worthwhile training and benefit workers who need it most.

To sum up

In sum, after a difficult adjustment period, Canada's economic performance has improved significantly over the past two years. The challenge for policy makers now is to ensure that these favourable developments are sustained in an uncertain external environment. In the absence of inflationary pressure, keeping easy monetary conditions would seem to be warranted in the near term. Fiscal policy should ensure that the high public debt ratio remains on a clear downward trend, while making adjustments to the tax system. Structural policies will also have to play a role in this respect, as unfinished business remains, notably in the financial sector, which is in a process of restructuring; in the labour market, where additional measures are needed to reduce structural unemployment further; and in the area of education and training, to facilitate the transition to a knowledge-based economy.

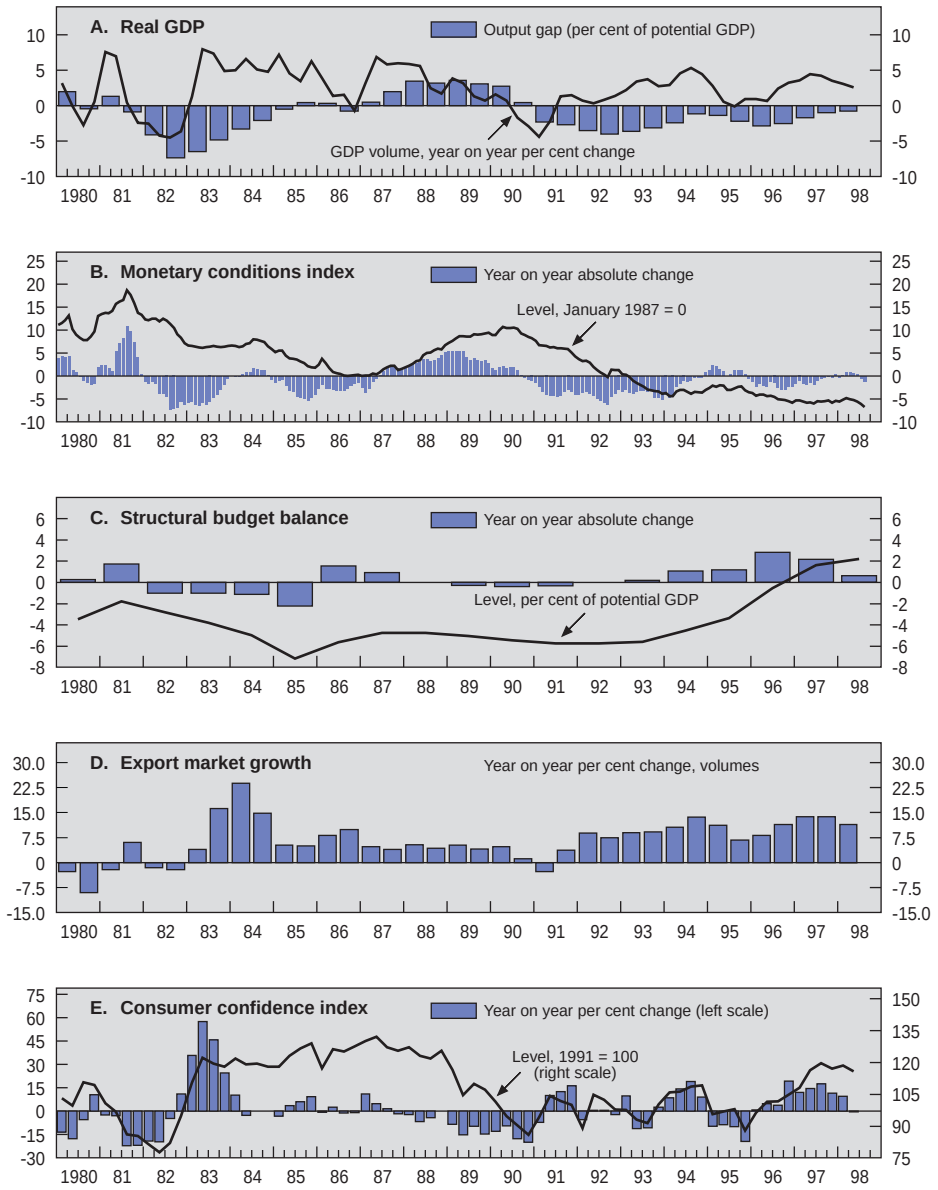
I. Macroeconomic performance

Forces shaping economic activity

The Canadian economy recovered very slowly from the recession in the early 1990s (Figure 1) as large fiscal imbalances entailed a mix of macroeconomic policies that was not conducive to growth. High public deficits and debt levels not only excluded an expansionary fiscal stance but also prevented policymakers from achieving the easier monetary conditions that would have been appropriate in view of the amount of excess supply prevailing in the economy. The monetary relaxation that did occur was mainly in the form of exchange rate depreciation, with real interest rates remaining relatively high. As a result, the recovery mainly relied on export growth while domestic demand stayed weak as consumer confidence remained at historically low levels. Moreover, Canada was poorly placed to withstand rising international interest rates and financial market turbulence in the mid-1990s, so that the tightening of domestic monetary conditions which took place at that time, along with the weakening of export demand, brought economic growth to a halt.[1]

The past two years, however, have seen a remarkable turnaround in the economic situation. Not only has growth resumed but it has also become more broadly based. While the strong performance of the US economy has helped by boosting export demand, the solid expansion in Canada of late reflects to a large extent domestic policy settings. The good inflation performance and rapid progress in budget consolidation paved the way for historically low interest rates from 1996. This led to a substantial strengthening of growth in domestic demand, despite sharp fiscal retrenchment, as consumer confidence rebounded (Figure 1). The following paragraphs review, in turn, three major forces shaping economic activity. Challenges policy makers will face in the period ahead are discussed at the end of this Chapter in the context of Canada's economic outlook.

Figure 1. **CONDITIONS FOR GROWTH**



Source: Statistics Canada; Conference Board; OECD Secretariat.

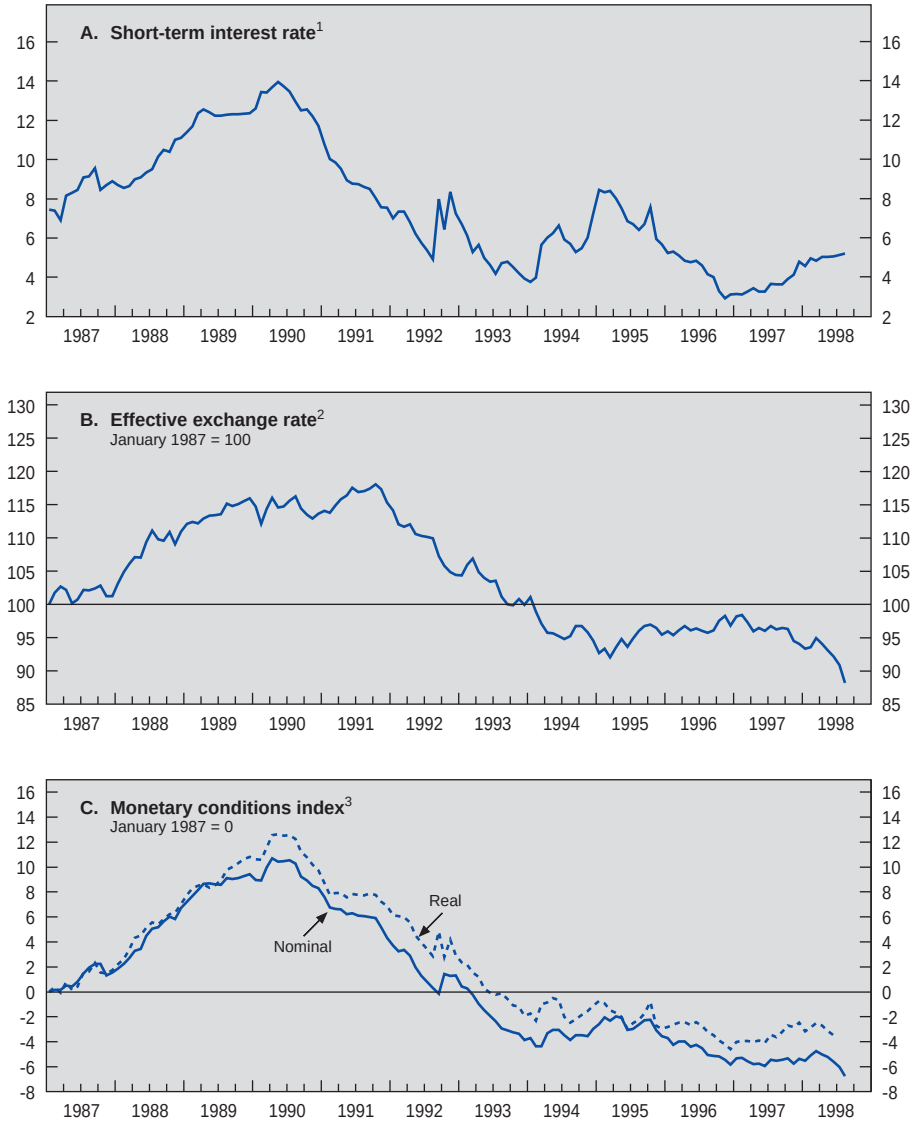
Monetary conditions

As outlined in more detail in previous OECD *Surveys*, the Bank of Canada aims at directing monetary conditions along a path that is expected to hold inflation inside the 1 to 3 per cent target range which has recently been extended until the end of 2001. The Bank monitors a number of indicators to assess future inflation trends, including estimates of the output gap, cost developments, the slope of the yield curve, and monetary aggregates. As an operational guide of monetary policy, the Bank uses a monetary conditions index (MCI) which combines movements in short-term interest rates and the effective exchange rate, these being the most important channels through which monetary policy can affect the economy. The authorities have emphasised, however, that the desired path of the MCI at any time is not precise, but rather a zone, and needs to be constantly reassessed in the light of new information and economic shocks. Indeed, the MCI is sensitive to the weight attached to the interest and exchange rate, which might vary over time, and the chosen definition of the effective exchange rate (see Annex I).

Given the lag with which policy actions affect the economy, economic developments over the past two years to a large extent reflect changes in monetary conditions up to the end of 1996. By that time, the MCI had fallen to its lowest level in recent decades. Contrary to trends in the first half of the decade, further monetary easing from mid-1995 was largely in the form of interest rate declines, with the effective exchange rate actually firming a little (Figure 2). With an expected rate of inflation of the order of 2 per cent, real short-term interest rates fell from around 6 per cent to about 1 per cent, providing considerable stimulus to domestic demand. Given increasing signs that the economy was responding to lower interest rates, the Bank sought to stabilise monetary conditions in the latter part of 1996. Indeed, continued interest rate reductions during that period were intended to offset the effect of a strengthening Canadian dollar rather than to loosen monetary conditions further.

While remaining supportive of economic growth, monetary conditions have tightened somewhat in 1997 and the first part of 1998. This is more apparent from the trend in the *real* MCI which is the more appropriate indicator for longer-term comparisons (Figure 2). Moreover, as shown in Annex I, the use of a broader exchange rate index (including currencies of Asian countries) would indicate a slightly more pronounced rise in the MCI in that period. More recently,

Figure 2. **MONETARY CONDITIONS**



1. 3-month corporate paper rate.
2. Canadian dollar against G10 currencies.
3. Weighted average of changes in the interest rate and in the effective exchange rate.

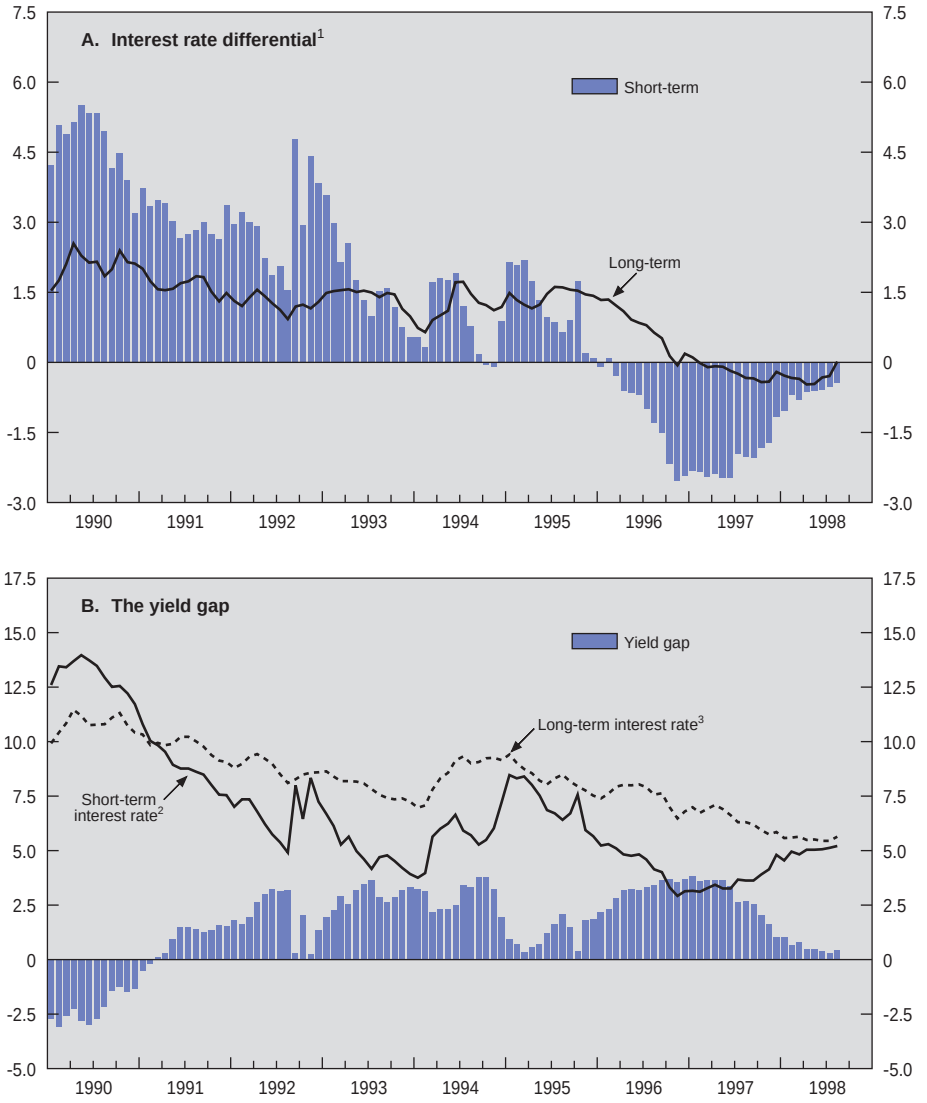
Source: Bank of Canada.

monetary conditions eased substantially as the Canadian dollar came under persistent downward pressure. As a result, at the time of writing, they were easier than in late 1996 as the impact on monetary conditions of the exchange rate depreciation appreciably outweighed an increase of about 2½ percentage points in short-term interest rates. The first rise in the Bank rate in late June 1997 still aimed at maintaining roughly stable monetary conditions in the face of recurrent currency weakness following a tightening of monetary policy in the United States. At the beginning of October, however, the Bank moved to moderate the degree of monetary stimulus, although measures of trend inflation (see Annex I) were running near the bottom of the target range, noting that there was growing evidence that the economy was expanding rapidly and absorbing unused capacity. As downward pressure on the Canadian dollar increased sharply toward the end of the year, reflecting fallout from the financial crisis in Asia and associated commodity price developments, the Bank raised interest rates further on three occasions from late-November to late-January 1998. This was motivated by the need both to counteract the easing in monetary conditions resulting from the depreciation of the exchange rate and forestall a potential loss of investor confidence in Canadian dollar assets. In the following months, the authorities refrained from further interest rate increases despite continued currency weakness as the Asian situation and lower commodity prices have led it to revise downward expected inflation and hence its desired path for monetary conditions. However, as the currency decline developed a momentum and medium and longer-term interest rates came under upward pressure, the Bank of Canada raised short-term rates by a full percentage point in August. The increase – which has been partly reversed since – aimed at providing support for the Canadian dollar in order to bolster confidence, while preventing an excessive easing of monetary conditions.

The MCI does not pay explicit attention to the relationship between short-term and long-term interest rates. The authorities have recognised this problem: when commenting on latest increases in the Bank rate, they pointed out that markets appear to have overlooked some positive developments for the economy such as significant declines in long-term rates. Indeed, since the beginning of 1995, the yield on government long-term bonds has fallen from over 9 per cent to about 5 per cent, the lowest level since the 1960s. In part this is the result of international developments. Furthermore, following with a lag trends in short-

Figure 3. **INTEREST RATE DIFFERENTIALS AND THE YIELD GAP**

Per cent



1. *Vis-à-vis* the United States.
2. 3-month corporate paper rate.
3. Over-10-year Government bond yield.

Source: CANSIM-Statistics Canada; OECD Secretariat.

term interest rates, long-term rates fell below corresponding US rates in early 1997 (Figure 3). The negative Canada/US spread that persisted until recently was unprecedented. It seems to reflect primarily two factors: the commitment to, and continued achievement of, inflation-control targets has strengthened investor confidence; and the rapid improvement in the fiscal situation has reduced concerns that the government might adopt inflationary policies to lighten its heavy debt load. Against this backdrop, long-term interest rates kept falling even as short-term interest rates firmed noticeably (Figure 3). With the turbulence in worldwide financial markets and investors' flight to US assets, long-term interest rates in Canada increased somewhat during the summer while US long-term rates edged down, before resuming their downward track again. The slope of the yield curve flattened progressively in 1997 and into 1998. In the past, such a flattening has often been followed by slower output growth. However, to the extent it reflects lower risk premia built into long-term interest rates, it provides a stimulus to economic activity.

The Bank has also taken account of the development of monetary aggregates in its assessment of the appropriate level of the MCI. The decision to reduce monetary stimulus in the second half of 1997 was partly motivated by concerns about the rapid growth in the measure of money used for transactions (M1). To a considerable extent, this was attributable to shifts of funds into business demand deposits associated with the introduction of attractive interest rates on such accounts. Nonetheless, M1 growth was considered unsustainably strong even after allowing for such asset shifts. As short-term interest rates started to rise and these portfolio shifts were nearing completion, narrow money growth slowed markedly. A renewed uptick more recently seems to reflect temporary factors. Underlying growth of M1 remains high enough to support robust growth of output.

The fiscal stance

By 1992, Canada's general government deficit (national accounts basis) had reached 8 per cent of GDP, more than double the OECD average, entailing a rapid rise in the public debt ratio to levels that were very high by international standards and involved interest payments equivalent to 5 per cent of GDP. The ensuing increased vulnerability of the Canadian economy to financial shocks and the adverse effects on economic performance more generally prompted a radical

change in policy orientation and, from 1994, budget deficits started falling rapidly (Figure 4). In 1997, the general government financial balance moved into a surplus of about 1 per cent of GDP while the rest of the OECD area, where budget consolidation was more gradual, was still recording a deficit of 1½ per cent of GDP. This impressive turnaround in Canada's fiscal position largely reflected a reduction in the public expenditure-to-GDP ratio, which declined by 8½ percentage points. In contrast, about one-half of the improvement in other OECD countries' government finances over that period resulted from an increase in the tax burden. Even though Canadian policymakers were constrained by the magnitude of the country's fiscal problem and the relatively high tax burden compared with the United States, this seems to confirm international experience that a focus on expenditure control is likely to be more successful than revenue raising as a means for reducing structural deficits.

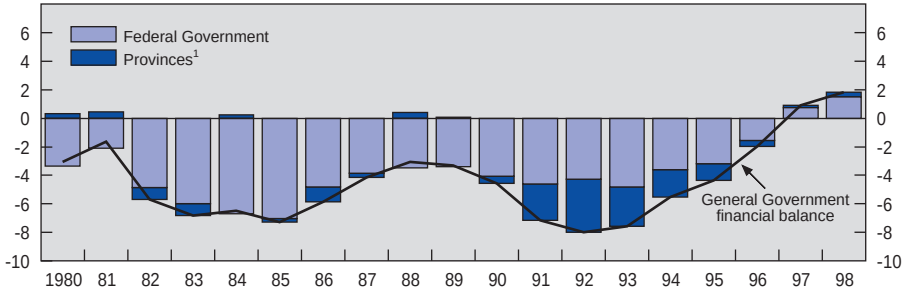
According to OECD Secretariat calculations, the bulk of the improvement in Canada's public finances from 1992 to 1997 can be traced to discretionary measures. Cyclical influences – that is, some narrowing of the output gap – account for less than one-fifth of the positive swing in the general government balance over this period. The structural budget deficit, which had been stubbornly high (in the 5 to 6 per cent of GDP range) since the mid-1980s, started falling gradually in 1994-95 but then declined by 5 per cent of GDP in 1996-97 (Figure 4). As noted, this massive withdrawal of demand did not prevent economic activity from rebounding during this period, as it was accompanied by an easing in monetary conditions and a rise in business and consumer confidence. Fiscal retrenchment has continued into 1998, but at a much slower pace, with an estimated negative direct demand effect of the order of ¾ per cent of GDP in the current year.

The federal and provincial governments have contributed to a similar extent to the improvement in public finances and are now both in a surplus position (on a national accounts basis). However, budget consolidation in the provinces started earlier (Figure 4) as their foreign debt exposure made them more vulnerable to adverse financial market reactions. Thus, initial declines in the general government structural deficit were largely attributable to provincial developments. By contrast, subsequent budget restraint has mainly occurred at the federal level as the Government sharply improved its fiscal position in 1996 and 1997.

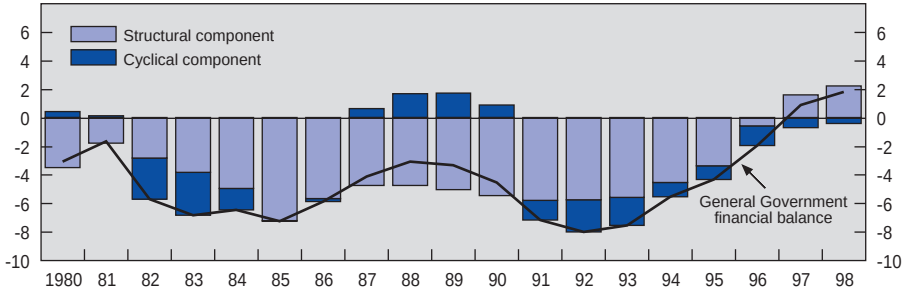
Figure 4. **DECOMPOSITION OF THE GENERAL GOVERNMENT FINANCIAL BALANCE**

Per cent of GDP (National Accounts Basis)

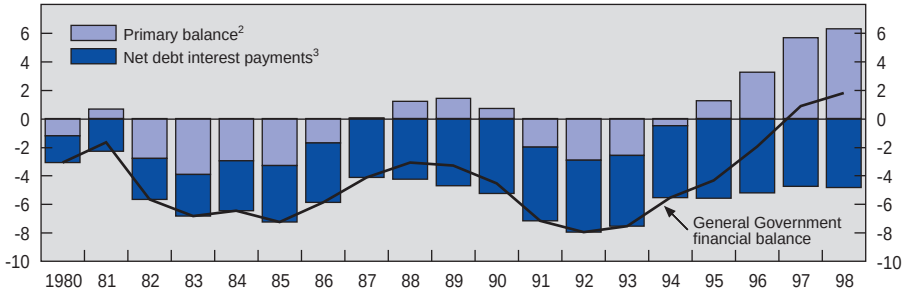
A. Federal and provincial components



B. Structural and cyclical components



C. Primary balance and debt interest payments



Note: 1998 data are OECD Secretariat projections.
1. Including local government, hospitals, Canada and Quebec Pension Plans.
2. Excluding debt service.
3. Interest payments are shown with a negative sign.

Source: CANSIM-Statistics Canada; OECD Secretariat estimates.

From the 1997 Budget, the federal government has commenced to provide some tax relief, primarily to low and middle-income earners (see Chapter II). On the expenditure side, new initiatives mainly aim at enhancing access to knowledge and skills (including the creation of the Millenium Scholarship Foundation, see Chapter III). Moreover, transfers to provinces (for health care, post-secondary education and social assistance) have been raised compared with previous budget plans, which will reduce fiscal pressures at the provincial government level. Despite these measures, the federal government is likely to continue to repay some of its debt, given that Budgets include ''prudence factors'' in the form of contingency reserves and projections based on conservative assumptions for the key macroeconomic variables. Nonetheless, recent initiatives imply that fiscal policy at the federal government level is gradually moving toward a neutral stance.

The outlook for the provinces is very similar. A majority now has balanced budget legislation and some have legislated debt reduction plans (see Annex II). The few provinces (notably Quebec and Ontario) which are still running deficits intend to eliminate them by the end of the decade. Again, with ''prudence factors'' included in many provincial budgets, fiscal objectives should be achieved or bettered, despite continued personal income tax reductions and targeted spending increases (mostly in the health and education areas). Thus, provincial budget developments are unlikely to modify significantly the trend in the fiscal stance at the federal level described above. Although fiscal policy is not expected to become expansionary, the gradual decline in effective rates of direct taxes on persons at both levels of government means that the growth in households' disposable income is no longer depressed by fiscal drag.

The external environment

With sales to the United States accounting for more than 80 per cent of total merchandise exports and almost 30 per cent of GDP, economic activity in Canada is strongly influenced by the conjuncture in its major trading partner. Moreover, although improved economic fundamentals have created some room for manoeuvre for Canadian policymakers, there are limits to the extent interest rate settings in the two countries can diverge. Over the past two years, developments in the United States have been very favourable to growth in Canada. As noted above, stable US interest rates have provided a ''window of opportunity''

for monetary policy to stimulate domestic demand, while strong US growth has been associated with buoyant import demand to which Canadian exporters were well placed to respond. The strong performance of the US economy has meant that Canada's export market has continued to expand relatively rapidly, despite a sharp fall in demand in Asian countries.

Although the fact that only a small proportion of Canada's trade is with the Asia Pacific region (8 per cent, including Japan) limits any direct effects of the Asia crisis, there are a number of indirect channels through which the economy has been adversely affected. The most important is the impact of the crisis on global commodity prices and Canada's terms of trade. Albeit falling, the share of commodities in Canada's merchandise exports is still relatively high (over one-third).[2] However, while declining commodity prices have led to a marked decrease in export prices, import prices have been dampened too, moderating the deterioration in Canada's terms of trade. Developments in Asia have also contributed to downward pressure on the Canadian dollar, both through their impact on global commodity prices and by generating flows into US dollar assets. In these circumstances, it has become more difficult for Canada to sustain negative interest rate spreads. Canada is also affected indirectly by damping effects of the Asia crisis on US growth. However, to the extent that this has contributed to lower US interest rates, it has reduced pressure on monetary policy and thus helped attenuate one of the potential risks facing the Canadian economy.

While Canada's overall exposure to the Asia crisis is limited, this is not the case for some industries and provinces. Close to 70 per cent of Canada's exports to Asia Pacific economies draw on the country's resource base. Sectors that export heavily to the region are, in particular, agriculture, forestry and mining. These sectors are those which have been most affected by falling world market prices, with declines being particularly pronounced for base metals, lumber, livestock and crude oil. Geographically, it is the Pacific coast which is most exposed to these developments. British Columbia accounts for over 40 per cent of Canada's exports to the Asia Pacific region, and one-third of the province's total exports (equivalent to 8½ per cent of its GDP) are destined for that region. Moreover, 85 per cent of British Columbia's exports to Asia Pacific economies are produced by resource-based industries and trade restrictions in this area (see Chapter II) limit the ability of Canadian producers to compensate for losses in the Asia Pacific by increasing shipments to the US market. Consistent with its greater

exposure to Asian markets, the British Columbia economy has weakened considerably, in marked contrast to the rest of the country.[3]

In sum, the Asia crisis and declining commodity prices represent a non-negligible negative external shock for the Canadian economy. However, the impact is concentrated in a few industries and regions. Moreover, the effects of the shock have, so far, been partly offset by positive developments both abroad and in Canada. In particular, partly in response to the crisis, long-term interest rates have declined, which has helped to boost the value of equities. This is likely to have contributed to sustain demand, as discussed in more detail below.

Recent developments

Supported by the favourable influences outlined previously, the Canadian economy has expanded at an annual rate of 3½ per cent since mid-1996, despite a slowdown more recently. This favourable performance has to be put in perspective, however. Even if the economy ends the decade without entering recession, as suggested by current OECD Secretariat projections, growth in the 1990s will have averaged just 1¾ per cent, a low rate by both historical (Table 1) and international standards. Chapter II will examine structural factors that may – in addition to the already mentioned macroeconomic aspects – help explain this outcome. The remainder of the present Chapter reviews recent economic developments before discussing whether improved economic performance is likely to be sustained over the short and medium run.

Demand and output

As noted, the strengthening of economic activity from mid-1996 reflects a broadening of the sources of growth as private consumption and investment responded to lower interest rates. In spite of a decline in public spending on goods and services, aggregate domestic demand expanded by more than 5 per cent in 1997, as compared with 1½ per cent on average in the 1990s (Table 1). In contrast, export growth, albeit relatively strong at 8 per cent, hardly exceeded its average performance over the current cycle. Moreover, with imports being boosted by the surge in domestic demand, the real foreign balance actually acted as a drag on growth in 1997. These broad trends have become less pronounced this year, with exports holding up better than domestic demand. As this demand

Table 1. **Demand and output**

Percentage change over previous period, seasonally adjusted annual rates, volume (1992 prices)

	1979/ 1973	1989/ 1979	1997/ 1989	1995	1996	1997	Q1 1998	Q2 1998
Private consumption	4.0	2.9	1.9	1.7	2.4	4.1	0.6	5.5
Government consumption	3.7	2.5	0.5	–0.4	–1.3	–0.1	–0.7	2.9
Gross fixed investment	5.0	5.3	1.0	–2.8	4.8	11.4	2.9	5.9
Public	0.2	4.9	0.9	–1.2	–3.7	–5.4	13.6	9.0
Private	5.9	5.4	1.0	–3.1	6.3	14.1	1.6	5.5
Residential	3.4	4.0	–1.5	–14.1	10.9	13.1	–0.6	–6.4
Non-residential	8.2	6.4	2.5	3.0	4.2	14.5	2.7	11.5
Final domestic demand	**4.1**	**3.3**	**1.4**	**0.4**	**2.0**	**4.5**	**0.8**	**5.1**
Stockbuilding[1]	0.1	–0.1	0.0	0.6	–0.9	0.8	–0.4	–1.8
Total domestic demand	**4.2**	**3.2**	**1.4**	**1.0**	**1.1**	**5.3**	**0.3**	**3.1**
Exports of goods and services	3.8	5.3	7.7	9.3	5.7	8.0	6.3	5.3
Imports of goods and services	5.2	6.8	6.7	6.7	5.2	13.3	0.1	6.1
Foreign balance[1]	**–0.1**	**–0.3**	**0.3**	**0.9**	**0.3**	**–1.6**	**2.3**	**–0.2**
Error of estimate[1]	0.0	0.1	–0.1	0.2	–0.1	0.1	0.8	–1.0
GDP at market prices	**3.9**	**2.9**	**1.6**	**2.2**	**1.2**	**3.7**	**3.4**	**1.8**
Agriculture	..	..	0.9	1.0	2.2	–0.8	–6.7	–6.4
Mining	..	..	3.4	4.1	3.0	4.4	–1.4	–1.2
Manufacturing	..	..	1.7	4.9	1.1	6.2	3.3	0.6
Construction	..	..	–1.3	–5.1	2.9	7.3	14.9	–6.6
Services	..	..	1.8	1.6	1.5	3.4	2.1	2.9
GDP at factor costs	**..**	**..**	**1.6**	**1.9**	**1.6**	**4.0**	**2.7**	**1.5**
Memorandum item:								
GDP per capita	2.7	1.7	0.3	0.9	0.1	2.3	..	..

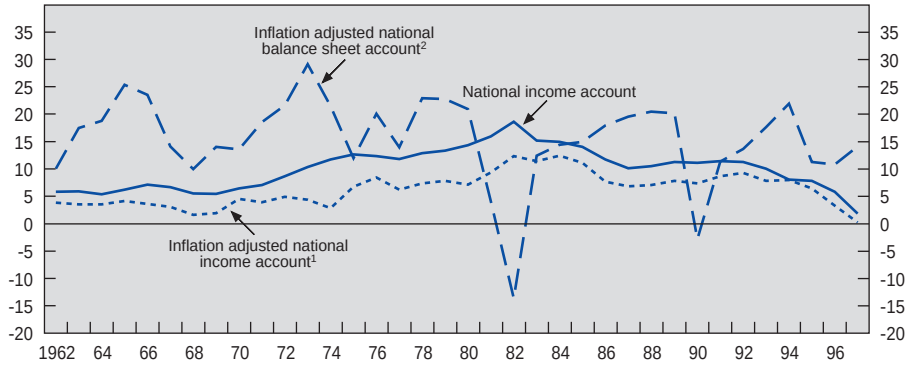
1. Contribution to GDP voume growth.
Source: CANSIM-Statistics Canada.

shift has been accompanied by slower import growth, net exports have picked up. Nonetheless, economic growth has slowed markedly in mid-1998, partly reflecting various labour conflicts.

Private consumption has been the engine of the recovery from mid-1996. In 1997, it expanded at the fastest pace in nine years, driven by the strongest burst of demand for durables since the recovery in the early 1980s. This is all the more remarkable as it occurred in the face of declining real disposable income (which was depressed by fiscal drag), implying a sharp fall in the personal saving ratio.

Figure 5. **MEASURES OF PERSONAL SAVING**

Per cent of disposable income



1. Both income and savings are adjusted to remove the portion of interest income which represents a decline in the value of assets due to inflation.
2. The change in net worth is adjusted to remove the portion which represents a decline in the value of assets due to inflation.

Source: Department of Finance; Statistics Canada.

Indeed, from mid-1996 to late-1997, the ratio dropped from 6 per cent to virtually zero, the lowest level on record. While an earlier downward trend in the personal saving ratio partly reflected the effects of disinflation, its recent drop cannot be explained this way (Figure 5). However, the National Accounts measure ignores important sources of saving coming from capital gains on assets (such as housing, stock, bonds and pension funds) as well as accumulation of consumer durables. When measured as the change in real net wealth, the personal saving ratio has actually increased since 1996 and is close to its long run average. This suggests that, although households are saving little out of the current flow of income, their overall financial position remains favourable. Besides growing real net wealth, the steep yield curve prevailing in late 1996 and the first half of 1997 (Figure 3) may have caused private consumption to outpace income growth by encouraging households to advance spending.[4] The flattening of the yield curve to a more normal slope since then has removed some of the impetus to consumption from this effect. However, at the same time, real disposable income has finally picked up (partly due to the rise in short-term interest rates). In the event,

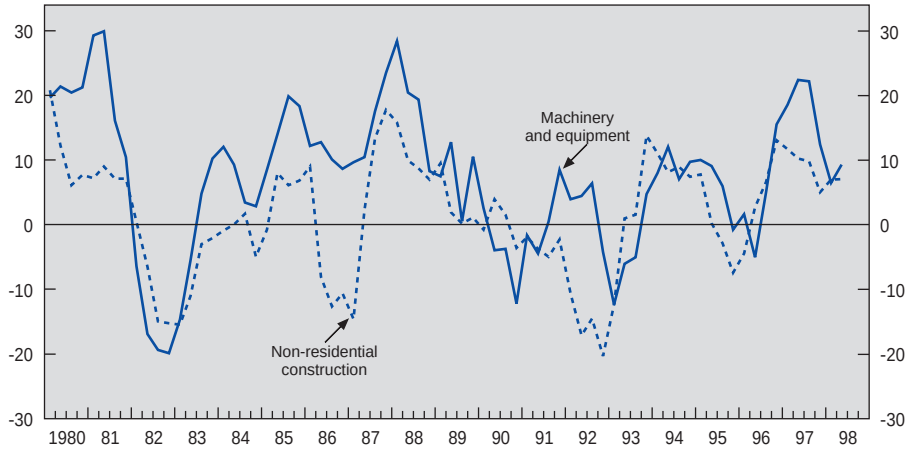
following sluggish growth in early 1998, private consumption has expanded at a healthy pace more recently (Table 1).

Residential investment has also recovered strongly, growing at double-digit rates both in 1996 and 1997 (Table 1) as buyers moved to take advantage of the lowest nominal mortgage rates in a generation before weakening recently (partly due to labour conflicts). Nonetheless, it has kept falling short of the levels recorded in the late 1980s, with housing starts running at the lower end of estimates of potential demographic requirements. This has puzzled analysts and led them to look for structural changes that may have lowered potential household formation, especially among the youth who are the key source of new housing demand. In particular, it has been argued that strong increases in post-secondary enrolment to meet a perceived increase in educational requirements in the labour market (see Chapter III), along with the inability of a growing proportion of new entrants to find career-oriented jobs, may have resulted in a substantial decline in the share of youth willing to start their own household. Moreover, immigrants – who now account for a larger proportion of population growth – seem initially to have a lower propensity to form households than other Canadians. Another explanation for lower than expected demand is that, while mortgage rates are historically low, in real terms they are still higher than in the late 1980s, when strong growth in housing prices resulted in a negative real user cost of housing.[5]

Business investment has responded vigorously to buoyant profits until recently, low financing costs and falling relative prices for machinery and equipment. Its expansion in 1997 was the strongest in nearly a decade. Spending on machinery and equipment has been particularly buoyant, rising by 31 per cent between the second quarter of 1996 and the third quarter of 1997 before tapering off more recently. However, the rapid increase in this investment component overstates its (short-term) contribution to economic growth because its import content is much higher than that of non-residential construction which has advanced at less spectacular rates (Figure 6). It is not a mere coincidence that the slowdown in machinery and equipment investment has been accompanied by slower import growth. Despite its recent surge, over the 1990s, business investment has expanded at a modest rate compared with the two preceding decades (Table 1). This has been one factor behind lower potential output growth, as discussed in Chapter II.

Figure 6. **BUSINESS FIXED INVESTMENT**

Year-on-year volume change



Source: CANSIM-Statistics Canada.

Inventory investment reinforced the spurt in activity, as it had added to the preceding slowdown (Table 1). Nonetheless, the inventory-to-sales ratio declined markedly through 1996 and 1997 as strongly rising final demand outpaced stockbuilding. Despite an up-tick in the ratio more recently, it has remained near record lows as stock accumulation has slowed noticeably. However, while there is no evidence of a substantial inventory overhang that would warrant correction, stockbuilding has ceased to be supportive of economic growth.

Strong growth in aggregate demand from mid-1996 has led to a considerable reduction in excess supply in the economy. According to OECD Secretariat estimates, the output gap has narrowed from around 3 per cent in the first half of 1996 to about 1 per cent, despite weaker activity growth recently. The swift absorption of economic slack reflects the fact that productive potential appears still to be expanding at a modest rate by historical standards. Secretariat calculations suggest that potential output growth has edged up gradually from 2¼ per cent in the mid-1990s toward 2½ per cent currently as a result of strong capital spending and some rebound in total factor productivity. However, although

preliminary evidence seems to lend some support to this view, the lack of recent data means that it is still unclear whether, adjusted for cyclical influences, total factor productivity is growing again after a protracted period of broad stagnation. Thus, it cannot be excluded that the economy has less ''room to grow'' before running up against its maximum sustainable output than indicated by the above estimates.

Labour market

With a lag, the faster pace of economic expansion has fuelled employment growth (Table 2) and, as a result, further inroads have been made into unemployment which has declined from nearly 10 per cent of the labour force in late 1996 to under 8½ per cent in recent months, a level last seen at the beginning of the decade. Nonetheless, some labour market slack has persisted as structural unemployment also appears to have declined. According to Secretariat estimates, the structural rate of unemployment has fallen from around 9 per cent in the first half of the 1990s toward 8 per cent in response to reforms of the unemployment insurance system and social assistance programmes. The impact of these

Table 2. **Labour supply and demand**

Per cent change over corresponding period

	1979/ 1973	1989/ 1979	1997/ 1989	1995	1996	1997	August 1998/ August 1997
Working age population	2.2	1.4	1.5	1.4	1.4	1.4	1.3
Labour force	3.2	2.0	1.0	0.7	1.5	1.3	1.3
Employment	2.9	2.0	0.8	1.6	1.3	1.9	2.0
Goods producing sector	1.6	0.6	–0.5	3.3	0.7	2.7	2.3
Service sector	3.6	2.6	1.3	1.0	1.5	1.6	1.9
Unemployment rate[1]	7.2	9.4	10.0	9.5	9.7	9.2	8.3
Participation rate[2]	61.7	65.8	65.7	64.8	64.9	64.8	64.9
Employment ratio[2]	57.3	59.7	59.1	58.7	58.6	58.8	59.5
Memorandum items:							
Labour productivity	1.3	0.9	0.8	0.5	–0.0	1.8	..
Total factor productivity[3]	0.8	0.5	–0.14	0.4	0.4	..	..

1. Per cent of labour force.
2. Per cent of population 15 years and over.
3. Business sector.
4. 1996/1989.

Source: CANSIM-Statistics Canada.

initiatives, and its timing, is, however, difficult to quantify and some analysts, including those at the Canadian Department of Finance, believe that the structural rate could already be lower.

In contrast to earlier declines in the 1990s, the recent fall in unemployment does not reflect a decrease in labour force participation. In fact, the participation rate has edged up a little (Table 2). However, it is still 2½ percentage points below the level prevailing in the late 1980s, and well below its estimated trend level, suggesting that there is a large pool of people who could return to the labour force if economic conditions continue to improve. A particularly positive aspect of the recent improvement in labour force participation is that it has been most pronounced for young people, whose participation rate had declined most over the 1990s (by almost 10 percentage points). Although that decline is in part attributable to rising school attendance as the economy becomes more knowledge based (see Chapter III), poor employment opportunities for youths also played a considerable role. Youth are the first to be fired in a recession and the last to be hired in an expansion. Historically, this has meant that there is some waiting period between an economic upturn and improved employment prospects for youth. This time, with a substantial output gap persisting during an unusually long period, it has taken much longer for youth to regain a foothold into employment.

Another positive feature of recent labour market developments is that virtually all of the additional jobs created have been full-time positions in the private sector. Public sector employment has stabilised as most of the planned cuts at the federal level have already occurred. In the private sector, some of the people involuntarily working part time (one-third of the total) have been able to get a full-time job. Moreover, dependent employment has started to grow again after stagnating over the two years to mid-1997. At the same time, with improving labour market conditions, the strong rise in self-employment over that period – which to some extent seems to have represented hidden unemployment (see Annex VI) – has been partly reversed.

Inflation

Reflecting conditions of excess supply, inflation has remained subdued, with annual consumer price (CPI) increases running near the lower end of the official 1 to 3 per cent target range in recent months (Table 3). Core measures of inflation

(Annex I) have also stayed within a stable and low range. Although the output gap has been closing steadily, the persistent slack in product markets continues to put downward pressure on inflation. At the same time, cost increases have been modest even though exchange rate depreciation has moderated the effect of falling world market prices. While overall profit margins rose before easing recently, increasing competition at the retail sale level[6] has led to widespread price discounting. The major offsetting factor working to hold CPI growth up appears to be inflation expectations that remain close to the 2 per cent mid-point of the target range. Despite sharp rises in certain areas, housing price increases for the country as a whole have remained modest. The broadest measure of inflation – the GDP deflator – has shown no annual growth recently. In addition to the factors lowering CPI inflation, this reflects the decline in primary commodity export prices which are included in the GDP deflator.

Table 3. **Wages, prices and profits**

Per cent change, annual rates

	1979/ 1973	1989/ 1979	1997/ 1989	1995	1996	1997	Q1 1998/ Q1 1997	Q2 1998/ Q2 1997
Wages								
Compensation per employee	10.8	6.8	2.8	1.7	2.2	3.4	0.5	–0.0
Hourly earnings in manufacturing	11.6	6.3	2.8	1.4	3.2	0.9	1.9	2.3
Major collective settlements	..	6.4	2.0	0.9	1.0	1.6	1.8	1.6
Unit labour costs								
Total economy	9.6	5.7	1.4	1.2	1.3	0.1	0.2	0.3
Manufacturing	9.4	4.9	0.6	0.3	3.9	–0.3	1.0	1.9
Profits								
Pre-tax	14.0	5.3	3.8	18.3	–3.4	16.2	–2.3	–0.8
After-tax	14.2	4.1	3.3	21.6	–9.4	15.7	–7.4	–10.9
Per unit of output	9.7	2.4	2.1	15.9	–4.6	12.1	–5.8	–3.7
Prices								
GDP deflator (current weights)	9.5	5.8	1.7	2.6	1.4	0.5	–0.4	–0.1
GDP deflator (fixed weights)	10.3	6.0	1.9	2.6	1.3	0.8	–0.3	0.2
Consumer price index	9.2	6.5	2.4	2.2	1.6	1.6	1.0	1.0
Private consumption deflator	9.1	6.2	2.2	1.4	1.5	1.7	0.9	1.1
Import price deflator	11.6	1.7	1.9	2.8	–1.2	0.4	1.3	0.1
New housing price index	..	..	–0.8	–1.2	–1.9	0.8	1.1	0.9

Source: CANSIM-Statistics Canada.

Wage increases have continued to be moderate. The slight pick-up in annual wage settlements (Table 3) reflects renewed pay rises in the public sector after several years of restraint. Survey evidence suggests somewhat higher wage growth for non-unionised employees. Thus, the underlying average wage increase in the total economy seems to be of the order of 2 per cent on a year-over-year basis. At the sectoral level, there have been reports of shortages and large pay rises for skilled personnel in the computer and other high-technology areas. However, these groups represent a small fraction of the workforce. The growth of labour income per employee was unexpectedly high in 1997 but has edged down since. This seems to have resulted from larger-than-normal bonus payments that were associated in part with productivity gains. Indeed, the cyclical pick-up in output entailed a recovery in productivity (Table 2) which broadly matched wage increases and thus implied largely stable unit labour costs.

External balance

Following persistent deficits since the early 1980s, the external current account had moved into a slight surplus in 1996 (Table 4). This was attributable to a number of influences that tended to strengthen the external balance, including Canada's weak relative demand position, improving terms of trade due to rising world market prices for commodities and restored international competitiveness owing both to exchange rate depreciation and low domestic inflation. The surplus has proved temporary, however, as many of these developments have gone into reverse over the past two years or so. For the first time in the 1990s, aggregate demand has been growing faster than in Canada's major trading partner countries, with a shift in demand toward import intensive goods (such as consumer durables and machinery and equipment investment). In addition, Canada's terms of trade have deteriorated because of a significant decline in the prices of primary commodities of which the country is a net exporter. Loss of overall competitiveness has not been a factor, since measures of relative costs have actually improved as the Canadian dollar has tended to weaken and unit labour costs have been broadly stable in domestic currency terms. Since late 1997, the current account deficit has narrowed somewhat as import growth has slowed while exports have held up relatively well despite sharply lower demand from Asian countries.

Table 4. **Balance of payments**

C$ billion, annualised

	1994	1995	1996	1997	1998 Q1	1998 Q2
	Seasonally adjusted					
Goods exports	228.2	264.8	279.7	301.1	310.3	312.9
Goods imports	207.9	229.9	237.8	276.8	291.9	296.4
Trade balance	20.3	34.8	41.9	24.3	18.4	16.5
Travel and other services, net	–11.7	–10.3	–9.3	–8.7	–9.5	–8.4
Investment income, net	–25.9	–30.8	–28.4	–28.9	–28.8	–25.5
Transfers, net	–0.5	–0.2	0.3	0.6	2.7	0.4
Current balance	–17.7	–6.4	4.5	–12.8	–17.1	–16.9
	Not seasonally adjusted					
Current balance	–17.7	–6.4	4.5	–12.8	–33.7	–17.9
Capital account	10.2	6.8	8.1	7.6	5.3	5.8
Canadian assets (net flows)	–49.0	–38.7	–72.1	–51.7	–52.3	–11.6
Canadian liabilities (net flows)	56.6	31.7	51.1	64.0	92.1	–12.6
Financial account[1]	7.5	–7.0	–21.0	12.3	39.8	–24.1
Statistical discrepancy	–0.0	6.7	8.4	–7.1	–11.4	36.2
	Seasonally adjusted					
Memorandum items:						
Current balance						
US dollars	–13.0	–4.7	3.3	–9.2	–12.0	–11.7
Per cent of GDP	–2.3	–0.8	0.6	–1.5	–2.0	–1.9

1. A minus sign denotes an outflow of capital resulting from an increase in claims on non-residents or a decrease in liabilities to non-residents.

Source: Statistics Canada.

The deterioration in the external balance over the past two years primarily reflects a pronounced decline in the merchandise trade surplus (Table 4). Although this can be largely traced to the above-mentioned factors (notably the strength and composition of domestic demand and turnaround in the terms of trade), recent trade performance can nevertheless be regarded as disappointing, given the historically favourable competitive position of the Canadian economy with an exchange rate well below measures of purchasing power parity. Strongly

rising import propensities in many sectors can partly be explained by the effect of trade agreements. But, while the latter have also boosted export propensity, Canada has apparently not been able to take full advantage of the recent surge in US demand. In contrast to the trade account, the invisibles balance has been broadly stable, with its major components changing little. Despite high (largely reinvested) profits of foreign-controlled enterprises, the investment income deficit has actually shown some improvement as the rollover of external debt at lower interest rates restrained servicing costs.

Short-term prospects

Central projections and underlying assumptions

Reflecting somewhat higher short-term interest rates and the adverse effects of the Asia crisis, economic growth is projected to moderate further, averaging around 2½ per cent over the next eighteen months or so (Table 5). With potential output expanding at a rate of just over 2½ per cent over that period, according to OECD Secretariat estimates, this implies no further narrowing of the output gap. Following its spurt in 1997, domestic demand is projected to grow at a slower pace. The expansion of private consumption is expected to stay below that of disposable income as households raise gradually their historically low saving ratio to a more normal level. Residential investment is projected to grow moderately, given improving household income and still favourable mortgage rates. Continued low long-term interest rates should also help sustain business capital spending, although it is expected to expand at less spectacular rates, as foreshadowed by investment intention surveys. As discussed, inventory formation is unlikely to make a contribution to growth in the period ahead. On the other hand, after five years of fiscal retrenchment, government spending will again give some support to domestic demand.

Export growth is projected to continue at a slower pace, reflecting several partly offsetting influences. In the short-run, continued robust US demand is limiting adverse effects of the Asia crisis; thereafter, improved competitiveness due to exchange rate depreciation should moderate the impact of slower US growth on Canada's exports. The current account deficit is expected to stay virtually stable as imports rise broadly in line with exports and the deterioration

Table 5. **Short-term projections**

Percentage changes, seasonally-adjusted annual rates, volume (1992 prices)

	1995 C$ billion[1]	1998	1999	1998		1999	
				I	II	I	II
Private consumption	460.0	2.9	2.4	2.4	2.6	2.2	2.5
Government consumption	171.1	0.5	1.0	0.2	0.5	1.2	1.3
Gross fixed investment	135.3	4.9	5.3	3.0	6.1	4.9	5.4
Public[2]	20.1	3.2	3.6	6.8	3.1	3.9	3.6
Private residential	36.3	2.2	3.8	0.1	4.5	3.4	3.8
Private non-residential	79.0	6.5	6.4	3.6	7.4	5.8	6.5
Final domestic demand	**766.4**	**2.8**	**2.7**	**2.0**	**2.8**	**2.5**	**2.8**
Change in stockbuilding[3]	8.2	0.2	–0.2	–0.3	–0.1	–0.3	–0.1
Total domestic demand	**774.6**	**2.9**	**2.4**	**1.7**	**2.7**	**2.2**	**2.7**
Exports of goods and services	302.2	6.8	6.2	6.3	6.2	6.0	6.5
Imports of goods and services	278.3	7.0	6.2	3.8	6.5	5.9	6.6
Change in foreign balance[3]	**23.9**	**0.0**	**0.0**	**0.9**	**–0.1**	**0.1**	**0.0**
Statistical discrepancy[3]	0.6	0.2	0.0	0.3	0.0	0.0	0.0
GDP at market prices	**799.1**	**3.0**	**2.4**	**2.9**	**2.6**	**2.3**	**2.7**
Inflation							
GDP implicit price deflator		–0.1	1.6	–0.6	0.8	1.7	2.0
Private consumption deflator		1.0	1.7	0.8	1.4	1.6	2.0
Industrial production		**2.6**	**3.2**	**0.8**	**3.0**	**3.2**	**3.3**
Unemployment rate[4]		**8.4**	**8.1**	**8.5**	**8.3**	**8.2**	**8.1**
Current balance (US$ billion)		**–12.6**	**–12.2**	**–11.8**	**–13.4**	**–12.5**	**–11.8**

1. Current prices.
2. Excluding nationalised industries and public corporations.
3. Contribution to GDP volume growth.
4. As a percentage of the labour force.

Source: OECD Secretariat.

in Canada's terms of trade comes to a halt. Although wage growth is projected to pick up somewhat as unemployment declines gradually toward its estimated structural rate, and despite the effects of exchange rate depreciation on import prices, inflation is expected to remain low – and consistent with the official target – during the projection period, given a continuing degree of excess supply in the economy and price competition from Asian producers.

The above projections, which are those published in *December 1998 OECD Economic Outlook*, embody the following assumptions:

- Real GDP in the OECD area grows by over 2 per cent in 1998 and less than 2 per cent in 1999, while the expansion of world trade accelerates from under 5 to 5½ per cent.
- With constant nominal exchange rates from October 1998, Canada's import prices rise at an annual rate of about 2 per cent in the period ahead.
- The average dollar price of OECD oil imports (cif) is estimated to average $13 per barrel in the second half of 1998 and projected to rise somewhat thereafter, averaging $14 during the second half of 1999.
- The cyclically adjusted general government financial balance improves by around ¾ per cent of GDP in 1998 and remains stable thereafter.
- Short-term interest rates decline gradually as their US counterparts fall. Long-term interest rates increase somewhat in line with corresponding US rates.

Risks and uncertainties

The outlook is subject to a number of risks, and uncertainties surrounding it are greater than usual. These relate mainly to events beyond Canada's borders, in particular the turmoil that has affected many Asian countries and has spread to other emerging economies where the situation is still evolving. A deepening crisis would carry the risk of slower growth in Canada, although, as discussed earlier, the relatively small share of Canadian exports absorbed by the Asia Pacific economies limits direct trade effects from a possible further worsening in prospects for that region. A more prolonged negative impact of the Asia crisis on commodity prices would adversely affect business profits and investment in Canada but, again, the damping effect on the overall economy should not be overestimated, given the declining share of the natural resource sector. Nonetheless, the fact that markets still perceive Canada as a commodity producer could complicate monetary management, being a major factor behind the recent weakness of the Canadian dollar and associated upward pressure on interest rates. However, given the close links between the Canadian and US economies, the most important risk on the external side would seem to be a marked deterioration in the outlook for the United States, whether resulting from turbulence in the world economy or other factors, such as a stock market crisis.

Domestically, the risks surrounding the outlook appear to be less important. Nonetheless, although sluggish activity in recent months reflects temporary factors (notably labour conflicts) it is possible that domestic demand could prove weaker than projected. Consumer spending, in particular, could, in view of the low level of the personal saving ratio, be subject to greater-than-expected retrenchment, especially in the case of a continued sharp stock market correction. Indeed, ongoing financial market volatility could adversely affect private sector confidence, which has been a major factor behind better economic performance in recent years. Continued sound economic policies should limit these downside risks, however.

The medium-term outlook

The baseline scenario

Although the outlook is not without considerable risks, Canada's medium-term prospects appear favourable. An OECD Secretariat central scenario (Table 6) is that, over the 1998-2003 period, real GDP growth could average 3 per cent, almost double the rate achieved so far over the 1990s and similar to the performance in the 1980s. This compares with 2¾ per cent growth in the OECD area (up from 2¼ per cent). Underlying this profile is a stronger expansion of domestic demand, which is expected to fall into line with longer-term trends following its weakness in the first half of the 1990s. This is true for all major demand components, except public spending which is projected to keep growing at slower pace than in the 1980s. Under the assumption of a constant real exchange rate from 2000, exports are expected to grow in line with markets, implying little change from past performance. Although higher import growth associated with stronger domestic demand leaves little room for an improvement in the external balance, the current account deficit should remain low by historical standards. With labour force participation rising to its trend level, unemployment is projected to fall only slowly, staying above its structural rate which is also expected to edge down in response to labour market reforms. The output gap is assumed to close only gradually, with actual output never overshooting productive potential, which is estimated to be expanding at a rate of 2¾ per cent per annum reflecting a rebound in total factor productivity. At the same time, inflation would remain low, converging toward the mid-point of the target range.

Table 6. **A medium-term scenario**

Average annual percentage change, volumes

	1989-1997	1997-2003
Private consumption	1.8	2.7
Government consumption	0.5	1.0
Fixed capital formation	1.0	6.0
Stockbuilding[1]	0.0	0.0
Total domestic demand	1.4	3.1
Exports	7.7	7.3
Imports	6.7	7.7
Real foreign balance[1]	0.3	–0.1
Real GDP	1.6	3.0
Output gap[2]	–1.4	–0.2
Potential output	2.2	2.7
Labour productivity	0.8	1.2
Employment	0.8	1.7
Participation rate (per cent)	65.9	65.9
Unemployment rate (per cent)	9.7	8.2
Private consumption deflator	2.2	1.6
GDP deflator	1.7	1.8
Private compensation per employee	3.2	2.8
Current balance[1]	–2.6	–2.4
Real exchange rate	–3.0	–0.2
Short-term interest rate (per cent)	7.4	5.3
Long-term interest rate (per cent)	8.7	6.0

1. Per cent of GDP.
2. Per cent of potential output.

Source: OECD Secretariat.

Policy challenges

Whether this favourable scenario can be realised will – apart from the external environment, which has become more uncertain – crucially depend on economic policies pursued in Canada, both in the macroeconomic and structural areas. Monetary policy, in particular, will play a central role, facing the challenge to ensure that monetary conditions are set on a path consistent with output converging smoothly to its potential level without triggering a significant rise in inflation, as assumed in the above scenario. In the near term, given the weaker and more uncertain outlook for the world economy, this implies that policy

tightening should be put on hold. Looking further ahead, though, when the closing of the output gap resumes, a reversal of such a stance would be needed at some point. Indeed, an overshoot of potential would require corrective action[7] to deal with resulting inflation pressures, leading to a more cyclical outlook for the economy and possibly lower growth on average over the medium term.

The inflation-targeting framework in place has contributed to a significant improvement in price performance in recent years, but has not yet been tested under conditions where the economy operates close to full capacity. Given the forward-looking nature of the framework, the problem facing policy makers in such circumstances is that they will have to reconcile the need for early action to deal with possibly emerging inflation pressures with the fact that there is a wide margin of error around the estimates of the major variables on which their assessment of future inflation relies. On the one hand, empirical findings for Canada that policy actions have their main effects on inflation with a lag of six to eight quarters suggest that the authorities should err on the side of caution. On the other hand, in the light of the successful US experience in this respect, it could be argued that, given the substantial uncertainties in the estimates of the economy's capacity and the structural rate of unemployment, monetary policy should probe to see whether, as a result of structural reform, the economy could reach higher levels of output and lower levels unemployment without inflation pressures.

The MCI has proved to be a useful conceptual and practical tool in the conduct of policy as it draws attention to the fact that, for a small open economy, the exchange rate is an important monetary transmission channel. However, because the appropriate level of the MCI varies in response to economic shocks, the market has had difficulties at times interpreting the Bank of Canada's policy actions. For instance, an exchange rate depreciation driven by a negative terms-of-trade shock, as experienced recently in Canada, might actually have a net contractionary effect on aggregate demand, in which case there would be no need for a compensating increase in interest rates. The challenge in such circumstances is to ensure the market understands why the Bank does not try to bring the MCI back to its original level. While it is important to allow flexible responses to economic shocks, it is not clear that market-generated changes in monetary conditions following a terms-of-trade deterioration are exactly those called for to achieve inflation targets. These limitations of the MCI as an operational guide are

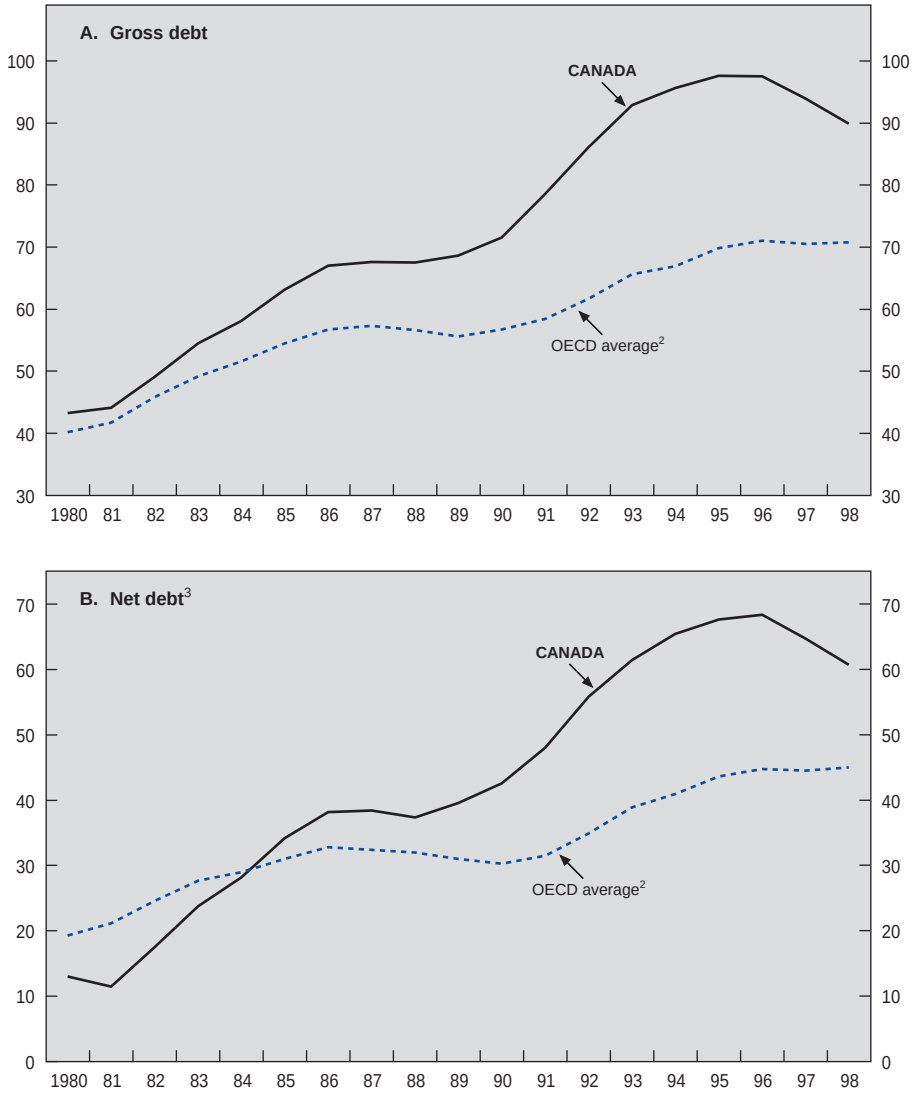
well understood by the monetary authorities who have tried to cope with the signalling problems involved.

Fiscal policy will also have an important role to play in ensuring that recent improvements in economic performance are sustained. The uncertain external environment heightens the importance of avoiding renewed budget deficits which might lead to upward pressure on interest rates. By reducing its demand on savings further the government would contribute to maintaining interest rates at the low levels required to achieve the higher rates of capital spending and potential output growth envisaged in the above scenario. Although Canada's approach of two-year fiscal targets combined with prudent planning assumptions (see previous Surveys) has been successful in eliminating budget deficits, the establishment of a medium-term anchor for budgetary policy in the form of a debt-to-GDP target might help in this respect. While there are no easy answers concerning the pace of debt reduction the government should choose,[8] bringing the public debt ratio in line with the OECD average over the medium term would appear to be a first minimum objective, as it should serve to reduce vulnerability to changes in financial market sentiment. As can be seen from Figure 7, despite its recent decline, Canada's public debt-to-GDP ratio is still high by international comparison. At the current pace of debt reduction it would approach the OECD average and US level – which are also expected to decline somewhat – toward the end of the medium-term projection period.

Running budget surpluses does not exclude tax cuts and expenditure increases, however, as the associated fall in the debt burden reduces interest payments. Targeted spending initiatives in some areas (such as human capital development) could serve to enhance economic efficiency. Nonetheless, priority should be given to tax reductions. As shown in Figure 8, over the past four years, tax collections have reduced disposable income growth by 1½ to 2 percentage points each year, with transfers providing only a small offset. While this has contributed to budget consolidation, the historically low personal saving ratio means that growth in *after-tax* income will be a key factor in the medium-term outlook. If average effective rates of personal direct taxes were not to decline significantly, it would be unlikely that the increase in disposable income would be strong enough to sustain the expected growth in household spending, despite the projected solid increase in employment.

Figure 7. **GOVERNMENT DEBT**[1]

Per cent of GDP

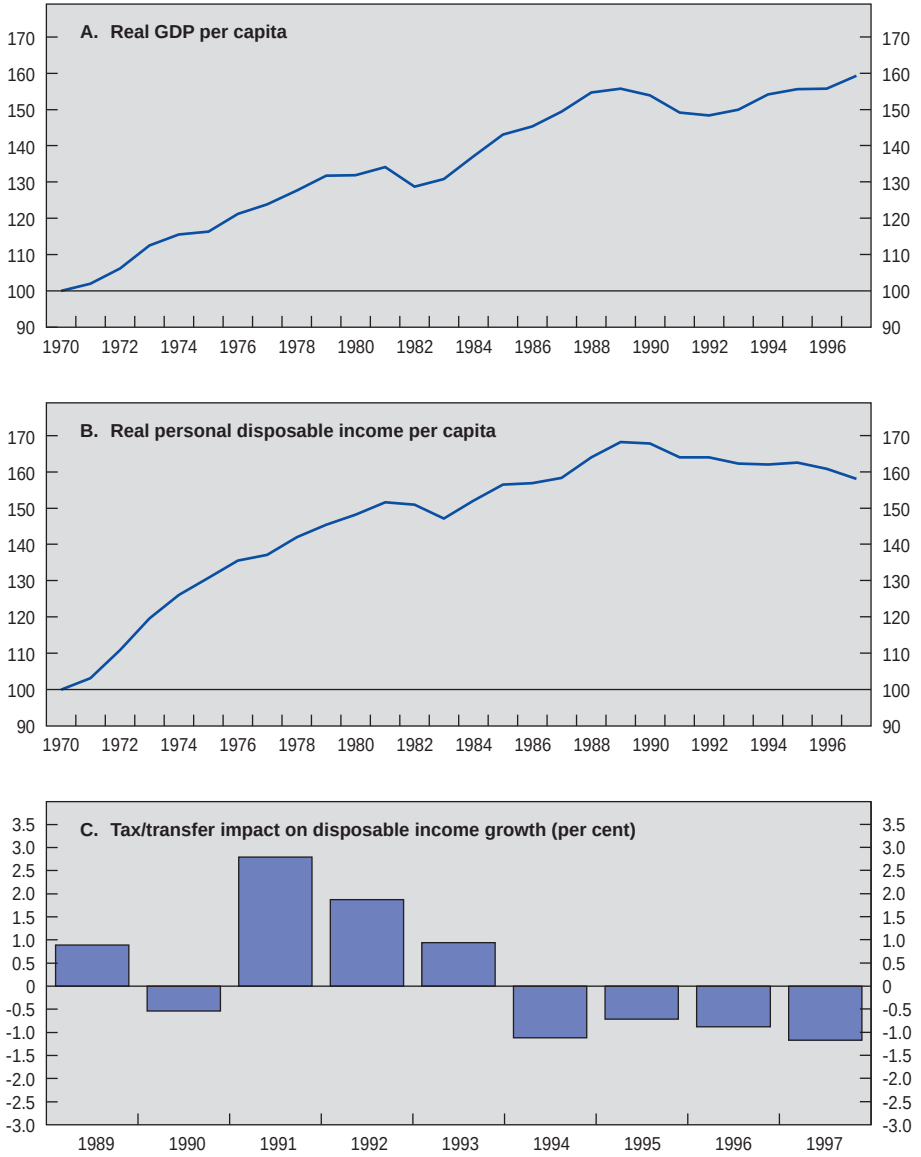


1. General government; *National Accounts* definitions. 1998 data are OECD projections.
2. Weighted average using 1991 GDP weights and exchange rates.
3. Financial liabilities less financial assets.

Source: OECD, *National Accounts*; Secretariat estimates.

Figure 8. **PER CAPITA INCOME**

1970 = 100



Source: Statistics Canada.

Reducing taxes is also important for competitive reasons. Although Canada's tax burden is not excessive compared with the OECD average, it is much higher than in the United States (see last year's *Survey*). This pressure point needs to be taken into consideration. As noted, the 1998 Budget provides some modest tax relief. Options for further tax reductions are discussed in the next Chapter. They include the elimination of surtaxes and re-introduction of effective indexation in the personal income tax system as well as an acceleration of reductions in employment insurance premia to compensate for the scheduled more rapid increase in pension contribution rates. In addition, it would be desirable to deal with several deficiencies and distortions in the corporate tax system.

Finally, Canada's growth prospects over the medium term will importantly depend on progress in structural policies. One important assumption underlying the above projections is that structural reforms have enhanced the efficiency of resource use. However, as noted, despite some encouraging signs, there is no hard evidence as yet that trend growth in total factor productivity has actually picked up. Hence, there is a risk that the structural reforms will take longer to produce the expected efficiency gains than assumed, implying weaker potential output growth over the projection period. Moreover, the above scenario assumes that some rebound in the labour force participation rate, along with a declining structural rate of unemployment, will allow stronger employment growth without leading to tightness in labour markets. Again, despite some rise recently, considerable uncertainty is still attached to the future development of the participation rate. In addition to a possible lack of productivity advancement, continued slow growth in labour inputs would limit the scope for sustained improvements in economic performance. Although it would seem reasonable to expect that past initiatives will increasingly bear fruit in the current more favourable economic environment, in the light of the above risks and uncertainties, there is no cause to relax efforts in the structural reform process in several areas, as discussed in the following Chapter.

II. Structural issues and policies

Over a number of years, structural policies in Canada have gradually moved away from direct intervention, toward providing framework conditions to enhance product and labour market flexibility and to encourage structural adjustment. As discussed in the previous Chapter, despite improvement more recently, economic performance considered over the longer term has been disappointing. This Chapter examines the impact of reforms undertaken in several areas, and then reports on a number of measures to boost future prospects. The discussion centres on issues deemed to have high structural policy priority, namely: future changes in the financial services sector; liberalising external and, more importantly, internal trade; taxation; pension reform; and the labour market.

Microeconomic reform to date: an overall assessment

In the OECD area, there has been, over time, a structural shift of value-added toward services and away from manufacturing, except in two countries: Sweden where the share of manufacturing value-added has increased slightly and in Canada where it has remained stable. Within the manufacturing sector, several mixed trends have emerged in Canada. Since the 1970s, there has been a small increase in the share of output accounted for by the so-called high- and medium-high technology industries, with a commensurate decline in the relatively lower technology sectors (Table 7). This compares with OECD economies, who have experienced a similar move (although the share of these industries in Canada is somewhat lower). Changes in employment parallel shifts in value-added, but Canada continues to employ comparatively more labour – and has a larger proportion of value-added – in the low-technology area. Even so, it has become increasingly internationally oriented in manufacturing, with both the export share of production and import penetration being considerably higher in the 1990s than

Table 7. **Indicators of structural change in manufacturing**[1]

	Canada			OECD[2]		
	1970-80	1980-90	1990-95	1970-80	1980-90	1990-95
Value added						
(as a share total manufacturing)						
High-technology	5.6	7.8	9.9	8.3	11.7	13.0
Medium-high technology	26.2	26.8	28.9	31.0	31.4	31.2
Medium-low technology	24.8	22.9	19.6	28.6	26.0	25.2
Low-technology	43.3	42.5	41.9	32.0	30.9	30.7
Employment shares						
High-technology	5.7	7.0	7.9	n.a.	10.0	10.2
Medium-high technology	23.1	24.1	25.0	n.a.	28.5	28.8
Medium-low technology	24.6	23.7	23.0	n.a.	25.0	24.8
Low-technology	46.6	45.2	44.1	n.a.	36.4	36.2
Export share of production						
High-technology	37.1	47.1	53.5	22.1	25.6	31.2
Medium-high technology	41.9	53.7	61.2	22.2	27.4	30.6
Medium-low technology	17.9	20.6	32.4	11.7	12.8	13.5
Low-technology	20.0	24.1	30.7	8.9	10.6	12.2
Import penetration[3]						
High-technology	51.1	62.3	66.1	17.7	23.4	30.0
Medium-high technology	51.9	60.0	65.7	16.1	22.5	25.9
Medium-low technology	17.3	19.8	32.3	10.7	13.5	14.2
Low-technology	12.6	14.2	20.7	10.1	12.5	14.9

1. The classification of industries comprising the manufacturing sector by level of technology is based on OECD Science Technology and Industry, scoreboard of Indicators. See Annex III for details.
2. Fourteen OECD countries: Australia, Canada, Denmark, Finland, France, Germany, Italy, Japan, Netherlands, Norway, Spain, Sweden, United Kingdom and United States.
3. Defined as imports as a percentage of total domestic demand (production plus imports less exports).

Source: OECD, *Main Industrial Indicators.*

in the 1970s. These developments are the result of a number of factors: the integration of Canada's production structure with its trading partners, as reflected in high levels of intra-industry trade (see the section on trade below); the competitiveness of the manufacturing sector at home and abroad; and the liberalisation of the trade regime.

All this suggests that structural change in Canada, while not overly dynamic, is at least comparable in scope to developments in the average OECD economy. In one important respect – openness to trade – Canada' structural progress has been more impressive. However, given the importance attached in the growth and productivity literature to international trade and the prospects for technological exchange and spillovers, it is somewhat surprising that Canada's

Table 8. **Indicators of structural performance in selected OECD countries**

	Canada	United States	Japan	Germany	France	Italy	United Kingdom	Australia	Ireland	Netherlands	New Zealand	Norway
Potential output growth (annual average)												
1980-90	**2.7**	2.7	3.9	2.2	2.3	2.4	2.3	3.3	3.3	2.1	1.7	2.3
1990-98	**2.3**	2.3	2.2	3.5	1.9	1.8	2.2	3.0	6.7	2.8	2.5	2.1
Output gap (period average)												
1980-90	**–0.3**	–0.6	–0.3	–1.0	–0.6	0.2	0.8	–0.2	–1.0	–0.2	–0.5	–0.3
1990-98	**–1.7**	0.0	–0.6	0.0	–1.4	–1.0	–0.5	–1.4	–0.3	0.4	–2.3	–1.3
Structural unemployment rate (period average)[1]												
1980-90	**8.6**	6.5	2.5	6.4	8.5	8.1	8.9	7.6	12.4	7.2	4.5	3.0
1990-98	**8.8**	5.7	2.7	8.9	9.8	10.4	7.8	7.8	12.7	6.1	6.7	4.8
Labour productivity growth (annual average)												
1980-90	**1.1**	1.1	2.8	1.7	2.1	2.1	1.9	0.9	3.8	1.3	2.2	1.8
1990-98	**1.0**	1.1	0.9	0.8	1.5	1.8	1.7	2.0	3.9	0.9	0.8	2.6
Total factor productivity growth (annual average)[2]												
1980-90	**–0.4**	0.8	1.7	1.3	1.7	1.2	1.8	0.7	3.8	1.2	0.9	–0.2
1990-96	**–0.4**	0.5	0.5	–0.1	0.8	1.3	0.9	1.6	3.7	0.8	1.3	2.7
Employment rate (period average)[3]												
1980-90	**68.4**	68.1	71.0	63.5	60.9	54.1	67.8	66.2	54.0	52.7	62.4	75.4
1990-98	**68.9**	72.3	74.3	63.1	59.3	51.7	69.4	67.6	54.8	57.3	59.3	74.4
Business investment (share of GDP)												
1980-90	**9.4**	9.9	15.4	11.8	11.9	10.6	11.0	14.7	11.6	12.0	9.8	18.9
1990-98	**10.4**	10.4	17.6	12.9	11.9	10.8	12.0	13.8	9.8	12.5	12.0	12.5

Table 8. **Indicators of structural performance in selected OECD countries** *(cont.)*

	Canada	United States	Japan	Germany	France	Italy	United Kingdom	Australia	Ireland	Netherlands	New Zealand	Norway
Openness to trade (share of GDP)[4]												
1980-90	**42.6**	16.2	18.7	50.7	46.3	32.2	46.3	30.6	99.8	94.2	47.7	68.2
1990-98	**61.3**	24.2	23.4	57.7	55.3	41.3	55.4	44.4	174.8	113.0	58.1	80.7
Business R&D investment (share of GDP)[5]												
1980-90	**0.6**	1.4	1.6	1.6	1.9	0.4	1.1	0.3	0.3	1.1	n.a.	0.7
1990-95	**0.9**	1.9	2.1	1.7	2.9	0.8	1.6	0.8	0.9	1.1	n.a.	0.9

1. The structural rate of unemployment is defined as the NAWRU, or non-accelerating wage rate of unemployment.
2. Total factor productivity is that of the business sector. For Italy, the United Kingdom, Australia and New Zealand the end date is 1995; for Germany 1993; and 1994 for Norway.
3. Total employment as a percentage of the working-age population.
4. The ratio of exports and imports to GDP.
5. For Canada and Italy the end date is 1997; for Germany 1993; and 1996 for Ireland.

Source: OECD Secretariat estimates; OECD Analytical Database; OECD STAN Database; and OECD Science and Technology Indicators.

economic performance – in underlying terms – is not altogether better. This is reflected in Table 8 which shows that, compared with the major and fast growing smaller OECD countries, Canada has not kept pace on several accounts over the 1990s: total factor productivity has not increased and instead has declined, an experience which is shared with no other country in the table;[9] structural unemployment remains relatively high and increased slightly in the early 1990s, before declining recently; and potential output growth is lower than a decade ago, having slowed – as estimated by the OECD Secretariat – from around 4 per cent to just above 2 per cent over the past thirty years (before rebounding to 2½ per cent recently). The latter is due, in large part, to the negative contribution from productivity developments and a substantial decline in the growth of labour inputs, with a somewhat smaller negative influence from a slower expansion of the capital stock (Table 9). Indeed, in the 1970s, the increase in labour supply contributed around 2½ percentage points to potential output growth but this fell to only 1 per cent in the 1990s, reflecting both demographic factors and lower labour force participation.

The lack of productivity advancement combined with both the slow growth of labour inputs and the capital stock, could, if ongoing, lead to a substantial decline in Canada's per capita income (measured in purchasing power parities), relative to the OECD average. Using current estimates of factor inputs, productivity and population growth (produced by Statistics Canada), a long-term scenario constructed by the OECD Secretariat highlights this possibility under the ''status quo'' (Figure 9, details of the scenario are outlined in Annex V).[10] In this scenario – where Canadian growth averages 2¾ per cent per annum from 1998-2003 and falls toward 1½ per cent in the year 2020 – the relative decline in Canada's real per capita income below the OECD average is driven by slow productivity growth and importantly, by falling labour inputs.[11] Indeed in a high growth scenario, where productivity growth is assumed to rise above that in the United States, the structural rate of unemployment to decline, and the growth in labour inputs not to fall-off so significantly, the Canadian standard of living is still projected to decline from the current level of around 110 per cent to just below the OECD average. In short, because of relatively stronger population growth in Canada than elsewhere in the OECD area, potential output growth must substantially exceed that in other countries if the relative Canadian level of real per capita income is to be sustained.

Table 9. **Contributions to potential output growth**[1]

Percentage changes at annual rates

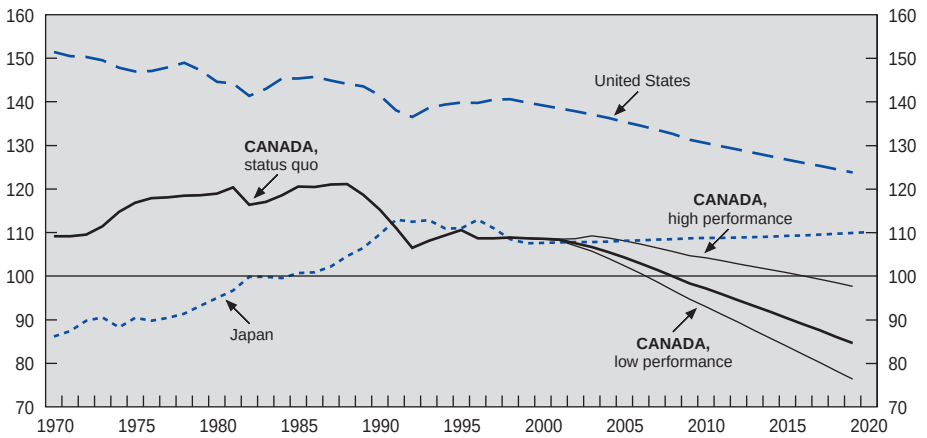
	1971/80	1980/90	1990/97
Trend productivity	0.1	–0.3	–0.1
Capital stock	1.8	1.7	1.3
Trend participation rate	0.7	0.5	0.0
Working-age population	1.6	0.9	1.0
NAWRU[2]	–0.3	–0.1	0.1
Potential output	3.9	2.7	2.2
Memorandum item:			
Real GDP	4.2	2.8	1.8

1. Details concerning the OECD Secretariat's method of calculating potential output can be found in Annex IV.
2. NAWRU is defined as the non-accelerating wage rate of employment.

Source: OECD Secretariat estimates.

Figure 9. **PROJECTIONS OF GDP PER HEAD IN SELECTED OECD COUNTRIES**[1]

Current PPPs, OECD = 100



1. See Annex VI for details of the assumptions underlying the projections.

Source: OECD, *National Accounts* and Secretariat estimates.

In light of the significant microeconomic reforms Canada has pursued since the mid-1980s,[12] it is difficult to understand such relatively weak performance, especially in the growth of total factor productivity. Nevertheless, a number of elements might have played a role in this regard:

- *First,* there is resistance to change in some sectors and the adjustment to new policy settings and economic incentives take time. How long it takes is uncertain, although the timing does not appear to depend upon the speed with which the structural environment shifts: radical reforms (such as in New Zealand) seem to yield results at about the same speed as those implemented progressively (such as in Australia).
- *Second,* over the 1990s there has been a wide margin of unused capacity in Canada and this has probably contributed to poor structural outcomes, given the deterioration of the stock of physical, technological and human capital that this entails. It is important to note that the extent of this under-utilisation of resources is not evenly distributed across the country, and because of regional linkages this might have implications for aggregate results. In large parts of Quebec and Atlantic Canada, both the mobilisation of resources and economic efficiency are poor, particularly in labour markets; employment-to-population ratios and labour force participation are well below the national average, while unemployment is above that average; and productivity levels show wide scope for catching-up (Table 10).
- *Third,* some of the conditions felt to be favourable to productivity and growth – such as high rates of capital accumulation, technological innovation and adaptation, and high levels of research and development – are less apparent in Canada than in better performing economies.
- *Fourth,* relative to other OECD countries in general, and to the United States in particular, Canada has a high share of small-and medium-sized enterprises and these appear to be less internationally oriented, less innovative and as a result less productive than their counterparts in other countries.

The above discussion points to a number of structural policy priorities for Canada and action required to arrest, or at least to slow, the anticipated decline in relative living standards noted in the above scenarios.

Table 10. **Indicators of provincial economic performance**

	Canada	New-foundland	Prince Edward Island	Nova Scotia	New Brunswick	Quebec	Ontario	Manitoba	Saskat-chewan	Alberta	British Columbia
Real GDP per capita (1992 prices, thousands)											
1981	22.11	13.20	14.12	15.54	14.87	19.68	24.12	19.25	17.82	26.54	24.67
1997	26.43	18.03	20.26	20.25	20.64	23.56	28.93	23.91	25.40	31.92	25.10
Growth of real GDP (1992 prices, average annual)											
1981-1989	3.2	3.1	3.1	3.7	3.4	2.7	4.3	2.6	1.8	2.2	2.2
1989-1997	1.5	0.6	2.7	0.9	1.6	1.1	1.2	1.4	3.2	2.8	2.1
Equalisation entitlement (share of GDP, period average)[1]											
1981-1989		9.6	10.1	5.7	6.9	2.8	0.0	3.2	1.0	0.0	0.0
1989-1997		9.5	7.9	5.5	6.2	2.3	0.0	3.8	1.8	0.0	0.0
Labour productivity (level)											
1981	48.2	41.3	36.8	40.6	40.5	46.1	49.1	42.3	40.2	51.0	52.1
1997	57.3	52.9	45.9	48.9	49.7	53.7	61.0	50.7	54.7	62.1	53.5
Growth in labour productivity (average annual)											
1991-1989	1.4	1.6	1.3	1.7	1.7	1.2	1.9	1.5	1.1	1.5	0.6
1989-1997	0.7	1.5	1.5	0.7	0.8	0.7	0.8	0.8	2.8	0.9	–0.3
Employment rate (period average)[2]											
1981-1989	45.6	32.0	39.7	39.2	37.0	43.0	49.4	45.3	44.4	49.5	45.1
1989-1997	46.1	34.2	42.3	41.2	40.6	43.8	48.2	46.0	45.4	49.9	47.2
Employment growth (average annual)											
1981-1989	2.0	1.6	1.6	2.0	1.7	1.5	2.4	1.1	0.9	1.9	2.4
1989-1997	0.6	–0.7	1.0	0.2	0.7	0.3	0.3	0.5	0.3	1.5	1.9

Table 10. **Indicators of provincial economic performance** *(cont.)*

	Canada	New-foundland	Prince Edward Island	Nova Scotia	New Brunswick	Quebec	Ontario	Manitoba	Saskat-chewan	Alberta	British Columbia
Participation rate (period average)[3]											
1979-1989	65.7	53.6	61.3	59.0	57.1	62.3	68.5	66.0	65.4	72.0	65.5
1989-1997	65.9	54.0	65.6	60.6	59.4	62.9	67.6	66.8	66.4	72.1	66.4
Unemployment rate (period average)[4]											
1979-1989	9.2	17.1	12.6	11.7	13.1	11.2	7.3	7.5	6.6	7.8	10.9
1989-1997	9.7	18.7	15.9	12.3	12.3	11.6	8.7	8.2	7.2	7.9	9.3

1. Equalisation entitlements are based on the extent to which per capita tax revenues that would be collected by applying common tax rates to a representative definition of tax bases used across provinces falls short of the per capita revenue that would be collected by following the same procedure nationwide.
2. Total employment as a percentage of the population 15 years of age and older.
3. The labour force as a percentage of the population 15 years of age and older.
4. The unemployed as a percentage of the labour force.

Source: Statistics Canada, Provincial Economic Accounts.

- To the extent that the fiscal situation continues to improve, consideration must be given to *rolling back the relatively high average tax burden* faced by Canadians *and lowering high marginal income tax rates;*
- the anticipated decline in the labour force caused both by the slowing growth of the source population and a fall in participation due to demographic changes may lead to a slowing of potential output growth. Thus, in Canada (as in many other OECD countries), *augmenting the future supply of labour* will be of utmost importance;
- increased productivity growth will also be essential to enhancing the standard of living, suggesting that the *acquisition of skills and training as well as innovation* must be the subject of ongoing attention;
- the persistence of regional disparities suggests that some consideration needs to be given to *bolstering the mobility of productive resources and of goods and services across provinces*;
- for an economy so heavily exposed to international trade, *maintaining wide and secure access to world markets* and to those of its major trading partners should be considered an ongoing concern; and
- policy makers must deal with the *changing nature of competition* in various sectors in response to globalisation and rapid developments in information and communications technology. Nowhere is this more relevant than in the financial sector, which faces a number of challenges stemming, for example, from industry restructuring, the provision of services across financial institutions and proposals for banks' mergers.

The following paragraphs discuss the extent to which recent structural policy developments have dealt with these concerns.

Recent progress

Financial sector

The nature and scope of past changes

During the late 1970s and throughout the 1980s, pressures built for a major legislative restructuring of the financial sector as a result of increasing competition within the industry (facilitated by technological change), the lack of up-to-date legislation governing the activities of non-bank financial institutions,

changes in ownership structures of financial institutions, failures of a number of establishments in the 1980s, initiatives by provincial governments and the globalisation of financial markets. In 1980, Bank Act revisions permitted foreign banks to establish closely held subsidiaries in Canada with powers similar to Canadian banks. These banks were defined to be Schedule II banks. In 1987, provincial legislation removed restrictions on the ownership of investment dealers by domestic financial institutions, and federal legislation permitted banks and other federally-regulated financial institutions to own investment-dealer subsidiaries. In 1988, the restrictions on non-resident ownership of investment dealers were removed. In 1992, major reforms to the legislation governing all federally-regulated financial institutions were introduced further blurring the distinctions among the traditional ''four pillars''.

In general, the *1992* changes covered four areas. *First,* financial institutions were allowed to broaden the range of services and products they could offer, either directly through enhanced business powers, or indirectly through the acquisition or establishment of subsidiaries. Financial institutions were allowed to provide all financial services, except some services specifically excluded and, on a limited basis, some non-financial services incidental to their operations. For instance, banks and insurance companies were not allowed to offer trust services; nor could banks and trust and loan companies sell most insurance products, or insurance companies take deposits. However, as investment powers were expanded, those excluded services could be provided through financial subsidiaries and networking relations. For instance, banks and life insurance companies were allowed to own trust companies and banks and trust companies to own insurance companies. As a result, Canadian financial institutions have been able to develop into conglomerates with involvement in a variety of financial areas, but because of limitations on investments in non-financial businesses they have not been able to become (German-style) universal banks. *Second,* although the 1992 amendments retained the 10 per cent limit for individual ownership of Schedule I banks, domestic Schedule II banks are allowed to be controlled by individuals or commercial entities for up to a period of ten years from the time of incorporation, after which they must become widely-held.[13] Furthermore, the 1992 amendments introduced the requirement for any federal financial institution with more than C$ 750 million in regulatory capital to, within five years of reaching this level, have 35 per cent of its voting shares widely held and publicly

traded on a Canadian stock exchange. However, in some cases, ministerial exemption from the 35 per cent rule is available under the Bank Act. Third, the 1992 reforms strengthened corporate governance through rules on self-dealing, requiring financial institutions to establish a Conduct Review Committee comprised of a majority of directors that are not affiliated with the bank to ensure that procedures are in place for compliance with the new rules; and requiring some representation of unaffiliated members on the Board of Directors, as well as enhancing the duties and responsibilities of the current board of directors. *Lastly*, the 1992 changes opened the financial sector to further foreign participation by allowing foreign banks to establish unregulated finance companies subject to certain restrictions (for example, no deposit taking).

In *1995*, the government introduced measures to improve the supervisory and regulatory environment for federally-regulated financial institutions, the federal deposit insurance system, and federal oversight of clearing and settlement arrangements. A number of further amendments were put into place in 1997 designed to strengthen consumer protection, ease the regulatory burden on financial institutions and to keep the legislation current with evolving trends (see last year's *Survey* for more details). The government also announced that it would allow foreign banks to enter Canada using branches in addition to the subsidiaries, that had been allowed since 1980.

At the end of *1996*, the government created a Task Force on the Future of the Canadian Financial Services Sector, to provide guidance on major public policies affecting the sector, and on the establishment of a legislative framework suitable for the 21st century. The Task Force reported its findings in September (see Box 1). Its specific mandate was to make recommendations to enhance:

- the contribution of the sector to job creation, economic growth and the new economy;
- competition, efficiency and innovation within the sector;
- the international competitiveness of the sector in light of the globalisation of financial services, while at the same time maintaining strong, dynamic domestic financial institutions;
- the ability of the sector the take full advantage of technological advances as they occur and to meet the competitive challenges resulting from the introduction of new technologies; and
- the contribution of the sector to the best interests of Canadian consumers.

Box 1. **Task Force on the Future of the Canadian Financial Services Sector: Summary**

In September, the Task Force on the *Future of the Canadian Financial Services Sector* released its final report. The main conclusion of the report is that Canada's financial services sector is fundamentally sound, but that ''a rapidly changing marketplace has created a pressing need for a host of modifications so that the sector can effectively meet the needs and expectations of Canadian consumers and make its full contribution to the Canadian economy''. It contains 124 recommendations that address four main issues: enhancing competition and competitiveness; empowering consumers; corporate conduct; and improving the regulatory framework. (These recommendations are currently being reviewed by two parliamentary committees. The government will consider the committee's finding before making a final decision on the report's proposals.) Within these four areas, the main recommendations include:[1]

Enhancing competition and competitiveness

- Allowing financial institutions to compete across the financial sector, for example, by giving powers to credit unions to become or form banks, creating a new definition of wide ownership that provides scope for strategic alliances, and the ability for federal deposit-taking institutions to offer insurance and auto leasing through branches.
- Granting wider access (for life insurance companies, mutual funds and investment dealers) to the Canadian payments system.
- Creating a new consistent set of ownership rules for all financial institutions, including banks, that will make it easier and more attractive for new banks to start up.
- Eliminating, or modifying special capital taxes on financial institutions in order to make them more competitive and to remove disincentives to increase their capital base.
- With regard to mergers, the Task Force suggests that all proposals be assessed on their merits and with a particular view to whether they are in the public interest. It also puts forward a review process with the following steps: review by the Competition Bureau, the Superintendent of Financial Institutions and the Minister of Finance; a Public Interest Review for large mergers, which would require the proponents to produce a Public Interest Impact Assessment; and a clear ability in law for the Minister of Finance to accept enforceable undertakings from merger proponents, with severe sanctions for non-compliance.[2]

Empowering consumers

- Require the disclosure of transaction-specific fees and commissions so that consumers have additional information to allow them to compare prices and services and introduce a comprehensive, legislated privacy regime that will require finan-

(continued on next page)

(continued)

cial institutions to adopt legally binding codes of conduct that include minimum privacy standards for the sector.

- Ban coercive-tied selling (that is, the practice of requiring the purchase of a package of goods or services).
- Establish a financial sector ombudsman office, to which all federally regulated institutions would be required to belong.

Corporate conduct

- To improve financing for small and medium-sized businesses, banks should decentralise decision-making further to the regional and local levels, and lenders should be prepared to make available more innovative financing packages, including appropriate pricing for higher-risk borrowers who may now be denied financing.
- Establish an annual report on the state of small business finance, which would be submitted to the Industry Committee of the House of Commons.
- Make greater efforts to ensure user-friendly access to basic accounts by increasing the use of direct deposit of government cheques and providing such things as low-cost identification where desired. Require institutions to post in their branches, the terms and conditions of basic transactions accounts as well as the identification needed to open one.

Improving the regulatory framework

- Strengthening the mandate of the Office of the Superintendent of Financial Institutions by, among other things: bolstering its role in consumer protection and its responsibility to balance competition and innovation considerations; improving its governance structure by adding a board of directors to the Office; and by delegating solvency regulation to the that Office wherever possible.

---

1. Report of the Task Force on the Future of the Canadian Financial Services Sector, September 1998.
2. As noted in the text, a review of two proposed bank mergers is also proceeding. The government will await the findings of the Competition Bureau and the supervisory authorities, expected before the end of the year, before making any decision.

Finally, at the end of *1997*, Canada was among the 70 countries which concluded a World Trade Organisation Financial Services Agreement that aims at liberalising over 95 per cent of the global financial services market. Ratification of the agreement by each country is to take place by the end of January 1999, and its provisions to take effect in March of that year. In this agreement, Canada

committed to: maintaining its open financial services regime; allowing foreign banks to establish through branch offices, after implementation of the new bank branching regime; and removing, once legislation is passed, the requirement for foreign bank subsidiaries operating in Canada and originating in a non-NAFTA country to seek authorisation before opening additional branch offices.

The resulting structural adjustment

In the event, what has been the microeconomic impact of the numerous and far-reaching financial sector reform measures adopted by Canada since the late 1980s? Ultimately, the impact is difficult to assess due to problems in disentangling cause and effect. Nevertheless, given the extent of the restructuring that has gone on, it is reasonable to speculate that the reform programme – as well as other factors, such as globalisation, technological change and macroeconomic influences – played a prominent role. Four features of recent sectoral developments stand out:

- *First,* vigorous structural adjustment and rationalisation had been occurring in the financial sector since the early 1980s, thus an important consequence of the reform programme was to reinforce existing industry trends. This is evident in Table 11, which shows that in the period 1984 to 1991 the number of financial institutions declined in every sector (with the exception of securities dealers). In the post-reform period to 1996, the number of institutions of all types (again except securities dealers) dropped even further. The shake-out was particularly dramatic for trust and loan companies (the number of which fell from 133 to 31) and for life and health insurers (whose numbers fell from 257 to 128). This development is probably explained both by previous over-capacity and by the shrinking share of these intermediaries in total financial sector assets due to competition from other service providers. As well, the slump in commercial real estate in the early 1990s resulted in a number of failures of these institutions. Consolidation in the banking sector was less dramatic, the number of banks declining from 72 to 56. In contrast, for securities dealers the trend was in the opposite direction as their numbers swelled from 70 in 1984 to 172 by the end of 1996; and this reflects in part the impact of the 1987 reform and the degree to which previous legislation probably impeded earlier restructuring of this sector.

Table 11. **Number of financial institutions by sector**

Institutions	31 December 1984	31 December 1991	31 December 1996
Banks[1]	72	64	56
Domestic		8	11
Trust and loan companies[2]	133	78	31
Life and health insurers[3]	257	155	128
Domestic		82	57
Property and casualty insurers[4]	314	300	218
Securities dealers[5]	70	118	172

1. All banks are federally regulated.
2. Numbers relate only to federally-regulated companies which account for 95 per cent of sector assets.
3. Numbers relate only to federally-regulated companies which account for 90 per cent of sector premium volume.
4. Numbers relate only to federally-regulated companies which account for 76 per cent of sector premium volume.
5. All securities dealers are provincially regulated.

Source: Office of the Superintendent of Financial Institutions and Investment Dealers Association.

– *Second*, despite the openness of the financial industry to foreign and domestic participation, the reforms did not lead to a burst of new entrants (of federally-regulated institutions). This development is, however, perhaps not too surprising given the backdrop of ongoing rationalisation and the uncertain state of competition. As it stands, over the period 1992-98, eight new banks were established but nine either withdrew or closed; 16 life and health insurers arrived but 31 departed; and for property and casualty insurers 18 new firms entered and 38 either withdrew from the sector or closed down their business (Table 12). For trust and loan companies, on the other hand, twelve new entrants appeared and only seven departed. (No figures for securities dealers are available.) Thus, much of the consolidation within the industry has taken place through

Table 12. **Developments in federally-regulated financial institutions**
1992-1998

Type of institution	New entrants	Withdrawals/closures	Mergers	Acquisitions
Banks	8	9	17	4
Trust and loan companies	12	7	13	26
Life and health insurance	16	31	7	17
Property and casualty insurance	18	38	11	5

Source: Office of the Superintendent of Financial Institutions.

mergers and acquisitions. Lastly, in the banking and life and health insurers' sectors, foreign participation (measured by the relative market share of assets) has fallen.

– *Third*, there have been some significant changes in the market shares of various types of participants in the financial sector. Between 1990 and 1996, the share of the industries' total assets accounted for by both securities dealers and mutual funds had increased from around 6.5 per cent to above 21 per cent, while those of trust, credit unions, and life insurers collectively declined by over 14 percentage points to 22 per cent (Table 13). Meanwhile, the overall shares each accounted for by the banks and property insurers remained roughly constant, although the banks' position in the deposit-taking sector jumped significantly from around 70 to 82 per cent (at the expense of the trust and loan companies).

Table 13. **Assets of financial institutions held in Canada**

C$ million

Sector	1990	1996	1990-96 (average annual growth, per cent)
Banks	473.5	807.4	9.3
Share (per cent)	53.6	53.3	
Trusts	134.6	72.3	–9.8
Share	15.2	4.8	
Credit unions	72.4	104.4	6.3
Share	8.2	6.9	
Life insurers	113.9	157.5	5.6
Share	12.9	10.4	
Property and casualty insurance	32.3	49.3	7.3
Share	3.7	3.3	
Securities dealers	20.8	113.1	32.6
Share	2.4	7.5	
Mutual funds	35.5	211.8	34.7
Share	4.0	14.0	
Total	883.0	1 515.8	
Memorandum item:			
Deposit-taking sector	680.5	984.1	6.3
of which: (shares, per cent)			
Banks	69.6	82.0	
Trusts	19.8	7.3	
Credit unions	10.6	10.6	

Source: Department of Finance, Canada.

– *Fourth*, in general, the thrust of (the scant) empirical evidence suggest that the financial sector has become more efficient and competitive and that profitability, although high relative to other business sectors (except utilities, pipelines and consumer goods), is not above the perceived ''fair'' rate of return.[14] In banking for example, where high concentration exists, one study concluded that there is no overwhelming evidence of excessive profits, nor high market valuations, or significant deviations from marginal cost pricing.[15] (It has even been suggested that the return on equity in the financial sector is considered high only because performance in the rest of the corporate sector has simply been poor.)

In sum, there is ample evidence to indicate a rather strong response from industry participants – particularly across sectors – to the significant financial policy initiatives that were put into place a decade ago. Further, these reactions, and subsequent rationalisation are probably still ongoing, although with much less vigour than even a few years ago.

The scope for further restructuring

Since the beginning of 1998, four of the country's largest chartered banks have announced plans to merge: the Royal Bank of Canada with the Bank of Montreal which had combined assets of more than C$ 452 billion (in late 1997), and Canadian Imperial Bank of Commerce with Toronto Dominion Bank which had assets totalling more than C$ 401 billion. If the mergers are approved, the concentration of assets among the top three banks would amount to around 68 per cent, which is somewhat below several European countries such as Finland, and the Netherlands, but around the same level as Denmark and Sweden.

The proposed mergers must be approved by both the Bureau of Competition Policy and the Minister of Finance before they can proceed. Under the provisions of the Competition Act, the Bureau is required to examine a merger (if the combined sale revenue or assets exceed C$ 70 million) to determine whether it substantially lessens or prevents competition. This includes an evaluation of the market shares of the merged entities using threshold tests established under the Merger Enforcement Guidelines (35 per cent market share by product and geographic region for the merged entity and 65 per cent market share for the top four firms in the sector). Where the thresholds are exceeded, the Bureau undertakes an assessment of other factors – for example, the availability of acceptable substi-

tutes or market barriers to entry – to judge if the merged entity can unilaterally exercise market power or engage in collaborative activities with other market participants. This stage of the process is expected to take some months. As for Ministerial approval, the Minister has wide latitude in weighing costs and benefits. He also has the power to over-ride any Bureau ruling on a bank merger based on his assessment of the best interest of the financial system or of the public interest. In any case, the present Minister of Finance has stated recently that no decision will be taken until the Competition Bureau and supervisory authorities have made their findings known and specific public hearings have been held.

The economic case for merging, and more importantly for allowing the banks to amalgamate, rests on the fundamental assumption that a financial institution must be large to remain competitive in the future (global and domestic) environment. In looking ahead, some argue that only large banks will be able to:[16]

- maintain the infrastructure (technological) costs required to deliver financial services in years to come;
- take advantage of economies of scale in some parts of their operations, and be competent enough to provide all or most types of services to its customers in a sort of financial supermarket; and
- handle the kinds of transactions and deliver the kinds of services demanded by some customers at the global level.

While these propositions are plausible, they can be challenged to some degree.[17] *First,* the infrastructure investments in new technology need not come from only one provider, or for that matter, only from financial service providers. They could be developed jointly with other financial institutions, or with other entities such as telephone or cable companies. *Second,* empirical evidence has provided no evidence that a bank has to be a mega-institution, rather than just large, to exploit economies of scale; moreover, some of these economies can be exploited by outsourcing or by purchasing services from other providers. *Third,* the notion of a financial supermarket may not fit with customer preferences, for example, for those who would rather shop in separate institutions for different kinds of services, who do not want to rely on a single supplier, or who would face potential conflicts of interest from receiving advice on say, mergers and acquisitions from the same bank which stands to gain from handling the associated transaction. *Fourth,* there might in fact be some potential for diseconomies

of scale and scope (and not economies of scale) which could eventually lead to loss in focus and increasing cost. Thus, it is not clear that the global mega-institution will be the best at providing the efficient, innovative, and flexible service environment necessary to maintain high profitability.

Along with these general points, there are several issues in the Canadian context which pose further concern. These relate variously to the mergers' impact on the cost and availability of financial services for consumers, small businesses and for those in rural communities; and the uncertain impact on the rest of the financial sector, given the dominant position the large banks currently hold. (Some commentators are also worried about the potential for adverse employment effects. This concern relates less to the competitive implications of a merger, however, than it does to broader public-interest considerations.)

As well as dealing with the immediate issue of bank mergers, the government should move to address certain areas of financial sector legislation which were not adequately dealt with in the past set of reforms. Many of these issues have been discussed in some detail by the Task Force. Those with particularly high priority appear to be: relaxing limitations on ownership to permit more players (domestic and foreign) to enter the market; opening up competition in the sale of insurance products and auto leasing; levelling the playing field (by moving towards equivalent institutional powers) for all financial institutions, and between regulated and unregulated financial service providers; the harmonisation of federal and provincial regulations whose overlap imposes needless costs for governments and the private sector and reduces the efficiency of the financial industry; and wider access to the Canadian Payments System for all (appropriately regulated) financial institutions.

External trade

Trade policy in Canada has been the centrepiece of overall structural reform, in recognition of the important role freer trade plays as a driver of good economic performance. Since the late 1980s, Canada's commercial policy agenda has been particularly active on a number of fronts:

- In the period 1986-94, Canada participated actively in the Uruguay Round of multilateral trade negotiations under the auspices of the then GATT (now World Trade Organisation, WTO), where it committed to a host of liberalising measures in both traditional (goods) and new areas

(services and agriculture), and to a more rules-based approach to trade relations. In subsequent negotiations, Canada agreed with a limited set of WTO-member countries to open markets for Basic Telecommunications and Financial Services (see above).

- In 1989, Canada and the United States entered into a comprehensive Free Trade Agreement (FTA). In 1994, Canada, Mexico and the United States signed the North American Free Trade Agreement (NAFTA) that incorporated by reference certain specific elements of the FTA. (Canada's policy objectives on entering NAFTA included: the need to improve and protect the benefits of the FTA by ensuring equal treatment with the United States in the Mexican market; increased access to Mexico's market; and to preserve its standing as an attractive location for investment.) In 1997, Canada concluded separate free trade agreements with Israel and Chile, although the former is more limited in scope and coverage compared to the FTA/NAFTA. At the same time, the latter is similar to NAFTA and is largely designed to spur Chile's accession to it. More recently, Canada is playing a leading role in pursuing (together with its NAFTA partners) a Free Trade Area of the Americas, which would include much of South America.
- Canada has reduced a range of tariffs on a unilateral basis, for example: on certain consumer goods, textiles and manufactures, and so-called nuisance tariffs, or those below 2 per cent.
- Effective 1 January 1998, the Canadian government introduced a new Customs Tariff which reduced customs duties on a wide range of goods. It also resulted in a simpler, more transparent and more predictable tariff system by eliminating a number of regulations, reducing the number of tariff provisions (from 11 000 to approximately 8 000) and consolidating the several different schedules of the old Customs Tariff into a single schedule.
- Together with its APEC partners, Canada has committed to freeing trade and investment in the Pacific region by the year 2010.

Of all these initiatives, the NAFTA remains the key instrument of trade policy, given both the overwhelming importance of Canada-US trade and economic relations, and the relatively slow speed with which barriers to trade and investment are declining in other fora.

North American trade liberalisation

The FTA calls for the phased elimination of tariffs for most goods in ten steps over nine years which began in 1989. Thus, 1998 marks the year in which Canada-US trade will be virtually tariff free. (And as Table 14 shows, this will increase further the gap in duties between US and non-US imports into the Canadian market; further effort in reducing trade barriers would be desirable, such as those facing agricultural products affected by tariff quotas, textiles and footwear.) The policy objectives lying behind Canada's pursuit of free trade with the United States in 1985 were fourfold:[18]

- *security of access* to the United States market, particularly by reducing the risks inherent in the US system of restrictive trade measures (*e.g.* quotas, surtaxes, etc.) and the constitutional powers of the Congress to pass protectionist legislation affecting Canada;
- *expanded access* to the United States market in order to provide Canadian industry with a market large enough to realise economies of scale and specialisation, and to carve out niches for speciality products;
- *a stable North American trading system* which would encourage *an orderly transition* in Canada towards an economy more competitive at home and in world markets, and provide increased incentives for investment from all sources; and
- *an ordered and more predictable system* for managing the trade relationship and resolving disputes.

In the event, the Canadian side achieved much of what they sought in the FTA negotiations:[19] the FTA probably provides for a degree of trade liberalisation, and for the removal of many more trade irritants than any other international trade agreement of its kind; and for Canadian exporters to the US market, it increases access, makes that access more secure, and promises to add to that access in future (which it has done so in a much wider context by creating incentives for other countries to pursue free trade with Canada). Canada was not able to obtain a key objective; an agreed code to govern antidumping and countervailing duty practices. Instead, what was agreed to was a binational dispute settlement mechanism which deals with such issues through its role both as a legislative watchdog (should these laws change) and through the replacement of domestic judicial review with binational review.

Table 14. **Tariffs on US imports**[1]

	Total imports (C$ million)	Share of duty free imports (per cent)	Average tariff on dutiable imports (per cent)	Average tariff on total imports (per cent)
1981	54 537	68.9	12.84	3.99
1985	74 377	72.8	10.44	2.84
1989	88 096	72.1	9.39	2.62
1993	114 094	80.8	6.70	1.29
1997	184 309	83.0	1.8	0.32
Memorandum item:	Tariffs on all other imports			
1981	24 944	61.5	13.85	5.38
1989	47 095	53.8	11.99	5.72
1997[2]	89 506	70.0	8.9	2.65

1. Trade statistics on a customs basis.
2. Excluding trade with Mexico.
Source: Department of Finance, Canada.

The impact of the Free Trade Agreement ten years on is difficult to assess, given the problems of disentangling FTA-related effects with those emanating from the myriad of other forces (such as the movements in exchange rates Canada has faced) and economic policies (such as the adjustment associated with trade liberalisation at the multilateral level) which have been acting at the same time. Although causality cannot be precisely determined, an analysis of Canada-US trade developments can at least be suggestive of the broad (qualitative) outcomes thus far. A brief analysis of this kind follows. (Given NAFTA's recent implementation, the focus of the discussion is the FTA.)

The value of Canada's total merchandise trade (exports plus imports) with the United States has been much higher than with all other trading partners, expanding in the former case on average by 8.4 per cent per annum over the period 1981-88 and increasing to 10.3 per cent between 1988-95, while in the latter case increasing only slightly from 5.2 to 6.5 per cent over the same time frame. As a first approximation, this in itself is a very positive outcome, since a necessary condition for reaping the gains from liberalisation is an increase in the volume of trade. However, in the manufacturing sector – where much of the gains from free trade are expected – the growth in both Canada's exports and imports with the United States was less dynamic than for overall merchandise

trade, with exports and imports of manufactures moving ahead on average by 8.5 per cent per annum and 6.8 per cent, respectively, over the first seven years of the FTA (Table 15). Perhaps predictably from Canada's perspective, import growth from the United States has been strongest in the low-technology (relatively highly protected) sector, while somewhat surprising is that export expansion to the United States in the high-technology area (which was comparatively open in the pre-FTA period) was the fastest of the entire manufacturing sector (at over 14 per cent per annum).[20]

There appears to be considerable rationalisation of the Canada-US manufacturing (tradable) sector. In the decade prior to the FTA, there had been an overall decline in the specialisation (or revealed comparative advantage) of Canada's exports of manufactures (Table 15).[21] This decline was most pervasive in the medium-high technology area (motor vehicles, electrical machinery and chemicals, excluding drugs). This situation has been reversed somewhat in the post-FTA period, with all areas of manufacturing experiencing an increase in export specialisation in the US market, although not all the way back to the mid-1970s level. On the import side, the experience has been different. Canada's import specialisation (*i.e.* US revealed comparative advantage in Canada) in manufactures has increased throughout the pre-and post-FTA periods in the high and medium-low technology industries (particularly, drugs and medicines; office and computing machinery; radio, TV and communications equipment; and transport, other than motor vehicles and aircraft). For low-technology industries, US revealed comparative advantage increased in the post-FTA period, hence experiencing the same phenomena as Canadian exports in this area. Comparing the pre-and post-FTA periods, the overall structure of intra-industry trade has remained roughly stable – at a high level – although it has increased substantially in high and medium-high technology industries. This evidence suggests that the Canadian-US higher-technology sectors have become increasingly integrated and that there are significant economies of scale at play. Indeed, Canadian data on large business establishments (those with more than 500 employees) indicate a shift over the 1990s toward larger plant scales.

Trends in foreign direct investment (FDI) into Canada and by Canadians abroad are mixed. Between 1989 and 1997, Canadian FDI abroad had grown twice as fast as FDI into Canada (10 per cent per annum on average compared to 5.5 per cent, respectively). And the strong growth abroad was directed not at the

Table 15. **Trade developments with the United States**

Manufacturing sector[1]

	1981-88	1988-95
Growth of exports to the US (average annual)		
High-technology	10.5	14.1
Medium-high technology	12.9	7.7
Medium-low technology	6.4	8.9
Low-technology	8.8	8.3
Total manufacturing	10.1	8.5
Growth of imports from the US (average annual)		
High-technology	8.0	7.1
Medium-high technology	7.5	5.7
Medium-low technology	6.0	8.1
Low-technology	6.4	11.1
Total manufacturing	7.3	6.8

	1976-78	1986-88	1993-95
Export specialisation[2]			
High-technology	26.0	26.2	27.6
Medium-high technology	123.6	78.7	88.6
Medium-low technology	53.8	51.3	60.4
Low-technology	93.4	76.5	81.3
Total manufacturing	86.6	64.0	68.4
Import specialisation[3]			
High-technology	36.2	43.2	48.4
Medium-high technology	132.8	144.4	131.1
Medium-low technology	101.0	105.2	124.5
Low-technology	78.2	54.8	71.2
Total manufacturing	99.8	98.1	97.9
Intra-industry trade[4]			
High-technology	64.0	70.9	86.8
Medium-high technology	80.0	94.7	99.4
Medium-low technology	94.2	84.8	87.2
Low-technology	63.3	50.9	63.7
Total manufacturing	93.8	95.5	92.5

1. The classification of industries comprising the manufacturing sector by level of technology is based on OECD Science Technology and Industry, Scoreboard of Indicators, 1997. See Annex III for details.
2. Export specialisation (or revealed comparative advantage) shows the extent to which a country's' exports are specialised in a particular industry in another country's market. For a given industry, it is defined as the share of exports of that industry in a country's total manufacturing exports, divided by the share of total imports of the industry in its partner's manufacturing imports.
3. Import specialisation is the reverse of export specialisation from the trading partner's perspective. In this case, the US's export specialisation in Canada's manufacturing import market.
4. Intra-industry trade is a measure of two-way trade within the same industry. The index varies between 0 and 100, a higher reading indicating greater intra-industry trade (*i.e.* if a country exports and imports roughly equal quantities of a certain product).

Source: OECD, *Main Industrial Indicators.*

FTA partner, nor the United Kingdom (the two largest recipients of Canada's FDI) but at all other markets, where the share of Canadian FDI expanded from just over 24 to more than 37 per cent. Over the same period, the share of Canada's FDI outflows to the United States declined by over 10 percentage points (although its share of the inward US FDI stock did increase). Meanwhile, the share of US FDI in total inflows into Canada moved ahead by 4 percentage points, while on balance, that of other foreign investors declined. The effect on the direction of FDI flows stemming from the FTA was uncertain from the start, since FDI could either complement or substitute for international trade. On the one hand, free trade would provide less incentive for foreign firms to invest in Canada in order to circumvent high tariff walls, so that bilateral FDI could ease. On the other hand, if free trade spurred rationalisation of the industrial structure, FDI could be encouraged. Thus, there are several possible explanations for the trends observed above: 1) the FTA offers more secure access to the US market for Canadian exports, so trade flows provide a substitute for FDI from Canada's perspective; 2) given the existence of the FTA, foreigners choose to undertake their North American direct investments in the United States, rather than the Canadian market; and 3) increased opportunities in developing markets, especially in Latin America and Asia, have diminished somewhat the relative importance of the United States in Canada's direct investment abroad.

Canada-US trade relations are without doubt the most intense compared with any other country pair. Two-way Canada-US trade amounted to around $327 billion in 1997, compared for example, with US-Japan trade of $184 billion and US-Germany trade of $69 billion. Given the breadth of the Canada-US relationship, trade conflicts inevitably arise although such problems affect only a small proportion of total cross-border flows of goods and services. The FTA (and NAFTA) provides two mechanisms for resolving trade disputes: one in the event of disagreement over its interpretation and/or implementation (FTA, Chapter 18); and the other for those arising from the application of each country's national laws governing antidumping and countervailing duties (FTA, Chapter 19). From the Canadian perspective, these dispute settlement mechanisms were, at the time the FTA was negotiated, regarded as the best available means of gaining more secure access to the US market.[22]

On balance, both aspects of the dispute settlement mechanism have worked reasonably well. With regard to Chapter 19 disputes, procedures have been

Box 2. **Issues arising from the Dispute Settlement Procedures of the Free Trade Agreement**

There were two mechanisms for resolving trade disputes in the Canada-US Free Trade Agreement (FTA): a general one to handle problems arising from the FTA's interpretation and/or implementation (Chapter 18); and a specific one for reviewing the application and adjudication of each countries' national laws in respect of antidumping and countervail (Chapter 19). On balance, both mechanisms have functioned smoothly and a large number of relatively routine disputes (especially under Chapter 19 provisions) have been dealt with expeditiously and in a fair manner. However, neither element of the dispute settlement machinery has proceeded without controversy, given that there are always opposing views in dispute settlement cases. The following examples highlight some of the issues involved.

Salmon and Herring from Canada (Chapter 18): In this case, Canada required that 100 per cent of Pacific salmon and roe herring caught in its waters had to be landed in that country as well. Canada justified this rule on the grounds of conservation and habitat management. The United States argued that the 100 per cent landing rule violated the FTA and requested a binational review panel to be struck to examine it. In the event, the panel recommended a compromise solution.

Lobsters from the United States (Chapter 18): Canada objected to a US measure restricting trade of under-sized lobsters, which the United States argued applied equally to foreign and inter-state sales in accordance with the national treatment provisions of Article III of the GATT, Article 407 of the FTA. Canada took the position that such a restriction contravened Article XI of the GATT (also Article 407 of the FTA). In a 3 to 2 vote, the panel decided the US restriction conformed to the Agreement.

Softwood Lumber from Canada (Chapter 19): The United States applied a countervailing duty on softwood lumber imports from Canada, determining that certain practices (timber pricing and log exports) constituted a subsidy to Canadian exporters. A binational panel reviewed the final subsidy determination made by the US Commerce Department; the US Commerce Department then made a ''no subsidy'' determination on remand. A bilateral Extraordinary Challenge Committee rejected conflict of interests arguments raised by the United States with respect to the participation of two Canadian members of the panel and upheld the binational panel's ruling. Duties the United States had collected were refunded. Following the conclusion of the Extraordinary Challenge Proceedings, the Coalition for Fair Lumber Imports (representing the US industry in the investigation and binational panel review process) filed a constitutional challenge of the Chapter 19 dispute settlement system in the US domestic courts. This constitutional challenge was ultimately withdrawn in December 1994 when Canada and the United States agreed to establish an industry-government consultative mechanism intended to resolve the long-standing dispute. In 1996, they entered into an agreement whereby Canada would limit exports of softwood lumber to the United States, in exchange for US undertakings not to take trade

(continued on next page)

(continued)

action against imports of softwood lumber from Canada while the agreement was in effect. However, the resolution of this latter dispute might point to some limitations with the binational process, which does not constrain the application of trade remedy laws in either country, but only ensures that such laws are applied in accordance with domestic jurisprudence.[1]

1. See for example, Clendenning, E. and R. Clendenning, ''Analysis of International Trade Dispute Settlement Mechanisms and Implications for Canada's Agreement on Internal Trade'', Industry Canada, Occasional paper Number 19, 1997 and Robert Howse, ''Settling Trade Remedy Disputes: When the WTO Forum is Better than the NAFTA'', Commentary No. 111, C.D. Howe Institute, Toronto, June 1998.

followed expeditiously (unlike in the past), both partners have been active in pursuing appeals, and importantly, outcomes have demonstrated a consistent and fair application of each country's trade laws, thus offering some measure of constraint in their use as potent non-tariff barriers to trade.[23] With regard to Chapter 18 disputes, only a handful of issues have been referred to date, indicating a high degree of compliance with the spirit and intent of the Agreement's provisions. Nevertheless, the adjudication of disputes under this mechanism has not been entirely free of controversy (see Box 2). It is important to ensure that the Agreement's dispute settlement mechanism functions effectively, in order to achieve the credibility and certainty it was intended to provide.

Finally, up to now, there does not appear to have been a significant improvement in Canada's structural performance (see Table 8), even in the presence of the FTA. While the FTA may have led to improvements in sectoral productivity or cost performance through rationalisation, economies of scale or specialisation, these gains have been apparently masked at the aggregate level. As noted earlier, productivity growth for the total economy has been lagging in Canada relative to the United States. But even in manufacturing – the front and centre of international trade and an area subject to increasing returns – the situation is even somewhat worse. (Although it should be pointed out that while productivity performance in this sector has not been as robust as previously expected, it may be the case that the situation would have been worse had the FTA not been put into place.) Table 16 shows manufacturing productivity in *levels* between 1987

Table 16. **Manufacturing labour productivity levels in major OECD economies**[1]

Value added per hour worked, leader country = 100

	Canada	United States	Japan	Germany	France	United Kingdom	Australia	Netherlands	Sweden
					1987				
Food, beverages and tobacco	**59.6**	**100.0**	32.3	75.3	65.3	46.1	45.9	95.4	57.3
Textiles, clothing and footwear	**54.6**	67.4	38.1	60.1	61.7	47.7	42.2	**100.0**	60.8
Wood products and furniture	**63.8**	69.5	15.6	50.2	52.4	38.1	32.7	**100.0**	64.1
Paper products and printing	**81.4**	97.2	47.5	61.2	65.0	64.7	53.2	62.7	**100.0**
Chemical products	**68.0**	80.8	52.9	60.1	58.0	59.5	44.9	**100.0**	72.4
Non-metallic mineral products	**75.1**	77.0	55.1	67.1	**100.0**	59.9	56.4	97.7	75.5
Basic metal products	**89.3**	94.4	**100.0**	80.3	77.0	74.2	57.1	80.3	93.3
Metal products	**70.1**	86.3	76.0	76.3	57.3	50.6	42.3	68.9	**100.0**
Machinery and equipment	**64.2**	99.0	85.6	73.8	**100.0**	65.4	61.1	59.1	66.5
Electrical machinery	**66.4**	**100.0**	82.7	67.6	90.0	51.3	35.8	93.7	75.6
Transport equipment	**69.7**	96.9	**100.0**	76.7	84.9	42.1	39.3	47.0	55.8
Other manufacturing	**58.3**	**100.0**	39.4	45.3	40.1	52.5	33.0	47.2	67.0
Total manufacturing	**76.0**	**100.0**	66.5	78.5	80.3	59.4	51.8	98.5	82.0
					1993				
Food, beverages and tobacco	**64.3**	**100.0**	35.6	88.7	87.0	41.7	51.1	96.6	72.8
Textiles, clothing and footwear	**46.3**	78.3	41.9	72.1	67.1	51.5	32.3	**100.0**	66.5
Wood products and furniture	**52.6**	56.0	17.6	55.7	55.3	28.1	27.1	**100.0**	71.5
Paper products and printing	**67.6**	85.0	49.7	59.0	64.3	76.4	53.7	64.5	**100.0**
Chemical products	**52.6**	66.9	52.6	55.4	56.9	79.7	39.8	**100.0**	89.4
Non-metallic mineral products	**78.4**	81.8	62.9	79.5	99.4	70.6	77.4	**100.0**	81.0
Basic metal products	**87.9**	76.8	78.3	72.9	63.3	61.4	56.8	70.4	**100.0**
Metal products	**54.8**	68.9	67.6	64.6	46.4	42.5	35.9	54.0	**100.0**
Machinery and equipment	**55.5**	**100.0**	67.4	49.2	67.3	47.9	46.4	34.6	45.2
Electrical machinery	**51.9**	80.3	89.0	49.9	78.9	48.2	28.0	82.2	**100.0**
Transport equipment	**71.9**	88.4	**100.0**	68.0	85.0	47.8	45.5	41.8	49.5
Other manufacturing	**33.5**	100.0	41.4	43.1	31.4	43.5	22.1	27.0	47.4
Total manufacturing	**71.3**	**100.0**	76.6	79.8	84.2	64.1	52.0	95.6	91.8

1. The productivity level of the leader country in each industry is indicated in bold. The sectors are those corresponding to OECD's STAN Database.

Source: Pilat, Dirk (1997), ''Competition, Productivity and efficiency'', *OECD Economic Studies*, No. 27.

Table 17. **Relative levels of unit labour costs in manufacturing**

USA = 100, based on 1990 PPPs

	1975	1985	1990	1996
United States	100	100	100	100
Japan	91	74	116	169
Germany[1]	104	71	144	166
France	148	96	154	163
Italy	107	60	114	101
United Kingdom	125	100	158	148
Canada	105	84	118	102
Australia	151	98	118	145
Belgium	167	75	135	156
Denmark	162	97	205	218
Korea	26	29	51	58
Netherlands	131	65	122	120
Spain	74[2]	49	108	100
Sweden	144	82	158	160

1. West Germany.
2. 1977.

Source: Durand, Martine *et al.* ''Trends in OECD countries international competitiveness: the influence of emerging market economies'', OECD Economics Department Working Paper No. 195, Table 3, 1998.

and 1993. Over that period, the United States remained the overall productivity leader, while Canada's performance slipped 5 percentage points to just over 71 per cent of the US level.[24] And this decline has been pervasive, in both highly protected and relatively less protected areas of the manufacturing sector. Perhaps not surprisingly, Canada's cost competitiveness has also performed poorly (Table 17). Relative to the United States, Canadian unit labour costs had moved from around 85 per cent of the American *level* in 1985 to nearly 128 in 1990, before declining to 102 in 1996, largely due to the depreciation of the Canadian dollar.

In brief, there is ample evidence to suggest that the FTA has provided net benefits to Canada, given the ongoing dynamics of Canada-US trade relations. However, there are two areas where outcomes, at least in qualitative terms, have not lived up to expectations.

– The FTA has provided for wider access to the US market and trade volumes have probably grown because of it. Yet the security of that access is an ongoing concern, given the uncertainty that sometimes exists in response to actions on both sides of the border.

– Rationalisation has occurred in the manufacturing sector and is probably continuing. Nevertheless, productivity improvements remain elusive, which is surprising given the widespread belief that international trade is a main source of productivity growth through technological exchange and spillovers. The high degree of intra-industry trade suggests, *ex ante*, that Canada would stand to gain considerable benefits from more intensive trade and investment relations but this has yet to appear in the aggregate statistics.

Future priorities

Although the NAFTA is an important trade policy instrument, the fundamental basis of Canada's trade policy is the WTO. In that context, Canada has a number of priorities:

– The expansion of the application of transparent, predictable and enforceable multilateral trade rules.
– Greater transparency in the shaping and conducting of negotiations and in the day-to-day conduct of the WTO'work.
– Full implementation of existing agreements and commitments.
– Playing an active role in WTO-accession negotiations to expand the application of transparent rules-based trade regimes and secure more open, non-discretionary access.
– Bringing more discipline and transparency to the use of trade remedies with a view to reducing uncertainty in the international environment.
– Improving access for trade in goods by reducing traditional barriers to trade – industrial tariffs, quotas, import restrictions and the use of technical barriers to trade.
– Strengthening the rules-based trading system for agriculture through a new round of negotiations in 1999.
– Further improving access for trade in services during the next round of comprehensive negotiations in 2000; and
– Furthering work on trade and competition, trade and investment and government procurement.

On the regional/bilateral front, Canada is leading a push for more forward progress in the APEC forum; is a strong supporter of the Free Trade Area of the

Americas initiative; and is exploring the possibility of a free trade agreement with members of the European Free Trade Association.

Internal Trade

The Agreement on Internal Trade (AIT) came into effect in July 1995. It states as its objective, the reduction and elimination, ''to the extent possible'', of barriers to the free movement of goods, services, persons and investment within Canada. This goal is to be achieved by adherence to a number of principles (to which all provinces and territories have agreed), specifically:[25]

- no new barriers;
- the equal treatment of persons, goods, services and investment, regardless of where they originate in Canada;
- the recognition and harmonisation of standards and regulations to provide for the free movement of goods, services, people and investment;
- ensuring that administrative policies work in concert; and
- the explicit allowance for derogations from national or reciprocal non-discrimination in pursuit of ''legitimate objectives''.

In its present form, the AIT can only be considered as work in progress as much remains to be done to implement it fully. A host of barriers and non-conforming measures have been removed since 1995, and substantial effort has been focused on liberalising procurement, and ensuring that the institutional machinery to monitor adherence and to facilitate negotiations is operating smoothly. Developments since the entry into force of the AIT include: the approval of measures to harmonise corporate registration and reporting requirements across the country; an internet-based electronic tendering system to advertise procurement opportunities; and the preparation of guidelines laying out a process for provinces and territories to mutually recognise occupational standards and to reconcile differences between these standards. Attention is now being focused on the extension of procurement disciplines to many more crown corporations, the completion of an energy chapter and improved disciplines on investment incentives. While negotiations in these areas are well advanced, agreements acceptable to all governments have not yet been reached. With regard to procurement all jurisdictions, save for British Columbia and the Yukon, agreed to include the MASH sector (municipalities, academic institutions, social service agencies and hospitals) under the AIT. On the one hand, the inclusion of the

Table 18. **The public sector procurement market**

1993, C$ billion

Sector	Covered[1]	Excluded/other[2]	Total
Federal government			
Departments[3]	9.82	6.44	16.26
Crown corporations	7.55	9.85	17.40
Sub-total	17.37	16.29	33.66
Provincial governments			
Departments	23.91	6.93	30.84
Crown corporations	17.33	25.16	42.49
Sub-total	41.24	32.09	73.33
MASH[4]			
Excluding British Columbia and Yukon	27.32		27.32
British Columbia and Yukon		2.98	2.98
Sub-total	27.32	2.98	30.30
Total	85.93	51.36	137.29

1. Covered refers to the procurement of entities subject to the provision of the Procurement Chapter (that is, those listed under Annex 501.1A) and entities listed under the non-intervention commitment (that is, those listed under Annex 502.2B).
2. Including entities excluded by the Agreement on Internal Trade (Annex 502.2A), procurement below threshold levels, plus for the federal government procurement on non-traded goods and services (such as utilities, purchases and rental property, and goods purchased for resale).
3. Does not include procurement by the Canadian International Development Agency.
4. In February 1998, all jurisdictions save for British Columbia and the Yukon agreed to cover the MASH sector (municipalities, academic insitutions, social service agencies and hospitals).

Source: Industry Canada.

MASH sector is an important milestone for provincial relations, given its relative size (nearly 20 per cent of total procurement, see Table 18) and the importance attached by provinces to procurement policies as a tool for economic development. On the other hand, the exclusion of some participants sends the wrong signal and opens up the unwelcome prospect of a future patchwork of commitments in other areas.

Despite this important forward movement, progress has stalled in most other areas of the Agreement (Table 19). For example, major obligations to implement and/or extend the AIT's scope and coverage are unfulfilled (agriculture and food products, and energy); deadlines for concluding negotiations have been missed, raising questions about the commitment or ability to reach consensus on objectives (including the coverage of more services and crown corporations under the AIT's provisions, specifying rules for investment, and the mutual recognition of

Table 19. **Agreement on Internal Trade: major outstanding obligations**

Chapter	Obligation	Status
Procurement (5)	By June 1995, extend coverage to the MASH sector (municipalities, academic institutions, social services and hospitals).	February 1998, agreed (except for British Columbia and the Yukon) to include the MASH sector under the AIT, but still subject to ratification by the respective provincial legislations. Provisions of this new annex expected to enter into force in 1999.
	By July 1998, review, harmonise, reconcile bid protest procedures.	No action yet.
	By July 1996, further negotiate reduction of excluded services (including professional, transportation, financial, health and advertising).	Negotiations have been suspended.
	By June 1996, accelerate and conclude negotiations to move government entities (mainly crown corporations) off the exclusion list.	Negotiations are continuing.
Investment (6)	By December 1996, make recommendations as to the retention, removal or replacement of local presence and residency requirement measures.	Recommendations approved in February 1998. Some local presence and residency requirements remain (Annex 604.4).
	Examine matters as directed by Ministers, including on interpretation of the Code of Conduct on Incentives.	Negotiations ongoing.
Labour mobility (7)	In a ''reasonable time'', seek compliance of regional, local and other forms of municipal governments or governmental and non-governmental bodies.	Consultations ongoing.
	Mutual recognition of qualifications and harmonisation of occupational standards (process started in July 1996).	Process continuing. (At the end of 1996 more than 400 occupational regulatory bodies invited to initiate work towards compliance with the Agreement.)
Agricultural/food products (9)	By September 1997, review scope and coverage of chapter.	Agriculture Ministers recognised that internal trade negotiations could be linked to the results of the forthcoming WTO talks on agricultural issues. With the exception of Quebec, all other Agriculture Ministers have decided to follow a process aimed at revising the scope and coverage of this Chapter.

Table 19. **Agreement on Internal Trade: major outstanding obligations** *(cont'd)*

Chapter	Obligation	Status
	Review supply management, Western Grain Transportation Act, and agricultural safety nets.	With respect to supply management of poultry and eggs, Agriculture Ministers agreed to aim for greater flexibility and better adaptation to changing market conditions. The WGTA was cancelled in the 1995 Budget. Agriculture Ministers also extended the existing satefy net programmes until March 2000 and agreed to develop a new framework agreement by the summer of 1999.
Alcoholic beverages (10)	Review a host of non-conforming measures.	Problems related to New Brunswick's and Quebec's differential fees or charges for beer remain ongoing. An *ad hoc* Committee has been formed to review exceptions, restrictions and grand-fathered measures with a view to their eventual elimination.
Natural resource processing (11)	By July 1998, review achievement of objectives, identify and resolve implementation issues and make necessary revisions to the chapter.	No action yet.
Energy (12)	By July 1995, include energy under the Agreement.	Negotiations ongoing (draft chapter under development).
Transportation (14)	By July 1996, extend coverage to municipal governments.	Transportation Ministers recommend deletion of obligation.
	By January 1996, reconcile regulations and standards-related measures.	Work ongoing (mostly completed).
	Phase-out non-conforming measures.	Ongoing negotiations.
Environmental protection (15)	Harmonise environmental measures.	Negotiations ongoing.
Institutional provisions (16)	By April 1996, report on effects of the Agreement.	Not yet available.
Final provisions (18)	Annual review of scope and coverage of the Agreement.	No review in place.

Source: Internal Trade Secretariat.

occupational qualifications and standards); and in some cases no timetable has been established to undertake action, underscoring the lack of credibility of the current process (ensuring compliance of lower levels of government with regard to labour mobility, review of issues in supply management, and the harmonisation of a number of non-conforming measures).

The slow progress being made in liberalising interprovincial trade in Canada is disappointing for a host of reasons. *First*, the extent of remaining barriers suggest there are significant gains to be made from freeing trade (Table 20). These barriers are of several types:[26] discriminatory practices (such as residency requirements, preferential standards, supply management); unharmonised practices and regulations (such as for consumer protection, investment incentives and grants, and for trucking); and the inequitable application of administrative practices (such as for procurement, and the sale and distribution of alcoholic beverages). As in the case of international trade, removing these barriers could help improve the allocation of provincial resources and create the potential for exploiting economies of scale and specialisation.

Second, internal trade growth had been particularly strong in the past, but has slowed dramatically more recently. Between 1984 and 1990, international and interprovincial exports expanded at the same speed, an average annual rate of 4.7 per cent. However, between 1990 and 1996, the latter advanced by only 2.2 per cent per annum, while growth in the former exceeded 10 per cent. Further, in every province, the share of international exports in GDP increased over the twelve year period, 1984-96 (although it remained stable in Saskatchewan, which in any case has the highest share in both years, Table 21); meanwhile over the same period, in every province the share of interprovincial trade in GDP declined (although it remained roughly stable in New Brunswick). For Canada as a whole, the decline in interprovincial trade as a share of GDP amounted to over 4 per cent, while that of international trade expanded from around 28 per cent in 1984 to over 38 per cent in 1996. The decline in interprovincial, compared to international trade likely reflects several factors, such as the strength of external relative to domestic demand over the 1990s, the impact of the Free Trade Agreement, and the liberalisation of global trade more generally. However, it also coincides with a period in which barriers to pan-Canadian trade may have been rising, or if not rising, at least not declining. Thus, a concerted and

Table 20. **Agreement on Internal Trade: remaining (major) barriers to trade**

Sector	Barriers
Procurement	Limited progress in extending both the scope and coverage of the Agreement's provisions regarding procurement has meant that both direct barriers – local presence requirements, content, payment preferences and value preferences – and indirect barriers – such as through the inequitable application of administrative practices – remain outstanding.
Investment	Residency requirements for ownership of assets (such as land); provision of competitive provincial industrial incentive programmes (*e.g.* grants, loans, tax incentives) and unharmonised policies, including taxation, incentives and grants; inconsistencies in expropriation/compensation laws.
Labour mobility	Preferential standards and practices established by provincial governments, self-regulating associations and labour unions continue to exist.
Consumer protection	Unharmonised regulations for packaging and product labelling, regulations, and licensing requirements for the sale of health care products.
Food and agriculture	Price and income stabilisation policies; supply management policies (restricting the flow of milk, eggs, chickens and turkeys); unharmonised federal and provincial regulations for food products (particularly, imitation and blended dairy products, meat grading and inspection, organic foods and tolerance levels for residues in food).
Alcoholic beverages	Unharmonised provincial control systems; unharmonised environmental protection measures or charges.
Natural resources	Restrictions in the forestry industry (including harvesting and movement of wood fibre and residues) and on the provision of power to that industry; provisions promoting the processing of mineral concentrates within provincial jurisdiction (through legal requirements and the structure of fiscal regimes); and restrictions on the movement of unprocessed fish.
Transportation	Unharmonised trucking regulations (including vehicle weight, height and length); and intra- and extra-provincial regimes including trucking regulations and licensing requirements.
Energy	Provincial control over transmission and distribution of electricity; provincial grid transmission systems limits intra-provincial imports; limited use of transmission lines by other utilities diminishes scope for third party providers; unharmonised regulations limiting access to provinces in the gas sector (licensing and rate making); unharmonised regulations regarding energy performance standards.
Financial institutions	Financial institutions exempt from the Agreement, thus remaining barriers due to: overlap and duplication in prudential regulation of trust and loan companies; unharmonised insurance industry regulations; unharmonised securities regulations and the registration and licensing requirements of financial institutions among provinces.

Source: Internal Trade Secretariat, *Internal Trade Barriers in Canada*, 1997.

Table 21. **Developments in provincial exports**

Percentage share of GDP

	1984			1996		
	International	Inter-provincial	Total	International	Inter-provincial	Total
Newfoundland	29.2	10.4	39.6	35.8	9.2	44.9
Prince Edward Island	15.1	30.9	45.9	16.6	27.3	43.9
Nova Scotia	18.4	22.7	41.0	21.7	21.6	43.3
New Brunswick	31.5	28.7	60.3	34.8	28.9	63.6
Quebec	22.1	23.7	45.8	34.3	20.0	54.3
Ontario	33.9	23.5	57.4	45.6	20.2	65.7
Manitoba	19.9	29.2	49.2	29.0	26.9	55.9
Saskatchewan	39.6	24.0	63.6	39.4	22.5	61.9
Alberta	24.2	36.7	60.9	38.0	24.2	62.2
British Columbia	28.9	13.8	42.7	30.8	12.9	43.7
Yukon/Northwest territories	18.4	22.8	41.2	24.7	21.5	64.1
Canada	28.3	24.3	52.7	38.4	20.0	58.4

Source: Statistics Canada, *Interprovincial Trade in Canada*, 1994-1996.

committed programme of liberalisation might work to restore the dynamism of internal trade, and result in the familiar gains that such trade may bring.

Third, compared to similar international agreements, the scope and coverage of the AIT is not particularly ambitious and as such, problems with pushing it forward do not reflect well on provincial attitudes toward free trade. The AIT falls considerably short of the degree of liberalisation, policy harmonisation or legal authority called for example, in the European Union (EU) or by the FTA/NAFTA (Table 22). The EU for instance, goes beyond the AIT in a number of basic ways – in allowing for the free flow of *all* goods, investment and *nearly all* services – which is an impressive achievement, when considering the diverse cultures, customs and economic policies which characterise the European Union. Likewise, FTA/NAFTA, for example, allows for the liberalisation of financial services, energy trade, and limits the actions of sub-national entities, areas which are exempted or otherwise not adequately covered by the AIT. In short, the provinces are urged to finish the job they started in 1995 and to complete the Agreement on Internal Trade, so that there will at least be some benchmark by which to evaluate its future merits.

Table 22. **The EU, NAFTA and AIT trade frameworks: an overview of the main features**

	European Union (EU)	North American Free Trade Agreement (NAFTA)	Agreement on Internal Trade (AIT)
Objective	*The establishment of a customs and monetary union.*	*The establishment of a free trade area.*	*The reduction and elimination, to the extent possible, of barriers to the free movement of goods, services, persons and investment.*
Goods	Since 1992, intra-area trade is free of tariffs and QRs.	Immediate phase-out of some tariffs; five to ten-year phase-out of all tariffs (15 years in sensitive sectors). Liberalisation of NTBs. Free trade in agricultural products within 15 years, but with certain exceptions for each country (such as in Canada's supply management sectors).	The Canadian Constitution prevents the application of tariffs on inter-provincial trade. The AIT sets out principles for reciprocal non-discrimination; right of entry and exit; the general prohibition of obstacles to trade; however, it does allow for derogations to fulfil legitimate objectives. Includes separate chapters covering trade in agriculture and food goods (dealing mainly with technical barriers to trade); alcoholic beverages; and natural resource processing (production and sale of the forestry, fisheries and mineral resources). Regional development, culture, taxation, measures affecting aboriginal peoples and the financial sector are excluded from the provisions of the AIT.
Investment	All restrictions have been lifted from the movement of portfolio capital among members. Provisions have been made for firms' freedom of establishment.	National treatment of investors. Mexican export performance, local content, and foreign exchange requirements are being phased out. Some restrictions remain in Mexico's basic energy and railroad industries, Canada's cultural industries and US airline and radio industries. It also provides for investor-state dispute settlement; and a negative list of reservations agreed to by the three parties.	Reciprocal non-discrimination for investment, removal of local presence and residency requirements; removal of performance requirements; derogations allowed for legitimate objectives; provisions do not apply to provincial monopolies or crown entities.

Table 22. **The EU, NAFTA and AIT trade frameworks: an overview of the main features** *(cont.)*

	European Union (EU)	North American Free Trade Agreement (NAFTA)	Agreement on Internal Trade (AIT)
Services	Trade barriers have been removed with the exception of energy, telecommunications and transportation which have undergone liberalisation to varying degrees. Exceptions are also made for leasing services, legal services, accountancy and postal services.	National treatment extended to services (services provisions do not apply to procurement, subsidies and financial services which are dealt with in other parts of the agreement). Nearly full liberalisation of telecommunication and land transportation services. Establishment of negative list of exceptions.	General principles applying to goods also apply to the provision of services. Specific provisions regarding trade in communication and transportation services are outlined in specific chapters devoted to such trade. Financial services (and institutions) exempt from the AIT.
Labour	Labour movement is almost completely liberalised. Some barriers still exist in the form of professional and vocational qualifications and limited portability of fringe benefits, pensions, and so on.	Includes provisions for the temporary entry of business visitors, traders and investors, intra-company transferees, and certain other categories of professionals. 5 500 Mexican professionals are allowed to enter the US under temporary entry provisions. Canada has not set a nominal limit with respect to Mexico. Established a separate North American Agreement on Labour Co-operation.	Instructs provinces and regional, local and other forms of municipal government and its governmental and non-governmental bodies to remove residency requirements; barriers created by licensing; certification and registration; recognise occupational qualifications; and harmonise occupational standards.
Antidumping provisions (versus competition policy)	Antidumping measures were eliminated within the area but common antidumping policy exists. Establishment of strong competition policy and enforcement mechanisms covering trade among members.	Continued application of antidumping laws but Mexican procedures have been brought closer to those of the US and Canada. Any amendments to trade laws may be subject to panel review for inconsistency with the object and intent of the Agreement, the WTO or the relevant WTO codes.	Not applicable under the AIT.
Subsidies	European competition policies have reduced state aid but level of subsidisation is still high. Community-wide subsidy for agricultural products persist (CAP).	No progress made in eliminating (all but agricultural) subsidies. Binational panel review of final antidumping and countervailing duty determinations made by administrative authorities in each country.	No specific provisions regarding subsidies.

Table 22. **The EU, NAFTA and AIT trade frameworks: an overview of the main features** *(cont.)*

	European Union (EU)	North American Free Trade Agreement (NAFTA)	Agreement on Internal Trade (AIT)
Regulations and standards	Mutual recognition of each country's regulations and standards (includes firms' rights of establishment and certification of professionals) unless minimum community-wide standards have been set (in health, safety, environment). Customs formalities abolished. Some differences in technical standards and indirect taxation persist. Working on harmonisation of technical product standards.	Certain national treatment obligations with regard to standards. Parties agree not to use standards-related measures as unnecessary obstacles to trade, and agree to co-operate and work towards the enhancement and compatibility of these measures in the free trade area.	Allows for adoption or maintenance of standards or standards-related measures to achieve legitimate objectives; however, move towards mutual recognition of standards (harmonisation) where appropriate and to the extent practical. Contains a specific chapter concerning consumer-related measures and standards to ensure legitimacy of maintaining such measures to protect consumers, but to avoid other barriers (licensing, registration, residency, local presence requirements and the like).
Government procurement	Public procurement of goods and services has been made subject to rules providing for transparency and free market access. Some monitoring problems exist.	Liberalisation of procurement by major goverment agencies in all three countries.	Liberalises government procurement above threshold levels (C$ 25 000 for goods and C$ 100 000 for services; C$ 100 000 for construction); extended to a list of included government agencies. Crown entities excluded. Municipalities, academic institutions, social services and hospitals will also be covered.
Dispute settlement	Disputes are handled by the Court of Justice. Binding dispute settlement mechanism.	Two dispute settlement mechanisms, one for issues arising out of the NAFTA's implementation and/or interpretation, the other for resolving issues related to antidumping and countervail. Disputes are binding but there are Extraordinary Challenge Procedures.	Disputes to be settled by consultation and negotiation; arbitration panel proceedings exist as well. Non-legally binding dispute settlement mechanism.

Table 22. **The EU, NAFTA and AIT trade frameworks: an overview of the main features** *(cont.)*

	European Union (EU)	North American Free Trade Agreement (NAFTA)	Agreement on Internal Trade (AIT)
Supranational institutions/ committees	Commission (administrative and technical body); Council of Ministers (main decision making body); Court of Justice (enforcement of common body of law); and the Assembly (parliament). Each of these organisations has power that is comparable to institutions in federal states.	Free Trade Commission (supervises implementation of agreement, resolves disputes). [Commission for Environmental Cooperation (commits members to improve and enforce domestic environmental protection laws) and Commission for Labor Cooperation (forum for consultation on domestic Labor standards) established in parallel with, but are not part of NAFTA.] In addition, eight committees and six working groups on various trade issues.	Committee on Internal Trade (cabinet-level representatives) to supervise implementation of the Agreement; assist in dispute resolution and to consider any matters relating to its operation. An Internal Trade secretariat established to provide administrative and operational support.

Source: The Agreement on Internal Trade and Hoeller, Peter *et al.* (1998), ''The European Union's Trade Policies and their Economic Effects'', OECD Economics Department Working Papers, No. 194.

Tax policy

While the emergence of a fiscal surplus discussed in Chapter I is unlikely to allow broad-based tax cuts in the near term, it has nevertheless, provided an opportunity to begin to reverse the tax burden Canadians face. At about 36 per cent of GDP, overall tax revenues in Canada are near the OECD average, but are considerably higher (by around 10 percentage points) than in the United States, the country's most important economic partner. This difference puts Canada at a competitive disadvantage in trying to prevent the emigration of its tax base, particularly to the south of the border.

Over the past four budgets, there have been targeted tax reductions to achieve economic and social objectives by providing assistance to education, child-care, charities and the disabled, and to improve tax fairness. The 1988 Budget, however, marks the beginning of general tax relief, in this instance for low- and middle-income earners. Two small measures are proposed which will become effective in July 1998:

i) An increase in the amount of income which can be earned tax free by low-income single Canadians by C$ 500, and by C$ 1 000 for a family. This measure is expected to remove 400 000 people from the tax rolls completely, and to reduce federal taxes for an additional 4.6 million tax filers.

ii) Tax payers earning between C$ 50 000 and about C$ 65 000 will see the 3 per cent general surtax (on basic federal taxes) reduced, and those earning up to about C$ 50 000 will see it eliminated altogether. This measure will lead to tax relief for 14 million people, or 90 per cent of all tax filers. (The general surtax was first introduced in 1986 as a temporary initiative to help reduce the budget deficit.)

The combined cost impact of these two changes on federal revenue is estimated to be C$ 880 million in 1998/99, rising to C$ 1 680 million in 2000/01, or around 0.3 per cent of GDP. In addition, the Budget provided for some tax relief for students and for those undertaking education and re-training; and a two-year tax holiday on employer employment-insurance premiums for additional young hires (those aged 18 to 25).

As the fiscal position continues to improve, there will be some scope for making adjustments to the tax system that are conducive to enhancing both

economic efficiency and fairness. (These issues were highlighted in last year's *OECD Survey*.) A number of arguments could be made to this effect. *First*, taking into consideration both the mounting surplus in the Employment Insurance (EI) account (which stood at more than C$ 16 billion in mid-1998), and the scheduled increases in Canada Pension Plan contributions as a result of pension reform (see below), there is a need to reduce Employment Insurance Premiums. Earlier this year, these payroll taxes were lowered to C$ 2.70 (employee rate) per C$ 100 of insurable earnings from C$ 2.90 in 1997, with a planned future decline to C$ 2.60 in the year 2000.[27] The main arguments supporting a payroll tax reduction are the following:

- Legislation governing EI requires that such premiums be set to ''assure that there will be enough revenues over a business cycle'' and the Chief Actuary has determined this to be between C$ 10 to C$ 15 billion. As such, the current surplus already appears adequate to finance the EI programme, if economic activity should turn downward.
- Given the potential for adverse short-run effects on employment of rising payroll taxes, due to increasing CPP contributions, there is a case for offsetting any such effects by lower EI premium rates.
- Also, with the EI surplus reaching such a high level, the fairness of the programme comes into question given the unequal incidence of contributions and beneficiaries across provinces. EI is not intended to be a mechanism for redistribution.

Second, the share of personal income taxes in GDP is around 4 percentage points higher in Canada than the United States, reflecting both higher marginal and average tax rates.[28] The former is an issue with regard to standard adverse effects of taxation (including those to avoid paying taxes), while the latter raises concerns for higher-income earners who are potentially the most mobile across international borders. Thus, tax policy should move to close this gap in the personal income tax area which is likely weighing on locational, entrepreneurial and labour market behaviour.

Third, the absence of full indexation for inflation has meant ''bracket creep'', whereby tax liabilities rise sharply because an increase in nominal income causes a higher proportion of incomes to fall into higher tax brackets. (In Canada, indexation occurs only on the difference of consumer price inflation above 3 per cent.) As reported in last year's *Survey*, over the ten years to 1998,

18 per cent of tax filers will have either become taxable or been pushed into a higher tax bracket due to only partial indexation, resulting in increasing average tax rates at all income levels (although proportionally more for low- and middle-income individuals). Now that fiscal surpluses are in prospect, the government is in a position to act to remove the threshold on indexing the personal income tax system to arrest the rise in the tax burden, even at the current low rates of inflation.

However, the likely size of the fiscal dividend available in the next few years means that choices will need to be made. In order to maximise national social and economic objectives, reductions in the burden of personal income taxes should clearly be the government's highest priority among possible tax reductions in this regard. This is justified on both economic and social grounds. On the economic front, the economic distortions arising from the impact of taxes on locational, entrepreneurial, and labour market activities are higher for personal income tax than for any other source of government revenue. On the social front, lack of indexation has particularly increased the personal income tax burden at lower income levels.

Fourth, the lack of sales tax harmonisation (outside of Nova Scotia, New Brunswick, Newfoundland and Quebec), has meant ongoing distortions on business inputs and traded products. However, there are no indications that provincial governments outside the harmonised sales tax areas are interested in pursuing this issue at this time. This is unfortunate since further sales tax harmonisation could benefit Canadian competitiveness and, importantly, lead to greater co-operation in the administration and co-ordination of tax policy between the federal and provincial governments, since co-operation will in fact be required in any reform of other areas.

Fifth, it would be desirable to make changes to taxation in the corporate sector. A report issued by the Technical Committee on Business Taxation, which was commissioned by the government, highlighted several deficiencies and distortions in the present framework. While the government views these issues as important, business taxation in Canada falls within both federal and provincial jurisdiction, so that moving forward will require sufficient time to allow all participants to review carefully both the issues and the report before reaching any decisions. (There are, after all, no advantages to be gained if one level of government moves ahead with an initiative which could be easily undone by

actions of the other.) The Committee made a number of recommendations (see Box 3) – many of which were already noted in last year's *OECD Survey* – to address what it believed to be the main deficiencies of the current business tax structure. These deficiencies can be summarised as follows:

- Canada maintains too high a corporate income tax rate, especially when compared with the United States. This results in less competitiveness and base erosion.
- There exists a number of distortions which create inefficiency, unfairness, and high compliance costs (such as varying tax burdens across different types of business activities; higher rates on services compared to other sectors; deficient international tax rules that effectively subsidise outbound investment to the detriment of domestic investment; the growing erosion of the corporate tax base due to planning techniques which flow otherwise taxable profits through non-taxable entities), and is impeding economic growth and job creation.
- There is a growing reliance on profit-insensitive taxes (payroll, property, sales taxes on business inputs and capital taxes), especially at the provincial/local level, which places a burden on business during periods of slow economic growth. (However, they can be efficient taxes if linked to public services used by business, but at present they are not.)

Thus, once pressures begin to ease in the high priority tax fields, both the federal and provincial governments must move in a co-ordinated manner to tackle the serious and highly complicated problems in corporate taxation.

Pension reform

The Canadian retirement income system combines elderly benefits designed to provide a basic income guarantee for seniors (through Old Age Security, OAS and the Guaranteed Income Supplement, GIS), an earnings-related contributory public pension plan (Canada and Quebec Pension Plans, CPP and QPP), and tax-assisted private pension plans (Registered Retirement Plan, RPPs and Registered Retirement Savings Plan, RRSPs). Over the past two years, the government has proposed substantial changes to the first pillar of the system (subsequently withdrawn), and has implemented changes to the other two.

In 1997, the federal government reached agreement with the provinces to reform the CPP, whose changes came into effect in 1998. The reform was

Box 3. **Report of the Technical Committee on Business Taxation: main recommendations**

The Technical Committee on Business Taxation was appointed by the Minister of Finance in 1996 with a mandate to review the business tax system in Canada and to make recommendations for its reform, under the constraint that such reforms be revenue neutral. The Committee reviewed the impact of the current business tax structure on economic growth and job creation, fairness and compliance. The Committee's findings were reported in December of last year and its main recommendations are as follows:[1]

Lower corporate income tax rates for all industries towards international norms, and at the same time *broaden the tax base* to bolster efficiency and fairness. Lower rates would help to improve the business tax system in three respects: allow Canadian firms to be more competitive; reduce incentives to shift income out of Canada and deductions into Canada; and to reduce distortions associated with the existence of multiple taxation rates. Addressing deficiencies in the current structure in relation to the taxation of international income and reducing distortions by scaling down tax preferences would broaden the tax base and increase revenues.

Moving towards the user pay principle for some taxes, since economic efficiency and fairness could be better served if businesses were charged at levels reflecting the benefits they receive from public goods or services, or the costs they impose on society. The Committee, therefore, recommends experience-rated Employment Insurance premiums for employers – which would provide incentives for employers to reduce employee turnover rates in order to lower contribution rates – and to consider restructuring the federal fuel excise tax (on gasoline and other motive fuels) to include other sources of pollutants.

To improve *compliance and enforcement* the Committee suggests: federal and provincial harmonisation of the structure and administration of certain taxes (notably capital taxes) to reduce compliance costs; simplifying tax legislation; applying commercial practices to settle disputes and collect taxes owing; and extend civil provisions in the tax legislation to advisors and promoters who are grossly negligent.

Working with the provinces to achieve a harmonised and balanced tax system. Specifically, the Committee recommends that both levels of government should: *i)* move toward using neutral common corporate income and capital tax bases; *ii)* extend the federal-provincial tax collection agreements to capital taxes and to all provinces; and *iii)* disallow the deductibility of capital taxes from the corporate income tax base. It is argued that these initiatives would further broaden the tax base for provinces and thus provide additional revenues to them. The Report also stresses that without a substantial level of co-operation by the federal and provincial governments, the overall thrust of the Report's objectives will not be achieved.

1. *Report of the Technical Committee on Business Taxation*, Submitted to the Honourable Paul Martin, Minister of Finance, December 1997.

motivated by the need to correct several deficiencies of the previous framework and to accommodate shifting demographics, life-expectancy and economic conditions. The previous system was deficient in three key respects. *First,* unforeseen demographic developments – lower birth rates and lower death rates – raised questions about its sustainability. Projections by the Chief Actuary of Canada showed the CPP Account Balance declining gradually and eventually vanishing by the end of 2015, requiring a shift to pay-as-you-go with an accompanying rise in the contribution rate to 14.2 per cent (from the 1997 level of 5.85 per cent) in the year 2030.[29] *Second,* since members of the younger generation over their working lives would have been required to pay much higher contribution rates for unchanged benefits as a percentage of pre-retirement earnings, the status quo was highly inequitable. *Third,* on the benefit side the fiscal soundness of CPP was compromised by the addition of social goals, such as disability provisions, in its implicit mandate. (This is clearly the case, for example, when eligibility conditions for CPP were broadened to include non-medical factors, such as regional unemployment rates.)

The new plan has four essential features:

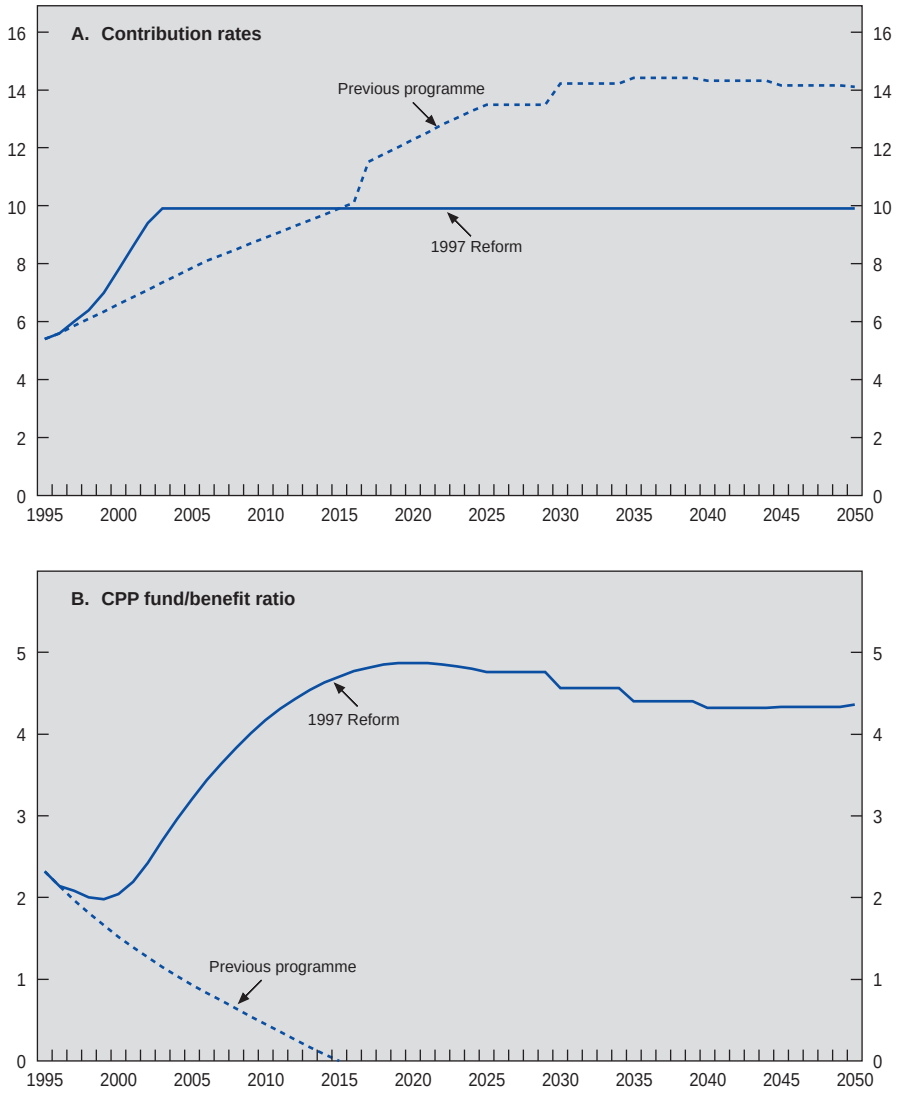
i) Beginning in January 1997, the contribution rate increased to 6.0 from 5.6 per cent in 1996.[30] This rate will rise quickly to 9.9 per cent in 2003, after which it will be capped at that level. Compared to the previous system, the rapid increase in contributions allow these rates to stabilise much quicker (in 2003 versus 2030) and to reach a significantly lower level (9.9 as opposed to 13.9 per cent) (see Figure 10, Panel A).

ii) The CPP Account/expenditure ratio of around 2 at the end of 1996 is projected to increase gradually to a level of nearly 5 by 2020, and to stabilise at around 4.3 further out (See Figure 10, Panel B). In contrast, under the previous plan as noted earlier, the Account balance would have been exhausted by 2015.

iii) A new CPP Investment board will be created to invest the CPP in a diversified investment portfolio at arm's length from governments. The real return on the CPP Account is expected to rise from the current rate of 2.5 to 3.8 per cent.[31] This together with the accelerated contribution rates is projected to reduce by 1.5 percentage point the stable contribution rate.

Figure 10. **CANADA PENSION PLAN REFORM: CONTRIBUTION RATES AND FUND/BENEFIT RATIO**[1]

Per cent



1. Under the previous programme, legislated contribution rate scheduled until 2016, then pay-as-you go.

Source: Canada Pension Plan, Fifteenth and Sixteenth Actuarial Reports.

iv) The strengthening of work attachment rules for disability benefits and revised guidelines for the determination of disabilities to put the emphasis back on their medical basis and to de-emphasise the use of socio-economic factors in allowing benefits.

The March 1996 Budget proposed replacing the OAS and the GIS with a single Senior's Benefit. OAS and GIS are paid out of general tax revenues and are primarily directed at alleviating poverty. They provide for a basic universal grant (OAS) to all Canadians aged 65 and over, together with an income tested supplement (GIS) whose level declines as other income grows. The OAS has a clawback of 15 cents for every dollar of income above C$ 53 215, while GIS benefits are reduced by 50 cents for each dollar of income from other sources. The primary motivation for the proposal was to slow the growth in programme costs in years after 2010 when the baby boom generation begins to reach age 65. The cost savings were projected to rise to 10.7 per cent of programme costs by 2030. The consolidated programme would raise slightly the current level of combined OAS/GIS benefits for single seniors, have full indexation with regard to the benefit and thresholds for taxation, and be tax free but would have a feature which taxes back the entire benefit for those single seniors with incomes above C$ 52 000 (and for couples above C$ 78 000).

The government has now decided not to proceed with the proposed changes to the OAS and GIS programmes. The decision is based on concerns raised in consultations and on the major improvement in the government's long-term fiscal position. A key concern that emerged in the consultations was that the proposed benefit reduction rate of 20 per cent of private income above a fairly low threshold would significantly increase marginal tax rates on retirement income and so discourage private retirement saving. At the same time, the government's much improved fiscal situation opened up new options for addressing the long-term demographic pressures on OAS/GIS costs. Rather than introduce a problematic reform of OAS and GIS, therefore, the government committed to apply a portion of the future fiscal savings coming from a steadily declining debt-to-GDP ratio to fully maintain the pension programmes.

The third pillar of the retirement income system, private pensions, were also affected by changes in the March 1996 Budget. These are either employer-sponsored (RPPs) or individually based (RRSPs). Contributions to these plans are on a voluntary basis and are tax deductible within prescribed limits. Pension

benefits, representing the accumulated value of contributions plus investment earnings are taxed when paid. In 1996, the annual RRSP contribution limit was frozen at C$ 13 500 until the year 2003, after which it will rise to C$ 14 500 in 2004 and C$ 15 500 in 2005. For the RPP, the current maximum pension limit will be kept at the current levels until 2004.

OECD Ministers have recently identified a number of principles to guide policy reform for maintaining prosperity in an ageing society.[32] Those directly affecting the structure of both retirement income and the retirement decision were elaborated as follows:

- Public pension systems, taxation systems and social transfer programmes should be reformed to remove financial incentives to early retirement, and financial disincentives to later retirement.
- A variety of reforms will be needed to ensure that more job opportunities are available for older workers and that they are equipped with the necessary skills and competence to take them.
- Fiscal consolidation should be pursued, and public debt burdens should be reduced. This could involve phased reductions in public pension benefits or anticipatory hikes in contribution rates.
- Retirement income should be provided by a mix of tax and-transfer systems, advance-funded systems, private savings and earnings. The objective is risk diversification, a better balance of burden-sharing between generations, and to give individuals more flexibility over their retirement decisions.

In relation to these principles, changes to the Canadian pension system appear to deserve praise. Recent moves should ensure the sustainability of the publicly-provided pension, and the strengthening of the multi-pillar approach is desirable to ensure the increasing flexibility and capacity of the system to respond to a wide range of future challenges. It is also generationally more equitable. Nevertheless, much more consideration needs to be given to the interaction of the retirement framework with the tax system, social transfer programmes and the impact on economic performance. At present, the trend to early retirement is not particularly marked in Canada, compared with the OECD average, nor is the retirement burden on the working population. However, given Canada's demographic profile, the retired-person dependency ratio is expected to increase dramatically over the next 30 years or so, and much faster than for the

OECD as a whole. (For Canada, from around 24½ to 52 per cent in 2030, as opposed to 28½ per cent for the OECD average which rises to 49 per cent thirty or so years on.[33]) This shift stands to have material implications for Canadian living standards.

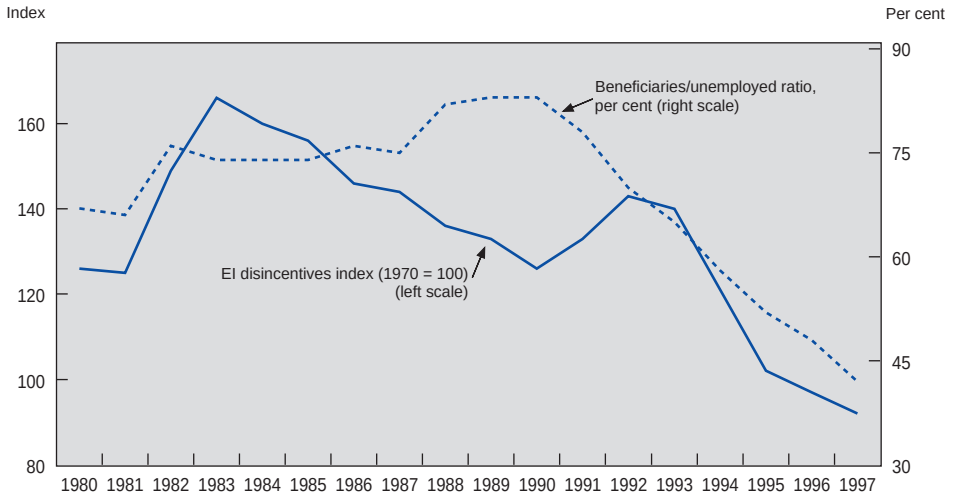
Thus, while the current emphasis is on ''early retirement'', the demand for ''postponed'' retirement will continue to grow. From a policy perspective, this will necessitate ensuring that longer labour force participation is less constrained, for example: by current Canadian laws and legislation requiring a mandatory retirement age; features of the pension system which claw back benefits and discourage recipients from earning income; early retirement features of the CPP/QPP (at age 60); and tax credits in the personal income tax system which are taxed-back at low income levels.[34] As for the ability of older workers to extend their working lives, Canadian policies – with regard to life-long learning and skills training – could also be strengthened as highlighted in Chapter III.

Labour market

Over the past twelve months or so, there have been no major policy initiatives in the area of labour markets at the federal level. This lack of action reflects the fact that major reforms were taken in 1996 to the Employment Insurance (EI) system and in the scope and delivery of active labour market policies (ALMPs). In 1996, EI replaced the previous Unemployment Insurance (UI) system which had become what many analysts believed to be one of the most important factors in driving up the structural rate of unemployment.[35] Since UI was introduced in 1971, the ratio of beneficiaries to unemployed (the B/U ratio) remained fairly high, reaching above 83 per cent in 1989-90, given the wide coverage and generosity of the programme (Figure 11). Subsequent changes to the system began to reduce the B/U ratio, and coincident with the introduction of EI it has fallen to an historic low of around 42 per cent. (However, it should be pointed out that not all of the decline in the B/U ratio should be ascribed to changes in the UI programme, since the slow recovery from the recession has also played a role, by *inter alia* increasing the number of long-term unemployed workers who would not qualify for benefits.[36])

As noted in last year's *OECD Survey*, EI represents the most fundamental reform of the unemployment insurance system since UI was initially introduced. The main features of the new system (so-called Part I measures) relative to the

Figure 11. **UNEMPLOYMENT INSURANCE INDICATORS**[1]



Source: Sargent, Timothy, "An Updated EI Disincentives Index for Canada", Department of Finance, mimeo, 1998 and Human Resources Development Canada.

previous programme, include (see the *1996 Survey* for details): lower Maximum Insurable Earnings limits; tightened eligibility requirements; reduced replacement rates and limited experience rating (of employees); and clawbacks. There were also two changes that have the potential for increasing benefits and coverage. These involve: the move to an hours-based system which will expand coverage by including those employees who were otherwise excluded because they worked less than the required time and earned less than the Minimum Insurable Earnings limits; and the Family Supplement, which works to increase the replacement rate (to 80 per cent by the year 2000) for claimants with dependants and earnings less than one-half the Maximum Insurable Earnings limit. Taken together, previous reforms to UI and the introduction of EI have led to a marked decline in the work disincentive (at the national level, Figure 11) coming from the unemployment insurance system in Canada, and according to one estimate, perhaps all the way back to the pre-1971 level.[37] (However, it should be pointed out that no regional differentiation of the UI system existed at that time so that the index overstates

the decline since then.[38]) The implication of this evolution is expectations of a further downward adjustment to the structural rate of unemployment. Ongoing provincial efforts to remove disincentive effects in the labour market (see Box 4) should buttress the EI initiative and lead to much improved labour market performance in the near term.

An integral part of the EI reform programme is a mandate requiring the federal government to offer the provinces options in participating more directly in all aspects of planning, designing and delivering active labour market policies (so-called Part II measures, ALMPs). (These policies are discussed in greater detail in Chapter III.) So far, eleven labour market development agreements (LMDAs) have been signed, with negotiations between the federal government and Ontario just beginning. As noted in last year's *Survey*, there are two types of LMDAs: either a full transfer of responsibility (five provinces and the North West Territories) or co-management (five provinces). Where co-management agreements are in place, the federal government will continue to directly deliver labour market benefits and measures. In contrast, where a full transfer agreement is in place, the provincial or territorial government assumes full responsibility for ALMPs using funds from the EI Account. (In either case, the federal government retains responsibility for providing benefits under Part I of the EI programme.) It is projected by the government that aggregate funding to support the labour market agreements will be up to C$ 1.95 billion in 2001/02.

Since EI is being implemented in stages (with full phase-in not until 2001), it is premature to make pronouncements on its impact thus far. Nevertheless, a preliminary analysis undertaken by Human Resources Development Canada summed up recent developments – comparing the twelve months prior to EI reform with the first twelve months under the new system – as follows:[39]

- With regard to Part I, Income Benefits, the amount of regular benefits declined by 8.4 per cent; the number of initial claimants declined from 2.13 million to 1.82 million (which in part reflects improving economic prospects); a somewhat smaller than expected decline in average weekly EI benefits due primarily to the ability of some claimants to prevent benefit reductions by securing greater employment; and a declining percentage of beneficiaries to unemployment ratio.

Box 4. **Social assistance reform and labour markets: the case of British Columbia**

Federal fiscal consolidation has been felt heavily at the provincial level, as cuts to the Canada Health and Social Transfer (CHST, which in 1996 combined the previous Canada Assistance Plan and Established Programs Financing) have been particularly sharp. CHST transfers federal resources to provinces to fund social assistance and services as well as for health and post-secondary education. The block funding of CHST resulted in a significant change to the financing of social assistance and social services, which had previously been cost-shared under the Canada Assistance Plan. In 1980/81, federal transfers to provinces under the pre-CHST mechanism amounted to 4 per cent of Canada's GDP, and this has since fallen to 3 per cent in 1997/98.

All provinces and territories are now either reforming or planning reform to their social assistance frameworks, as a result of both the decline and shift in the nature of CHST transfers and because of provincial fiscal pressures. These reforms are based on a move away from ''passive'' to ''active'' assistance which is premised on helping recipients integrate into the labour force. Nowhere is this shift more pronounced than in British Columbia.

In late 1995, British Columbia revamped its social safety net, BC Benefits. The main components of that programme that impact on the labour market are as follows:[1]

BC Family Bonus: The BC Family Bonus provided the model for a similar federal programme, the Child Tax Credit, which came into effect in July. The Bonus is an income-tested benefit given to all low-income and welfare-dependent families with children, and is motivated by the need to remove previous inequities whereby families in the latter case were better-off than those of the former (since income assistance provided levels of support above low-wage employment) and since this group also lacked incentives to move off state support. The Bonus has a maximum annual benefit of C$ 1 236 per child under the age of 18, up to a family net income of C$ 18 000, above which benefits are reduced by 8 per cent of net income for families with one child and 16 per cent for those with two or more. It disappears completely at C$ 33 540 for families having one or more children. The Family Bonus is administered by Revenue Canada on behalf of the province, using information from income tax returns. As long as a family files a return and meets eligibility, a monthly cheque is sent automatically (there is no need to fill any applications). Early indications are that the Family Bonus is having desirable effects: a reduction in poverty gaps, reduced welfare caseloads, and a shift towards employment for those previously on assistance.

Youth Works: Under this policy, welfare is no longer being provided to those aged 19 to 24. Instead, young people who qualify receive a living allowance as long as they continue to actively participate in each of three phases of the programme:

(continued on next page)

(continued)

independent job search (seven months duration), with assistance provided for career exploration and planning materials, labour market information, referrals, and the requirement to complete job search reports; assisted-job search (two months duration) which includes career planning, skills plan development, and job finding clubs; and in the third phase, employability programmes are offered, such as basic education, work-place training in the private sector, and post-secondary training. (As yet, there have been no evaluations of the impact of *Youth Works* on employment.)

Welfare to Work: This is a similar measure to Youth Work, for adults 25 and over. However, under this policy welfare recipients who participate are not guaranteed the active assistance provided to youths, nor access to work experience or training as these aspects of the programmes remain subject to availability. As such, if resources are available, adult recipients of welfare can follow the same three phase assistance noted above. (As yet, there are no formal evaluations of the impact of this policy). These reforms are to be welcomed as they create incentives for self-sufficiency, either by reducing welfare dependence and/or by making social assistance dependent on participating in initiatives which improve labour market prospects. They are also supported by sanctions ranging from benefit reductions to the suspension of benefits.

1. British Columbia also maintains the highest level of minimum wages (which applies to all workers) in Canada which were increased in April to C$ 7.15 per hour from C$ 7.00. The country-wide average is C$ 5.85. The government of BC deems that such a level is required to ensure a modest standard of living for low-wage earners, and does not believe that it has significant adverse impacts on the job prospects for young, or low-skilled workers.

– With regard to Part II, Employment Benefits and Support Measures, the numbers of individuals receiving such assistance increased from 230 000 to 305 000, while expenditures declined from C$ 1.7 billion to C$ 1.2 billion, primarily due to the transition required to implement the new measures; the types of employment services provided shifted away from long-term interventions (such as training) toward increased short-term ones (such as providing information and counselling), a shift reflected in the average cost per participant declining; and during the first twelve months of operation, it is estimated that 55 per cent of participants found employment within twelve weeks of completion of the assistance.

Thus, tentative evidence suggests that EI reform is having some positive and intended effects. Nevertheless, this does not suggest that there are no further changes which could make the system better. The major criticism of the previous

UI programme was twofold: 1) the propensity to encourage frequent use; and 2) its regional differentiation. Several aspects of the reform deal with the first, namely, an ''intensity rule'', which reduces the replacement rate according to previous use of the system and an income tax clawback which is greater for those with more than 20 weeks of benefits in the last five years. However, as noted in the *1996 Survey*, the scale of changes is relatively modest, allowing scope for tightening provisions somewhat more. With respect to regional generosity, a reduction in the maximum duration of benefits from 50 to 45 weeks in high unemployment regions is a direct attempt to deal with this problem. Unfortunately, the retention of a variable entrance requirement (a rule that allows benefits to be determined by a range of weeks in the divisor) which is linked to the unemployment rate, and the subsequent introduction in 1997 of Adjustment Projects in 29 of the 54 EI regions, has effectively reversed any initial reduction in regional disparities brought about by the 1996 changes, thus retaining a problematic feature that discourages labour mobility across regions.

A final issue concerns experience rating, not from the point of view of employees but employers. The extent of employer premiums for EI is substantial, and as noted by the Technical Committee on Business Taxation (see above), has the potential for being a drag on employment and growth, given its profit-insensitive nature, and is unfair (both across industries and regions). Since employer contributions (at 1.4 times employee contributions) are not linked to the costs of providing benefits to employees, employers with less stable employment can free ride on the EI system, while those businesses which tend to minimise layoffs through various means are penalised. Creating a more direct link between benefits and contributions would, therefore, increase the overall neutrality of the business tax system, and more importantly could improve both the functioning of the labour market and individual employers' management of their employment.[40] If an employer's contribution varied directly with its employment practices, the employer would have incentives to reduce instability in its employment which would reduce average unemployment. The federal government is now exploring the possibility of adopting experience rating EI, and model simulations under alternative scenarios (and EI revenue neutrality) show positive benefits for the economy under a range of schemes considered. Hence, further progress toward determining the feasibility of such a policy is to be encouraged.

Remaining problems

Taken together, microeconomic policy reform in Canada since the mid-1980s would appear conducive to both economic efficiency and growth. Yet, the experience over that period has been somewhat disappointing, as underlying performance has remained below the benchmarks provided both by Canada's pre-reform history, and that of some other OECD economies more recently. Total factor productivity, the supply of capital and, in particular, labour inputs has declined, structural unemployment has remained high, and as a result potential output growth – the measure of sustainable economic activity – has been nearly halved since the 1970s. These developments are difficult to explain, although a number of elements seem to have been working to offset the beneficial effects of the structural policy initiatives. In contrast, the implication of ongoing weakness is much clearer: without a pickup in underlying economic performance the present high level of Canadian per capita income, relative to the OECD average, is unlikely to be maintained.

Nevertheless, there are forces which give rise to optimism about future prospects:

- After a comparatively deep recession in 1990-91, adjustment toward price stability, and a period of rapid fiscal consolidation, macroeconomic conditions are now highly favourable. With the recent expansion of activity, the economy might be expected to benefit from the long period of restructuring and greater factor utilisation, especially labour, which could lead to a period of above-average productivity growth. Moreover, a strong fiscal position will allow resources to flow to areas which could enhance investment and hence productivity, and in providing much needed scope for reducing the tax burden.
- Since 1995 there has been a significant shift in the share of employment into industry, away from the public sector and, to a lesser extent, agriculture. Public sector employment has declined by more than 112 000 (nearly 1 per cent of total employment). To the extent the business sector is better able to generate value-added, this shift (which is still ongoing, particularly at the sub-national level) could bolster underlying economic performance.

– Significant labour market (Employment Insurance) and social (assistance) policy reforms are now more or less in place at the national level and in most provinces. These reforms will go some way toward improving labour market attachment, lessening dependence on government support and enhancing work incentives, and thus should contribute to better labour market outcomes.
– The increasing openness of the Canadian economy to international, and especially US trade (the manufacturing productivity leader), should allow for a strong exchange of technology and R&D spillovers – as previously noted, a factor which is widely cited to be an important source of productivity growth. In addition, Canada is moving to address the ''innovation gap'' (or slow up-take of new technology) identified in previous *OECD Surveys,* the low levels of R&D spending, the adoption of flexible organisational structures, and management and human resource strategies, which are associated with improving economic efficiency and productivity. Indeed, research by Industry Canada points to strong growth in recent years of foreign patenting in Canada, national patenting, and Canada's patenting abroad compared with other major OECD countries.[41]

Despite these positive developments, there is no room for complacency as a number of structural policy challenges remain. *First,* the mobilisation of resources could be further bolstered by making required changes to the tax system, as the fiscal situation permits. Canada's first priority should be to reduce the relatively high level of public debt, but even in the near term this should allow scope for some tax cuts ahead of any increase in expenditures (see Chapter I). Such cuts might be directed, in the first instance, at the following areas: reducing payroll levies at a faster rate than presently envisaged; making an initial start toward reducing the high marginal and average personal income tax rates in Canada, compared to those faced in the United States; eliminate the increased tax burden caused by the partial indexation of the personal income tax system; and co-operating with the remaining provinces to harmonise value-added taxation, in order to remove ongoing distortions on business inputs and traded products that the dual sales-tax regime currently entails.

Second, in various sectors, and in financial services in particular, policy makers must remain responsive to the evolving nature of technological and market forces. While careful consideration will no doubt be given to the impact

Table 23. **Implementing the OECD Jobs Strategy – an overview of progress**

Jobs Strategy proposal	Actions taken since 1996	OECD assessment/recommendations
I. Reform unemployment and related benefit systems		
– Increase extent of experience rating. – Reduce regional dependence on unemployment insurance. – Monitor eligibility to disability benefits. – Increasing threshold of allowable earnings to social assistance benefits. – Make passive income support conditional. – Increasing the working income supplement to low-income working families.	– In 1996, Unemployment Insurance (UI) system revamped under the Employment Insurance (EI) Act; EI system increases experience rating; tightens eligibiilty requirements; and reduces both the extent and duration of benefits. – Social assistance levels cut in Ontario (as well as in other provinces); and introduction of a mandatory workfare programme for recipients of assistance deemed employable. – A two-step doubling of the working Income Supplement for low-income working families.	– The EI system should help lessen work disincentives; however, the variation in regional generosity still largely remains and experience rating could be further strengthened. – Make sure that workfare will add to human capital accumulation and does not undermine ALMPs. – Ensure that the announced overhaul of the child benefit system reduces welfare traps.
II. The wage formation system		
– Grade minimum wages by age.	– In 1998, the minimum wage in BC was increased.	
III. Improve labour force skills and competence		
– Address adult literacy problems. – Reduce barriers to mobility for workers. – Develop the Pan-Canadian Indicators Program.	– Work under way to recognise occupational qualifications and competencies across provinces.	– Agreement on Internal Trade which includes labour mobility has been in place since mid-1995 and negotiations seemed to be bogged down. A deadline for the completion of this work would help impart a sense of urgency.
IV. Active labour market policies		
– Target ALMPS. – Allow provinces greater responsibility for ALMPs. – Investigate other forms of employment assistance. – Maintain the operation of a ‘‘‘one-stop’’ shop for ALMPs.	– Since 1996, 9 provinces have opted to accept more direct responsibility of ALMPs; negotiations with Ontario are in progress. – In 1997, the government announced the Youth Employment Strategy; in 1998 funding for this programme was increased. – In 1997 it also announced measures to help persons with disabilities make the transition to employment.	– Potential for substantial gains as ALMPs should now be better tailored to regional needs and as duplication of efforts is eliminated; however, provinces must now adopt a more rigorous evaluation of the system than heretofore.
V. Enhance product market competition		
– Continue implementation of the labour mobility chapter of the Agreement on Internal Trade.	– Implementation of the labour mobility chapter of the Agreement on Internal Trade is ongoing.	– See point III above; the Agreement has been extended to include the MASH sector (although BC and the Yukon have opted out of this provision), meanwhile progress in many other areas remains stalled.

Source: OECD.

of proposed bank mergers on competition, at the same time, attention must also be devoted to the wider impact on the entire financial sector. In this regard, the decision to go forward with the mergers should depend upon what further reforms to federally regulated financial institutions will be forthcoming, or in fact, be taken ahead of these developments, such as: opening the banking sector to all comers (foreign and domestic); removing restrictions on foreign ownership; levelling the playing field for all financial institutions; and granting wider access to the Canadian Payments System. These and other issues were dealt with by the Task Force on Financial Services which presented its finding in September. Given this timing, the government is justified in delaying its decision on the mergers. However, once that report is tabled, quick action will be required to inform market participants of its intentions in order to remove uncertainty about the future policy environment.

Finally, in the area of labour markets, not enough is being done to deal with the problem of the under-utilisation of labour resources in several regions on the one hand, and the longer-term issue of a declining supply of labour on the other. In the former case, a number of disparate measures (many of which were previously noted in the *OECD Jobs Study*, Table 23) could be taken: a further tightening in both eligibility requirements and benefits for unemployment insurance and implementing changes to that system which effectively diminish (if not eliminate) its regional generosity; introducing experience rating on employer premiums for Employment Insurance as a means to enhance job creation and to deal with its inequities across industries and regions; and making a real and determined effort to finish the work set out in the Agreement on Internal Trade, so that it may begin to deliver the benefits of pan-Canadian free trade. In the latter case, consideration needs to be given to strengthening labour force participation, through, for example: easing laws and regulations requiring a mandatory retirement age; reducing features of the pension system which tax-back benefits and discourage recipients from earning income; making the early retirement aspect of the public pension plan less attractive; and extending the working lives of Canadians by a variety of active labour market and training measures as discussed in the next Chapter.

III. The labour market in a knowledge-based economy

A consensus is emerging that Canada – like other developed countries – is facing a new economic reality brought about by technological change (especially associated with the ‘‘information revolution’’) and increased global competition. In this view, the economy is entering a phase in which growth is more closely tied to the embodiment of knowledge in both human and physical capital. At the same time, new information technologies are seen as closely linked to expansion in world trade and competition. Because this model of growth emphasises the role of knowledge, it is usually referred to as a shift toward a ‘‘knowledge-based economy’’.

These trends have received considerable policy attention because they could turn out to be an opportunity for greater economic growth if Canada positions itself to take advantage of new technologies and new markets. Alternatively, there is concern that the country could be ‘‘left behind’’ if it is not adjusting to the new reality. In labour market terms, correct positioning is taken to mean having a workforce that is both skilled and capable of adapting to new technologies, one that would not have to compete directly with low-skilled and cheaper labour from other parts of the world. Even so, there are likely to be costs in the requisite adjustments which may arise in the form of fundamental changes in the nature of work as well as reduced opportunities for those unable to adapt.

Against this background, the goal of this Chapter is to examine three sets of questions. First, is there evidence of a shift toward a knowledge-based economy in the Canadian labour market, or, more directly, of increased relative demand for skilled versus unskilled workers and of a change in the nature of work? Second, what policies have been instituted in relation to skill development in the past decade? Third, based on the answers to the first two questions, what new policy directions ought to be considered? Before turning to these issues, the

implications of a knowledge-based economy for labour market developments and policies are examined in more detail.

The knowledge-based economy: what it means for labour

The term knowledge-based economy (KBE) is used in different ways. In very general terms a KBE can be defined as one in which the creation and implementation of knowledge is at the centre of employment decisions and economic growth. More specific definitions vary by what is included in the knowledge deemed to be at the heart of the KBE. Basically, one can distinguish two types of knowledge, each embodied in associated skills: physical skill-related knowledge (*e.g.* crafts type skills); and cognitive skill-related knowledge (*e.g.* managerial type skills). One main KBE definition emphasises skills of all kinds (both physical and cognitive) while another, more restrictive one, includes only cognitive skills.

Proponents of both definitions argue that developed countries such as Canada are in the midst of a shift toward being more predominantly KBEs. Despite some overlap, the two main models of this shift, each associated with one of the above-mentioned KBE definitions, have somewhat different implications for labour market outcomes and policy options, as discussed below.

The General KBE Shift Model

This model emphasises a shift toward all types of knowledge, which is assumed to arise out of an increase in demand favouring skilled workers over unskilled workers. the relative demand shift is argued to have been going on for a long period of time.[42] Its exact source is not a matter of consensus, though the two main factors are typically taken to be a skill biased technical change and the opening up of competition through increased globalisation.

The empirical implications of this model are:

– Rising real wages for more skilled workers of all types and falling real wages for unskilled workers in the short run. The implied increase in earnings inequality may, however, be diminished or reversed by supply responses.

– Rising employment for more skilled workers and falling employment levels for the unskilled. This holds whether or not there are offsetting supply effects.
– With the source of the demand shift not agreed upon, it is uncertain how long these patterns have been going on.
– There are no obvious implications concerning employment stability, though less skilled workers may face increased probabilities of layoff.

One main policy implication of a general skill biased demand shift is that increasing the supply of skills by moving individuals from the unskilled to the skilled category is very desirable. This raises the number of people who benefit from the involved easier access to employment and (possibly) higher wages. At the same time, it reduces the supply of unskilled workers, which acts to mitigate negative wage effects for that group. The issue for public policy makers is then one of deciding to what extent there are failures in the market for human capital and whether there are externalities in the generation of human capital that can be addressed using policy interventions.

The Accelerating KBE Shift/Information Technology Model

In this case, the demand shift is seen as arising because of a technological change associated with the emergence of Information Technologies (ITs). This means that it favours cognitive skills specifically, rather than skills of all types. IT makes it easier to ‘‘codify’’ existing knowledge so that it can be incorporated in more production processes more rapidly. Thus, investments in knowledge can lead to a ‘‘virtuous cycle’’ of knowledge creation and economic growth (OECD, 1996). Accompanying this is increased competition in output markets for both goods and services as new technologies are spread to competitors more rapidly.

The empirical features of this model are:

– Rising real wages for workers with cognitive skills related to new technologies and falling real wages for other workers. Some skills, even for university-educated workers, may be devalued by the introduction of new technologies. One should observe differences in outcomes even for workers with the same levels of education and training. Again, the implied increased earnings inequality may not be directly observable if there are offsetting supply responses.

- Rising employment levels for workers with relevant cognitive skills and falling employment for other workers. This holds whether or not there are offsetting supply effects.
- With the demand shift related directly to the rise of ITs, observed labour market effects should begin only in the early 1980s. Moreover, with the hypothesised ''virtuous cycle'' of growth and knowledge production, the ITs should spread and become increasingly integrated in economic activity over time. This implies that shifts in wages and employment should be accelerating over time.
- Several characteristics of the new technologies combined with an increasingly competitive market place point to more instability in labour market outcomes.[43] One should see a decrease in the proportion of workers in full time jobs, declining job tenure, and increases in the rates of self-employment and multiple job holding. Also, as firms shift toward a core of stable jobs and a periphery of contracted out services, the dispersion of hours worked should rise. These trends should be more obvious for less (cognitively) skilled workers but should still be observed for more skilled ones over time.

The policy implications from this second model are in some sense harsher. Again, encouraging training and education may be desirable. In this case, however, the relevant skills are more narrowly defined. Also, the case for government intervention is stronger because of the assumed growth externalities. At the same time, though, with direct and strong growth effects from firms flexibly adapting to new technologies, policies designed to protect workers from negative labour market trends have more strongly negative efficiency implications. Policy would need to be directed more toward encouraging and enhancing lifelong learning than trying to cushion negative technological impacts through measures such as minimum wages, employment stability laws and passive income support programmes.

Is there a shift toward a KBE in the Canadian labour market?

Empirical evidence

With the above implications of models of shifts toward a KBE in mind, developments in the Canadian labour market have been examined with a view to

establishing whether the hypothesised trends have occurred and, if they have, how substantial they are. It has been attempted to answer this by inspecting, in turn, trends in employment, unemployment and participation rates, sectoral employment patterns, the share of employment that is full year/full time, job durations, the proportion of the workforce who are self-employed, and the distribution of earnings and hours worked. The detailed results of the analysis are presented in Annex VI. The key findings are as follows:

Employment, unemployment and participation rates

- Since the 1960s, high-school-educated men have suffered large declines in employment outcomes both expressed as levels and as a ratio to the population of high-school-educated men; this is true both for given cohorts over time and across successive cohorts (Figure A5).
- University-educated men have experienced upward trends in numbers employed but downward trends in the employment to population ratios. Thus, while labour demand appears to have increased for this group, supply appears to have grown at an even faster rate. University education provides better opportunities than a high school education but overall employment rates for university-educated males have still been declining.
- Women of all education levels have experienced increases in employment, though these have slowed down or reversed in the 1990s (Figure A4). The ratio of the employment rates of more to less-educated women has actually declined over time.
- Evidence from participation and unemployment rates is mixed. Less-educated males have been experiencing declining participation rates relative to those with a university education since the late 1970s. On the other hand, the ratio of the unemployment rate for high-school-educated to university-educated males has increased for some age groups and decreased for others. Females generally have experienced large rises in participation (Table A5) and mixed unemployment outcomes over the last twenty years.

Sectoral employment patterns

- Structural change across industries has not been increasing. Rather, it is continuing at a rate similar to that observed since at least the early 1970s.

– Rapid employment growth has occurred in industries with a variety of knowledge intensity levels (Table A6). Increased demand for more skilled workers is an intra- rather than inter-industry phenomenon.
– High knowledge intensity industries have grown somewhat in importance but remain a relatively small component of total employment.
– There is less evidence of a quickening shift toward high knowledge intensity industries than a continuation of an ongoing trend.
– By international comparison, there appears to be both scope and reason for Canada to increase its degree of knowledge-based activities.

Employment instability measures

– There is no evidence of a pervasive increase in job instability in Canada. High-school-educated males, and to some extent younger workers, have suffered a decline in stability, however.
– Full-year/full-time rates and average job tenure have declined for less-educated males over the last two decades. Both have been stable for university-educated males, leading to an increased educational differential in these measures (Figure A7).
– Full-year/full-time rates have increased for females, with the increase being greater for the high school than university educated (Figure A6).

Self-employment rate

– While self-employment as a proportion of total employment has risen in Canada in the 1990s, it is not clear that this arises from a change in the nature of work (Figure A8).
– There is greater incidence of self-employment among workers with both low and high education levels.

Earnings inequality

– The educational earnings differential has not increased for males or females overall in Canada in the last twenty years (Figures A9 and A10). This is often explained as being due to offsetting increases in the relative supply of university workers masking relative demand shifts. However, the only age group that did experience increased educational differentials

is workers under age 30 who should have been the group most directly affected by the supply increase.

- Increased differences in earnings between younger and older males over time have not been related to increased returns to experience within cohorts. Rather, they are due to declining fortunes of younger males. With the education deficit of older workers diminishing, the age premium has tended to rise.
- Earnings dispersion measured in levels has not increased for groups of males with the same education and experience. Successive generations of males have done worse in terms of earnings regardless of their education level.
- Females have not experienced the same patterns of increased earnings inequality as men, nor the decline in earnings for successive generations.

Policy implications

Such evidence does not always fit easily with the view that the Canadian economy is experiencing a skill biased demand shift partially offset by supply responses. In that case, labour market outcomes should reflect greater relative advantage for more skilled workers over time. While there is some evidence that this is occurring, and increasingly so, it is still far from being a general phenomenon. In particular, differences in employment and full-year/full-time rates between university and high-school-educated women have actually declined over time. Also, there is no evidence that the overall male wage structure has been dominated by a general shift in demand favouring more skilled workers.

To the extent that such a skill biased demand shift arises from technological change, those parts of the labour force without cognitive skills could be expected to have weaker connections with specific firms. This could show up in increased part-time work, increased self-employment and declining job tenure. However, there is little evidence of pervasive changes of this type affecting workers. Women, and in particular older ones, have experienced rising average job tenure and increased access to full year/full time jobs. Some groups of more educated men have also experienced increasing job tenure. The main group suffering negative trends in these measures of job outcomes is less educated men, who have had dramatic declines in their employment levels, employment rates, access to full-year/full-time jobs and, to a somewhat lesser degree, average job tenure.

These declines have a substantial generational component, with more recent generations of high-school-educated men faring particularly poorly.

Although events in the Canadian labour market cannot be neatly grouped together as reflecting a shift to a knowledge-based economy, this does not mean that such a shift is not going on, with the technological change leading to significant modifications in work arrangements. The percentage of workers using computers on the job increased from 35 per cent in 1989 to 48 per cent in 1994, with 93 per cent of those in natural sciences and engineering and 76 per cent of those in managerial occupations reporting use of computers on the job. th ere appears to be a strong correlation between education levels and computer use, though clerical workers also reported a usage rate of 70 per cent in 1994 (Lowe, 1997). This has the potential to widen gaps between skilled and unskilled workers. The rising importance of software and computer services has been reflected in rapid growth of the information and technology sector in the 1990s. Nonetheless, the sector still accounts for only about 3 per cent of total employment.

The lack of evidence of a *dominant* skill biased demand shift also does not imply that differences in labour market outcomes by education level are not an important, ongoing feature of the Canadian labour market. Employment to population ratios for 25 to 30 year old high-school-educated individuals were approximately 72 per cent for men and 58 per cent for women in 1993. This contrasts with employment rates of 89 per cent and 83 per cent for university-educated men and women, respectively. Further, as can be seen from Figure A9, average weekly earnings (measured in 1980 dollars) of high-school-educated men aged 40 in 1993 were approximately C$ 420 while those for university-educated men of the same age in the same year were approximately C$ 620. Also, as shown in Annex VI, low-educated males have experienced dramatic downturns in employment levels, employment rates and earnings across successive generations.

Thus, in some sense Canada has been a knowledge-based economy for some time and continues to be one. For this reason, it is reasonable to argue that education and training deserve special attention. In particular, raising the average education level of males would mean a labour force in which fewer male workers faced the high unemployment rates associated with the low educated and more workers had access to better earnings opportunities. It might also be expected to reduce earnings differentials as the supply of workers with post-secondary educa-

tion increases and the supply of lower-educated workers declines. This, in turn, would lead to reductions in earnings inequality.

Beyond these issues of labour market and social policy, there is also a question of how to position Canada optimally in terms of international competitiveness. Increased freedom of trade due to successive GATT rounds and NAFTA means that Canada will face stronger competition in goods markets dominated by low skilled production processes. At the same time, Canada's knowledge intensity sector appears to be relatively small by international comparison. Thus, it may be reasonable to emphasise training and education as part of a national agenda that attempts to shift Canada to being more of a KBE, as this could turn out to be an opportunity for a greater economic growth.

The analysis presented in Annex VI suggests, however, that skill development cannot be a panacea. For example, raising education levels will not in and of itself reverse adverse trends in male earnings inequality to the extent the latter is related to increased differences in outcomes across generations rather than educational differentials. Improving the education level of low-educated males may move them *out* of the group showing the strongest declines in labour market outcomes but it still moves them *into* a group (university-educated males) who are also experiencing substantial earnings losses across successive generations. Nonetheless, given the substantial difference in employment and earnings outcomes between more and less educated workers, there is no doubt that education and training policies are crucial to improving labour market performance. But, as discussed below, whether such policies have beneficial impacts importantly depends on how any added education and training is delivered.

Improving workers' skills: the state of current government programmes

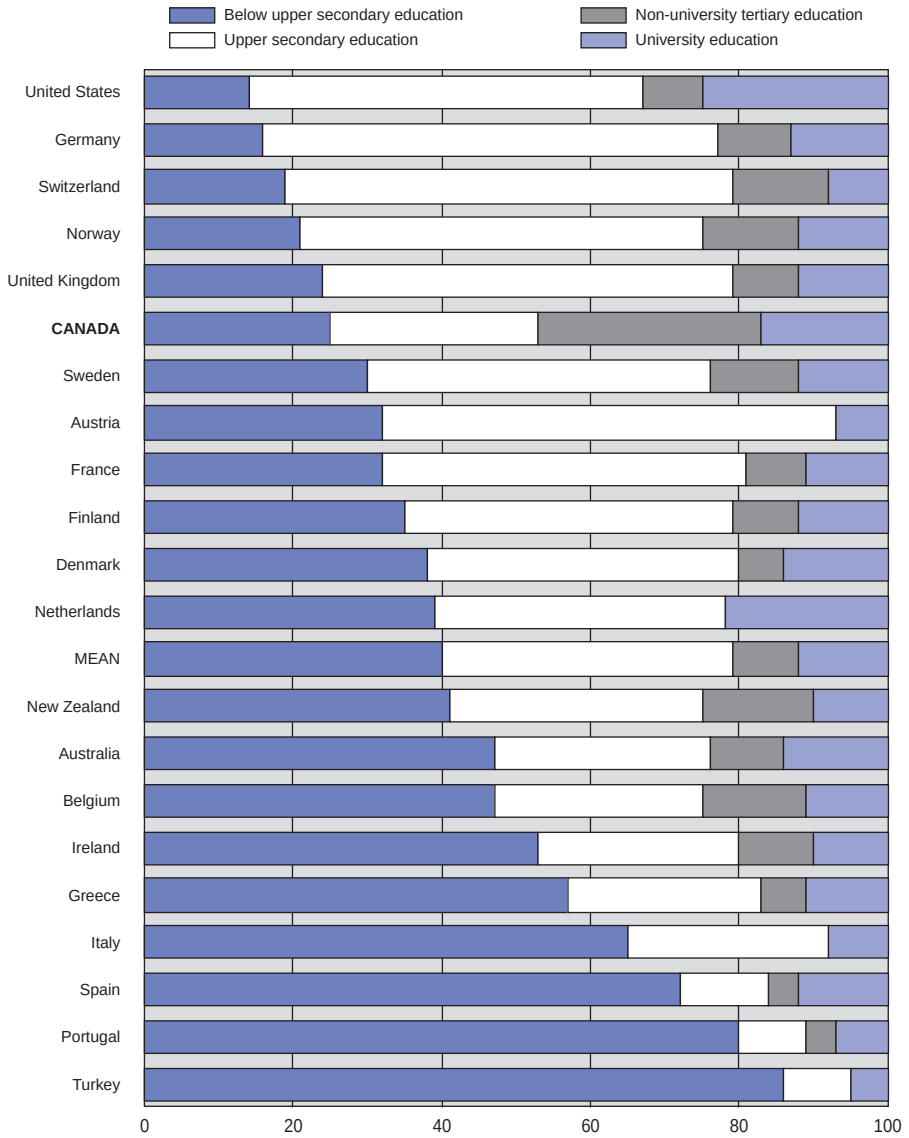
With a view to assessing their contribution to alleviating problems in the Canadian labour market, the following sections examine what policies are currently in place in the fields of education and training and who is taking advantage of them. These policies are discussed separately, but it will be seen that there is considerable overlap in take-up patterns between the two areas.

Education

Canada has one of the most highly-educated populations among developed countries, with 17 per cent of the adult population aged 25 to 64 having a university level education in 1995 (Figure 12). This compares favourably with the OECD average of 13 per cent but pales beside the US rate of 25 per cent. In part, though, this latter deficiency is made up by the higher proportion of adults with a non-university, post-secondary education in Canada (30 per cent) compared with the United States (8 per cent). On the other hand, 25 per cent of Canadians aged 25 to 64 had not completed high school in 1995, compared with only 14 per cent in the United States. The Canadian rate is still below the OECD average, however, and similar to that in the United Kingdom and Sweden. Overall, Canada's education levels exceed those in most OECD countries but fall short of those in the United States (OECD, 1997). The latter difference has been declining over time, though, as Canadian enrolment rates have overtaken those in the United States. In 1979, 33 per cent of Canadians aged 15 to 24 were enrolled in school full time compared with 40 per cent of Americans in the same age group. By 1995, the Canadian full time enrolment rate for that age group had surged to 50 per cent while the US rate had increased somewhat to 46 per cent. Newer generations of Canadians are staying in school longer and completing their education in a less intermittent pattern (Card and Lemieux, 1997). They are likely to be among the most educated in the world.

The high relative levels of educational attainment in Canada are consistent with spending on education. Expenditures for educational institutions made up 7.2 per cent of GDP in Canada in 1995, the highest rate in the OECD area. Canada ranked fourth, behind the United States, Switzerland and Sweden in expenditures per student at the university level (OECD, 1997). In spite of these relatively large expenditures, however, several of Canada's educational outcomes are less than impressive. As already noted, Canada has a much larger proportion of its adult population with less than a high school education compared with the United States. This may in part be due to the fact that, at least up until 1989, there was almost no difference in the earnings distributions of adult males whose highest level of education was completing high school and those who had not done so in Canada, but a substantial difference between the earnings distributions of these two groups in the United States (Donald *et al.*, 1997). If this is true, then increased school expenditures may have little impact on high school completion

Figure 12. **EDUCATIONAL ATTAINMENT IN OECD COUNTRIES**
Percentage of the population 25 to 64 years of age
that has attained a specific highest level of education (1995)[1]



1. Countries are ranked by proportion of the population attaining educational levels below upper secondary.
Source: OECD, *Education at a Glance*, 1997.

until youth perceive a greater return to completing high school (as current trends in participation indicate they may be doing).

But whatever the cause, a correlate of the large number of less-educated individuals in Canada has been a relatively high proportion of individuals with poor literacy skills. In the International Adult Literacy Survey, Canada has a mean score near the overall average for the twelve participating countries in terms of tests for prose, document and quantitative literacy. However, it also has a wider dispersion of scores on these tests than almost any other country. In particular, the 5th percentile of the test score distribution on both the prose and document literacy tests is lower in Canada than any other country except Poland and the United States. Given that Canada has higher educational expenditure rates than these countries, this is somewhat surprising. Moreover, this poor performance cannot just be blamed on lower high school completion rates in Canada. Indeed, Canada ranks eighth out of the twelve participating countries in terms of the proportion of adults who have not completed upper secondary education who score at levels 3 and 4/5 (with the top level being 5) on the document literacy scale. Only 27 per cent of Canadians who have not completed high school reach these higher literacy levels compared to 37 per cent in the United Kingdom and 59 per cent in Sweden (OECD and HRDC, 1997). Overall, Canada's poor performance in terms of literacy scores appears to result from a combination of low educational attainment rates among older people and the deterioration of skills of those who do not work in literacy-rich environments.

The combination of high spending levels and poor relative results in some literacy outcomes has led to calls for reform in the education system (Riddell, 1995). These, in turn, have resulted in a more ''back to the basics'' trend in several provinces, including the introduction of province-wide testing to help ensure that schools are meeting minimum standards. There have also been attempts to integrate school curricula more effectively with requirements from employers as it is possible that more ''job relevant'' material in schools will hold students' attention to a greater extent and help them to acquire higher levels of literacy (Packer, 1993). Whether such policies will work in changing literacy and ensuing labour market outcomes is not yet clear since experimentation in this direction is only beginning. Since international data on literacy have just become available, there is little current understanding of why countries, such as the United Kingdom, outperform Canada in this regard in spite of spending less

money on education. However, the creation of internationally comparable data opens the possibility of gaining valuable information to guide policy.

The rapid rise in enrolment discussed above has been accompanied by declines in real spending levels at the university level associated with fiscal retrenchment by both the federal and provincial governments. This has resulted in a marked fall in expenditures per student. While some part of this decline may have been made up for by increased productivity in the university sector, there has also been an increase in student/teacher ratios and worsening educational infrastructure in terms of laboratories and libraries (Riddell, 1995). At the same time, the number of applicants turned down by universities has risen and tuition fee levels have generally increased, raising concerns about accessibility and affordability. Expanding school enrolment rates for the young suggest that these changes have not so far altered accessibility for students significantly. It is also not clear that reduced spending per student has had significant harmful effects on the quality of education and/or labour market outcomes for students so far, and Canada still ranks high in terms of expenditure per university student. Nonetheless, these developments are of potential concern and have prompted policy action.

The 1998 federal budget attempts to address the issues of accessibility and affordability of post-secondary education through the creation of the Canada Millenium Scholarships programme. The scholarships will be granted at the rate of 100 000 per year, starting in the year 2000 and continuing for ten years. They are currently predicted to provide support at an average rate of C$ 3 000 per full-time student recipient per year. The scholarships will be targeted using a combination of needs and merit-based criteria and will be available to individuals of all ages at virtually all types of post-secondary educational and training institutions. Moreover, the Budget announced additional assistance for up to 25 000 students under the Canada Study Grants Programme, tax relief for interest on student loans, an education tax credit for part-time students, and tax incentives to save for education.

The new scholarship fund and the Canada Student Loan programme are both ways for the federal government to have some direct influence over education, which is constitutionally a matter of provincial jurisdiction. In fact, there are a number of arguments in favour of a policy role in this field at the national level. For instance, there are considerable spill-overs of the benefits of education

from one province to another. Allen (1997) shows that British Columbia ranks second among provinces in the proportion of its adult population having a university degree but last in terms of the number of university degrees awarded per capita; although the shortfall in the production of university graduates in British Columbia is partly made up through international immigration, it is largely met through inter-provincial migration. In effect, provinces like Nova Scotia have the reverse gap: they are exporting university graduates. The tendency for provinces to free ride off the educational expenditures of others has been compounded by fiscal pressures and cuts in federal transfers. Less generous funding has led many universities to raise tuition fees, reducing affordability for some students. As some provinces want to keep tuitions low, there have been threats to discriminate against students from other provinces who might be attracted by lower fees. Federal presence in the education field can help alleviate adverse effects stemming from such developments by reducing financial barriers and other obstacles that stand in the way of acquiring knowledge. A national strategy aimed at encouraging a shift toward a KBE requires, however, the co-operation of the provinces, which are responsible for educational institutions, curricula, standards, and quality.

Training

While education policy is a powerful tool for altering the skill level of the labour force, its impact will generally be longer run in nature. This is true because only a small percentage of individuals over age 25 are students in the formal education (*i.e.* colleges and universities) sector (Riddell and Sweetman, 1997). Moreover, with the passing of the baby boom beyond their 20s and the smaller size of subsequent generations, increasing the education levels of the latter can be expected to change the stock of skills in the labour force even more gradually. This suggests an important role for post-school training in any attempts to alter the skill level of the workforce.

Gauging post-school training

Post-school training is typically delivered in one of three forms in Canada: government funded programmes targeted at individuals (most often individuals receiving government transfers in the form of unemployment insurance or income assistance); employer-related training (which may be partly subsidised

through government assistance); and personal training acquisition by individuals. Exact measurement of the amount of post-school training is difficult, in part because informal training often associated with on the job training can be counted in different ways. The result is large variation in estimates of the amount of adult training in Canada, depending on the definition of training used (Betcherman, 1993). This also, not surprisingly, makes it difficult to decide on Canada's exact ranking in terms of level and intensity of training internationally. However, as part of the International Adult Literacy Survey, data were collected on the level of participation of those surveyed in ''organised adult education and training'' (*i.e.* education and training amounting to at least six hours in the year by individuals aged 16 to 65, excluding those 16 to 24 year olds who were full time students), permitting some comparisons across countries. According to this survey, in 1994/95, 38 per cent of the Canadian working age population participated in adult education and training. This places Canada in the middle/lower part of the countries taking part in the survey. the corresponding figure for Sweden was 53 per cent, while the United Kingdom recorded 44 per cent, the United States 40 per cent and Belgium 21 per cent. Canada's relatively low level could partly reflect recent high formal education enrolment rates for young people since those individuals staying in full time schooling might have been enlisted in training had they been part of the types of education/training systems in place in other countries. In terms of length of time spent on training, Canada had the third longest average training duration among the countries taking part in the Survey for trainees aged 26 to 35 and the longest average duration for trainees aged 46 to 55. However, training durations are much longer for non-employer sponsored training programmes (OECD and HRDC, 1997). this seems to confirm previous findings that Canada in fact performs relatively poorly in terms of employer sponsored training (Betcherman *et al.*, 1994).

The training captured in these measures is funded from a mix of sources. Thus, 26 per cent of men and 20 per cent of women participating in adult training and education in Canada in 1994/95 reported receiving funding from employers. This compares with 10 per cent of men and 14 per cent of women who relied on government sources (OECD and HRDC, 1997). The proportion of Canadian trainees who received funding support from employers is toward the bottom of the countries in the survey. Combined with Canada's middle to low level of overall adult training and education, this would seem to fit with claims that

Canada under-invests in job-related training relative to other developed countries. As Riddell (1995) points out, however, whether lower levels of investment in Canada can be termed ''under-investment'' or are in fact rationally justifiable is not clear.

It is important to note the gender differences in training patterns. Women are less likely to get funding from employers and are more dependent on other sources. Training acquisition is also not balanced across education levels. In fact, it rises uniformly with the education level from 15 per cent of individuals with only a primary school education to 60 per cent of individuals with a university education (OECD and HRDC, 1997). This educational imbalance is particularly evident in employer-based training and is also reflected in the fact that more training is carried out in high skill occupations (Betcherman *et al.*, 1994 and Crompton, 1994). Thus, it is the more highly skilled and educated and those with greater existing relative advantages in the labour market (*i.e.* men versus women) who get training (particularly, job-related training).

Government training policy

Government spending on labour market training in Canada is much lower than in continental Europe but higher than in other OECD countries. It must be noted, however, that the figures in Table 24 understate Canada's training effort because they only include federal spending. As in most other OECD countries, government expenditures on training have declined relative to GDP. Training and adult education-related government policies in Canada are delivered in several forms. Before discussing the major programmes, it is worth noting that, as with education policy, this is an area where there is a clear federal interest but the constitutional powers are at the provincial level. However, in contrast to education, where the federal government appears to be trying to define a role for itself, it has recently agreed to withdraw from direct training policy from June 1999 wherever provincial governments request it to do so (see also Chapter II).

- Transfer-related training delivery

One main route through which individuals come in contact with government training programmes is unemployment insurance (UI) and income assistance recipiency. Canada has made significant strides in moving from an unemployment insurance system which focused on what is often termed passive support to the unemployed to more active UI-related programmes. In particular, the new

Table 24. **Government expenditure on labour market programmes**

Per cent of GDP

	Labour market training		Total active measures	
	1993/94	1996/97	1993/94	1996/97
Canada	**0.31**	**0.17**	**0.66**	**0.48**
United States	0.04	0.04	0.21	0.17
Japan	0.03	0.03	0.09	0.10
United Kingdom	0.15	0.09	0.57	0.42
	1993	1997	1993	1997
Germany	0.42	0.36	1.37	1.25
France	0.45	0.36	1.28	1.32
Italy	0.01	0.01	1.88	1.08

Source: *OECD Employment Outlook*, June 1998.

Employment Insurance (EI) system which replaced the old UI scheme includes both a component focused on delivering benefits and one concentrated on delivering training and other assistance to speed up re-integration into the labour market. The new system includes three types of ''active employment benefits'' already in place and two others that have been proposed and are being tested. Already implemented are: targeted wage subsidies; self-employment assistance; and job creation partnerships. The *targeted wage subsidies programme* aims at creating incentives for a specified employer to hire and, hopefully, provide some form of on the job training to an EI claimant. Thus, this is a means of encouraging private sector provision of training for a group that might not otherwise have access to it rather than having the government provide the training directly. *Self-employment assistance* includes financial support and advice for claimants who want to start a business. *Job creation partnerships* are plans in which claimants receive EI benefits while developing or maintaining skills in the context of community projects. The two programmes still being considered or in an early stage of implementation are: targeted earnings supplements, and skills loans and grants. *Targeted earnings supplements* would deliver support to workers rather than employers in an attempt to get individuals to take jobs that they might otherwise see as too low paying. The idea is to help individuals facing barriers to labour force entry, such as the daycare costs of single parents, acquire work experience. *Skills loans and grants* make money available to individuals to

purchase the training they believe they need. The programme is already in place in some areas of the country, and in project status or design phases in others.

The programmes introduced or being proposed under EI can be seen as the current endpoint in some long-run trends in federal training policy. Prior to the early 1980s, the latter tended to be more classroom based and centrally designed. Evaluations of these earlier training schemes were clear in pointing out that institutionally-based training was much less effective than employer-based/experience-related training in generating improvements in employment outcomes and earnings (Riddell, 1995). The introduction of the Canadian Jobs Strategy (CJS) in 1985 involved a significant shift in the direction of federal training policy toward greater employer involvement and some decentralisation of decisions related to training to regional offices.

Evaluations of the component programmes in the CJS in the late 1980s reinforced the findings regarding earlier schemes: the most successful were those that involved employer-based training. In particular, programmes involving subsidies to employers for training, an employer-employee trust fund to finance training, and, to a somewhat lesser extent, work experience generated positive results. Few of the programmes entailed increases in wage levels for participants but many led to increased employment and hours worked. The one place where institutional training seemed to have some positive effect was for young labour market entrants and female labour market re-entrants. The evaluations also pointed to an important tension in training policy, however. The most successful programme was a wage subsidy to employers providing training in skills deemed to be in short supply (Riddell, 1995). This programme, however, had a greater tendency to help workers who already had some skills than other programmes targeted specifically at individuals having labour market difficulties. In some sense, this is not surprising: even with a subsidy employers are more likely to want to provide training to individuals who have already shown a strong attachment to the labour market and an ability to acquire skills. Thus, a key ingredient in programme design must be a decision on whether the goal of a given training policy is to develop a specific skill base (perhaps related to the KBE) or to help individuals with poor labour market histories.

Based on the evaluations in the late 1980s, a further shift in policy occurred in the early 1990s with the introduction of the Labour Force Development Strategy. This tended to discard the less successful classroom-based training

programmes and further enhance employer involvement in training delivery. The new initiatives were gathered together under the title of the Employability Improvement Programme (EIP) in 1991. The three main training elements of the EIP were a programme which offered a wage subsidy to employers providing training, one integrating class room training, on-the-job training and work experience directed by project co-ordinators, and one offering financial assistance to individuals wishing to purchase institutional training to upgrade their skills. A government evaluation of the EIP in 1995 found that all three programmes increased annual weeks worked, with the wage subsidy programme again providing the largest gains. Earnings gains associated with the programmes appear to have come almost entirely from increases in weeks worked rather than wages. Again, there was some indication of greater benefits for those who were (somewhat) better off before going into the programmes. Thus, older workers and workers with a high school education obtained better results than younger workers and those with less than a high school education. The programme offering assistance for individual retraining in particular had participants who tended to have previous skills and experience (HRDC, 1998). Thus, it appears that through a process of trial and evaluation the federal government ''learned its lesson'': developing training programmes that are integrated with work experience and emphasise employer involvement. It is possible to focus on the failures associated with classroom-based training in the past and argue that federal training policy has a poor record (see, for instance, Boessenkool and Robson, 1997). However, the most recent form of training initiatives appears to deliver some real employment benefits.

The programmes introduced under EI can be seen as continuations of the more successful elements of the EIP. This is the case, in particular, for the targeted wage subsidies programme which follows in the footsteps of a series of earlier ones described above. The proposed skills loans and grants resemble earlier programmes helping individuals finance retraining which have also been relatively successful. The self-employment initiative seems promising given some US experience in this respect (Meyer, 1993). Thus, there appears to be reason to be optimistic about the outcomes of these programmes. Again, though, one must evaluate outcomes relative to the goal of the policy. If the objective is to help shift the economy toward a KBE, then the employer wage subsidy and skills loans and grants programmes may prove useful. If, on the other hand, the

goal is to help the long-term unemployed, then other approaches, such as basic classroom training and job search training, may be more appropriate. These types of programmes may not register as dramatic returns as the wage subsidy type schemes but they may be a necessary first step to re-integrating these individuals.

An important feature of the EI-based programmes is that they are available to individuals who have received UI benefits at any time in the previous three years. This extends their scope potentially well beyond the current set of recipients and increases the potential impact of the programmes on the labour market as a whole. Finally, under an offer made by the federal government in 1996, all or any subset of these programmes can be taken over by a particular province in its jurisdiction or run jointly with the federal government. The division of powers and funding responsibility are being set in a series of Labour Market Development Agreements between the federal government and individual provinces. These vary from the immediate handover of all responsibility to Quebec, to an agreement with British Columbia that stipulates that the federal and provincial governments will play an equal role in designing and delivering training (but leaves many details to future consultation).

Training is also delivered directly to income assistance recipients in most provinces. The Strategic Initiatives Programme is a federal-provincial cost sharing initiative that funds innovative programmes related to training and helping to re-employ social assistance recipients. A number of projects are underway within this programme, some of which, such as the Nova Scotia Compass programme, have elements that are very similar to those just mentioned under EI. Another example is the Self Sufficiency Project being run in British Columbia and New Brunswick as a joint federal-provincial initiative to try to help move single mothers into the workforce by offering an earnings supplement in much the same way as the targeted earnings supplement will apparently attempt to help similar groups move off of EI receipt. Published evaluations of provincial programmes are few, so that it is difficult to assess their success.

- Direct purchase of training

Before the agreement to devolve training powers to the provinces, the federal government purchased blocks of training seats from provincial and private education and training providers. These were then either used directly for UI recipients or offered as a means of subsidising employer investment in worker

training. This is one element of labour market training from which the federal government is withdrawing.

- Programmes directed at youth

One area of stated policy concern for the federal government in recent years has been access of youth to work experience. Addressing youth labour market outcomes could help in responding to the inter-generational equity issues set out in the description of earnings trends earlier. The federal initiatives aimed at youths consist of three main programmes: Youth Internship Canada; Youth Service Canada; and Student Summer Job Action. The Youth Internship Canada initiative is aimed at unemployed and underemployed individuals under the age of 30. Through it, the federal government provides funds to set up internships with businesses, community and government organisations that are intended to provide youth participants with experience in growth areas of the economy. The Youth Service Canada initiative is targeted at youth who are having more difficulty entering the labour force and is in the form of similar types of internships but related to community service. The Student Summer Job Action has several components, with the one most directly related to training being wage subsidies to employers to create ''career-related'' summer jobs for secondary and post-secondary students. In the 1998 federal budget the government commits itself to expanding the size of these initiatives, with particular emphasis on youth aged 20 to 24 who have not completed high school. The Budget also announced an EI premium holiday to firms for youth hired in 1999 and 2000.

It is worth noting that these types of programmes are not new. The Career Access programme in place in the mid 1980s also offered a wage subsidy to employers to help young people gain work experience (Davies, 1985). Evaluations of programmes to aid integration into the labour market under the Canadian Jobs Strategy found the same type of results as those specified for the more general training programmes discussed above: initiatives involving work experience delivered improved employment outcomes. In contrast to the programmes targeted at the more general population, however, institutional training also yielded some positive outcomes for – especially female – youth (Riddell, 1995). The federal policies just listed are aimed mainly at generating work experience, in the hope that this will improve access to jobs in the future. The evidence from earlier initiatives suggests that this may be a successful route to pursue: youth appear to be one group where there is a return to training and experience

programmes which do not end up just helping those who are already better advantaged in terms of skills.

One point of concern with youth is that those with less than a high school education might end up in a cycle of dependency on Income Assistance, engendered by poor labour market outcomes. For this reason, several provinces have programmes specifically aimed at generating work experience for young, low-educated Income Assistance recipients. For example, the Integrated Training Centres for Youth in Alberta attempt to give individuals aged 16 to 20 who have left school a co-ordinated programme of work experience and occupation-specific skill training. The training is to be targeted at occupations specified by an employer committee as being in demand in the local economy. The programme also includes financial incentives for participants to meet training goals. In spite of this, the programme experienced relatively high drop out rates at the outset. Similarly, in Nova Scotia, a programme aimed at Social Assistance recipients aged 18 to 30 attempts to link the participants to local employers and offers a wage subsidy in order to encourage training. Both programmes thus make use of the most successful elements of earlier schemes: employer involvement and work experience.

- Programmes promoting training for older workers

The federal government has also introduced initiatives aimed at helping older workers upgrade their skills. Thus, the 1998 budget announced a programme under which individuals will be able to make tax free withdrawals from Registered Retirement Saving Plans (RRSPs) of up to C$ 10 000 per year to fund enrolment in full-time training or higher education for at least three months. In addition, as noted, tax credits previously available only to full time students will be extended to part-time students. Since the recently-created ''Millenium Scholarship fund'' has no restrictions in terms of age of participants, it might also play a role in helping some older workers to go back to school. Again, though, these programmes seem unlikely to help the least skilled. Individuals must have RRSP savings or enough earnings to make responses to benefits delivered through the tax system worthwhile.

Key findings

The key findings regarding education and training performance and policies can be summarised as follows:

- Canada has a highly-educated workforce by international standards.
- Canada's expenditures are high by international standards both as a percentage of GDP and on a per student basis.
- Nevertheless, the International Literacy Survey has revealed some surprisingly poor literacy outcomes among less-educated Canadians in particular.
- Education has become a policy priority. However, with potential spending power rising more rapidly at the federal than the provincial level but direct powers over education being at the provincial level, there is a tension in the setting and funding of education policy that has not been adequately addressed.
- Canada is not a leader in the amount of training acquired by adults.
- Training, and particularly employer-based training, is heavily skewed towards those who are already doing well in terms of their skill levels: more educated men in skilled occupations.
- Government training programmes appear to have become more effective but it is difficult to evaluate the current status of policy since the federal government is withdrawing from labour market training, which is considered an extension of education, a provincial responsibility.

Scope for further action

The preceding analysis has generated inputs to a discussion of possible future policy directions related to education and training. Its key conclusion is that, while in some sense Canada has been a KBE for some time, the labour market has not to this point been dominated by a skill biased demand shift. To reiterate, that is not do say that such a shift is not going on. Trends in the labour market do have a significant educational element to them, suggested mainly by the fact that the worst outcomes are observed for low-skilled males. However, there are also important gender and generational elements that do not appear to be directly related to changes in returns to education and training. For women, educational differentials in labour market outcomes have actually declined, and successive generations of men have done worse in terms of earnings regardless of their education level. Thus, in addition to the gradually emerging KBE, other

forces, such as demographic supply shifts and institutional factors, seem to have been driving the labour market and deserve attention.

This has implications for the design of policies. In particular, it suggests that further efforts are necessary to lower structural unemployment, to which skill mismatch is only one contributing factor. As discussed in the *Jobs Strategy Chapter* of the *1996 Economic Survey of Canada*, the enrichment of unemployment insurance and social assistance in the 1970s and 1980s reduced the cost of unemployment noticeably, leading a growing number of people in some regions to rely on public transfers as a regular form of income supplement. Recent reforms of the unemployment insurance system and changes to provincial social assistance (including those stemming from the new Child Tax Benefit) have reduced work disincentives significantly. However, as noted in Chapter II, more could be done to lessen labour market rigidities, such as a reduction of regional differentiation and introduction of experience rating in unemployment insurance as well as further adjustments to the tax and social benefit system along the lines of the Child Tax Benefit.

While these would seem to be the most effective policy options to improve labour market performance in the short run, there is no doubt that the increased policy attention that education and training have received recently is appropriate. As noted, skill-related differences in labour market outcomes are substantial and, even though the pace of change has been gradual in Canada, trends in technology and trade suggest that the economy will be becoming more and more knowledge based. Moreover, by international comparison, Canada appears to be an lagging in terms of the size of the knowledge intensive sector. Thus, it seems reasonable to emphasise human capital development as part of a national policy agenda that attempts to accelerate the country's move toward a KBE. This is indeed current government policy, both at the federal level and in some provinces, and Canadians have been responding to calls to achieve higher education levels in record numbers, given the high returns involved.

Although Canada has been successful in terms of higher proportions of young individuals entering and staying in school longer, there still are some disturbingly poor educational outcomes given the high levels of expenditure by international standards. The school drop-out rate remains substantial and literacy levels, in particular, continue to be surprisingly low compared to other countries. These are longstanding problems to which there is no obvious solution. Increased

public expenditures may have little impact until youth perceive a greater return to completing high school. More emphasis on vocational courses, in which few students participate, may make the system more attractive and help lower the drop-out rate. More job relevant curricula may also hold students' attention to a greater extent and help them to acquire higher levels of literacy. A comparison with other, better performing countries suggests, however, that, to some extent, the lack of success in raising literacy levels reflects failures in the education system. On the positive side, local autonomy allows for experimentation and sometimes adoption of innovative approaches. On the other hand, the system has led to wide inconsistencies in the commitment to enhancing literacy and in the way programmes are run. Not surprisingly, in poorer provinces, where literacy rates are lowest, governments have been the least well prepared to fund sizeable remedial programmes. This suggests the need for a co-operative effort across the country to ensure that education levels correspond to the requirements of a KBE.

With education being a provincial jurisdiction, the federal government has taken indirect routes to being involved in human capital development. The Millenium Scholarships, for instance, attempt to address the issue of accessibility of post-secondary education. The desired effect on educational affordability could be frustrated, however, to the extent that provinces curtail existing student assistance programmes and/or universities increase tuition as provinces reduce education funding. Raising instead federal transfers which are provided in the form of block grants runs the risk of having the money spent on other provincial priorities. This could be envisaged, however, if it helps to reinforce the federal government's role as a co-ordinator and setter of national standards and establish a national education policy. Acknowledging that education is a national concern with significant externalities across provinces, such a policy should – through consultation between federal and provincial governments – define standards on accessibility, spending per student and educational outcomes and provide for pan-Canadian recognition of qualifications. As with health care, failure to meet established standards could involve a reduction in federal transfers.

In contrast to education, where the federal government appears to be trying to define a role for itself, it is withdrawing from labour market training in favour of direct support for individuals with a view to improving accessibility and affordability. Training closely tailored to local conditions and opportunities could be more successful and flexibility, accountability and integration with related

services (such as welfare and education) better served by provincial control. There are, however, a number concerns related to the devolution of training policy to the provinces. In the absence of inter-provincial recognition of qualifications, this move could reduce labour mobility within Canada, by adding to the many existing inter-provincial barriers. Ensuring that similar training standards are used across the country would require an expensive bureaucracy. Moreover, devolution raises the possibility of some provinces acting as free riders, as seems to be the case in university education. Finally, it has been pointed out that assessing training needs and evaluating policy outcomes requires considerable research infrastructure and expertise which many provincial governments do not currently possess. Recent policy changes in Canada are at variance with developments in other OECD countries which have emphasised the advantages of a ‘‘one-stop shop’’ where benefit payment, employment placement and referral to active labour market measures are all handled. In any case, it will be important that the handover of responsibilities for training policy to the provinces be accomplished in a way that ensures that potential benefits are realised and the above concerns are addressed. In particular, provinces will need to implement the objective of developing a more pervasive ‘‘evaluation culture’’ towards training programmes and build on the experience of those which have proven successful in the past.

The current set of federal training policies, which has evolved through a process of trial and evaluation, appears to deliver positive results by emphasising employer involvement and integration with work experience. One feature of the current system is that, based on experience with past successful programmes, it tends to help those who already have some skills. It thus exacerbates the gap in skills between the more and less educated, creating a virtuous circle of investment in human capital for those initially well-endowed: initial investments in education are further augmented by investment in training. While such a policy is more likely to produce results, it needs to be complemented by programmes aimed at developing more basic skills to help those not in a position to take advantage of more advanced schemes. Furthermore, firm-based training of existing employees tends to focus on people who already have good skills. In addition, although employer-sponsored training is the most important component of total training funding, it is low by international standards. This suggests that ways need to be found to encourage employers to increase their training efforts.

Options include a training tax/tax credit scheme or the obligation to spend a minimum amount of the payroll on training. The risk to bear in mind with these measures is that resources can be wasted on ineffective training and there is no guarantee that the workers who receive the training would be the most in need. Another possibility is government assistance to joint firm/worker trust funds; previous experience suggests that the fact that such funds are financed and administered jointly by workers and management leads to worthwhile training initiatives.

In summary, further action to improve the functioning of the labour market is desirable as the economy is becoming more knowledge based. There are some problems in the areas of education and training but, in general, their solution seems to require getting better value for money rather than additional spending. In this respect, federal/provincial co-operation is crucial. Canada's decentralised system allows skill development to be more responsive to local needs. However, promoting national standards is also important to ensure that Canada has a highly skilled workforce with portable qualifications that correspond to the needs of a KBE.

Notes

1. For a more detailed discussion of the factors behind the sluggishness of economic activity during that period, see Charles Freedman and Tiff Macklem, ''A Comment on The Great Canadian Slump'', Canadian Journal of Economics, August, 1998.
2. 35 per cent in 1998 as compared with almost 60 per cent in 1980, according to the Department of Finance's preferred definition of ''commodities''.
3. Provinces other than British Columbia are nonetheless affected by the fall in commodity prices (*e.g.* Alberta by weak oil prices, Ontario and Quebec by the weakness in metal prices, and Ontario and Eastern Canada by the depressed prices for pulp and paper).
4. By lowering the current price of consumption relative to future spending, a steepening of the yield curve is likely to induce households to consume more now and less later.
5. As measured by the difference between the nominal interest rate and the rate of change of house prices.
6. This partly reflects restructuring in the retail sector (*e.g.* department stores) but also the downward price pressures from the depreciation of Asian currencies (particularly for motor vehicles).
7. As discussed in previous OECD *Surveys*, Canada is one of the few countries where there is evidence of the presence of ''asymmetric'' inflation effects which could make corrective policy action very costly in terms of output losses once excess demand emerges.
8. See the *OECD Economic Outlook 61* for a discussion of these issues, as well as Table 11 (an updated version of which will be contained in the *OECD Economic Outlook 64*) for details of the fiscal strategies pursued by OECD countries.
9. Productivity growth in Germany has been weak over the early part of the 1990s (the data in Table 2 extend only to 1993 for Germany) but has since picked up significantly. See *OECD Economic Surveys: Germany, 1998*.
10. A note of caution is warranted in considering such a long-term scenario, however. The projections for the US, Japan and the OECD are taken from previous work *(OECD Economic Outlook 62, December 1997)*, and as such may not reflect current thinking.
11. A similar scenario constructed by the Bank of Canada suggest a ''medium'' growth profile of potential output of 3 per cent per annum over the period 1996-2001; 2.4 per cent for the period 2001-2011; and 1.7 per cent for the years 2011-2021. See John Kuszczak and Richard Dion, ''Potential output growth: some long-term projections'', *Bank of Canada Review*, Winter 1997-1998.

12. A summary of these reforms is contained in Table 11 of *OECD Economic Surveys: Canada, 1997*.

13. A Schedule I bank (as defined in the Bank Act) is where no person or entities controlled by the person beneficially own more than 10 per cent of any class of shares, while a Schedule II bank refers to all other banks, that is, where a person beneficially owns more than 10 per cent of any class of shares, including all foreign bank subsidiaries and bank subsidiaries of eligible Canadian institutions.

14. See Booth and Laurence, ''Competition and Profitability in the Financial Service Industry'', in J Mintz and J. Pesando (eds.), *Putting Consumers First: Reforming the Canadian Financial Services Industry*, CD Howe Institute, 1988.

15. Booth (1988).

16. For a more complete analysis of this argument see Charles Freedman and Clyde Goodlet, ''The Financial Services' Sector: Past Challenges and Future Prospects'', Technical Report No. 82, Bank of Canada, 1998.

17. The following discussion is based on Freedman and Goodlet (1998).

18. Quoted from the *Report by the Minister for International Trade to the Prime Minister of Canada*, 26 September 1985.

19. For more details see Richard Lipsey and Robert York, ''Evaluating the Free Trade Deal: A Guided Tour through the Canada-US Agreement'', Policy Study No. 6, CD Howe Institute, 1988.

20. A description of the classification of manufactures, which is based on the OECD's *Science, Technology and Industry Scoreboard of Indicators*, is given in Annex 5.

21. For the definition of export specialisation see footnote 2 in Table 9.

22. See, J. Schott and M. Smith (eds.), *The Canada-United States Free Trade Agreement: The Global Impact*, Institute for International Economics, Washington, 1988; and M. Trebilcock and R. York (eds.), *Fair Exchange: Reforming Trade Remedy Laws*, CD Howe Institute, Policy Study 11, Toronto, 1989.

23. There have been 45 Chapter 19 proceedings filed with both sides initiating roughly the same number of them. Of these cases, only one (involving fresh, chilled and frozen pork from Canada) has led to the reversal of a national agency's (US International Trade Commission) final determination. In three cases, that challenged the US Commerce Department's finding of injury in countervailing duty determinations (red raspberries, fresh, chilled and frozen pork and new steel rails), the binational review panels upheld, but modified the Commerce Department's final decisions.

24. It should be noted that the choice of time periods may exaggerate differences in performance somewhat; given both the different cyclical position of Canada and the United States and the comparison between periods of excess demand and excess supply.

25. *Agreement on Internal Trade*, 1995

26. Internal Trade Secretariat, *Internal Trade Barriers in Canada*, March 1997.

27. The employer rate is 1.4 times the employee rate.

28. This gap is substantially wider if Canada's goods and services tax is included in such measures.

29. Chief Actuary, *Canada Pension Plan, Sixteenth Actuarial Report, September 1997*.

30. The legislation modifying CPP was adopted in 1998 but made the first increase in contribution rates retroactive to the previous year.

31. The real return of 3.8 per cent is projected by the Chief Actuary of Canada as a constant weighted average of the real unchanged rate of 1.5 per cent on the CPP Operating Balance and of the real rate of return of 4 per cent, taking the following factors into account: the average real yield on the QPP (which has always been invested in a diversified portfolio), the average real long-term yield on the funds of a sample of the largest private pension plans in Canada, and the real average long-term yield of a hypothetical portfolio invested equally in different areas. See Chief Actuary, 1997.

32. See ''Maintaining Prosperity in an Ageing Society: Summary'', in Meeting of the OECD Council at Ministerial Level: Documentation for Ministers, OECD, Paris, April 1998.

33. The retired person dependency ratio captures the combined effects of demography, age-specific activity rates and trends in retirement. For details, see ''The Retirement Decision'', in *OECD Economic Outlook 63*, pp. 179-192.

34. See, for example, the discussion in, Gunderson, Morley, ''Flexible Retirement as an Alternative to 65 and Out'', Commentary, CD Howe Institute, Toronto, 1998.

35. The introduction of UI in 1971 is estimated as having added as much as 2 to 3½ percentage points to the structural rate of unemployment. For a survey of the literature of the impact of UI, see Annex I in *OECD Economic Surveys: Canada, 1996*.

36. See for example, Sargent, Timothy, ''The BU Ratio: Prospect and Retrospect'', Department of Finance, mimeo, 1998.

37. See for example, Sargent, Timothy, ''An Updated EI Disincentives Index for Canada'', Department of Finance, mimeo, 1988. Sargent constructs an index which assumes that the key parameters affecting workers' decisions are the earnings replacement rate, the minimum number of weeks required to qualify for benefits, and the number of weeks of benefits one receives if one works the minimum number of weeks.

38. Figure 22, in *OECD Economic Surveys: Canada, 1996* shows that the work disincentive effects of UI across regions varies widely, although the extent of such variation has declined over the last few years.

39. As part of EI reform, the government made a legislative commitment to monitor and assess the effectiveness of the programme on an annual basis. The results cited are from the first of such reports. See Canada Employment Insurance Commission, *The 1997 Employment Insurance Monitoring and Assessment Report.*

40. See *Report of the Technical Committee on Business Taxation*, December 1997, especially Chapter 8.

41. See for example, M. Rafiquzzaman and L. Whewell, ''Recent Jumps in Patenting Activities: Comparative Innovative Performance of Major Industrial Countries, Patterns and Explanations'', Industry Canada mimeo, May 1998.

42. Katz and Murphy (1992), and Juhn, Murphy, Pierce (1993), for example, conclude that relative demand shifts favouring university-trained workers have been going on since at least the early 1960s.

43. In particular, the new technologies are often assumed to simultaneously replace or make easier some routinised types of tasks and require a particular set of skills for creation and implementation. This could imply a core of more stable jobs filled with workers who know how to create and implement new technologies – workers whom the firm might find difficult to replace – and a periphery of jobs that might once have required skills but now are routinised and easily filled by any workers. Those in these latter jobs would face unstable work patterns characterised by short-term contracts. Even workers in the core jobs should expect to have to adapt, as new technologies come on line. This is sometimes referred to as the need for ''lifelong learning''. These tendencies toward instability are enhanced by the increasingly competitive marketplace, causing firms to have to constantly re-invent themselves. In doing so, firms will need to lay off workers not trained for the new technology and hire others with relevant skills.

Bibliography

Addison, J.T., ''The US Labor Market: Structure and Performance'' in *Structural Change and Labour Market Flexibility,* H. Siebert, ed., Tubingen: Mohr, 1997, pp. 187-222.

Allen, R.C., ''The Demand and Supply of Post-Secondary Training and Education in British Columbia'', Department of Economics, University of British Columbia, 1997.

Baldwin, J.R., T. Dunne, and J. Haltiwanger, ''A Comparison of Job Creation and Job Destruction in Canada and the United States'', Statistics Canada, Analytical Studies Branch: Research Paper 64, 1994.

Bar-Or, Y., J. Burbidge, L. Magee and L. Robb, ''The Wage Premium to a University Education in Canada, 1971-1991'', *Journal of Labor Economics,* 1995, Vol. 13, No. 4, pp. 762-794.

Barrett, G. F. and M.I. Cragg, ''An Untold Story: The Characteristics of Welfare Use in British Columbia'', *Canadian Journal of Economics,* February 1998, pp. 165-188.

Beaudry, P. and D.A. Green, ''Employment Outcomes in Canada: A Cohort Analysis'', presented at the IRPP Conference on Adapting Public Policy to a Labour Market in Transition, Montréal, April 1997*a*.

Beaudry, P. and D.A. Green, ''Cohort Patterns in Canadian Earnings: Assessing the Role of Skill Premia in Inequality Trends'', NBER Working Paper 6132, August 1997*b*.

Betcherman, G., ''Research Gaps Facing Training Policy-Makers'', *Canadian Public Policy*. March 1993, pp. 18-28.

Betcherman, G., K. McMullen, N. Leckie and C. Caron, *The Canadian Workplace in Transition,* Kingston: IRC Press, 1994.

Boessenkool, K.J. and W.B.P. Robson, ''Ending theTraining Tangle'', CD Howe Institute Commentary, February 1997.

Borjas, G.J. and V.A. Ramey, ''Foreign Competition, Market Power and Wage Inequality'', *Quarterly Journal of Economics,* November 1995, pp. 1075-1110.

Bound, J. and G. Johnson, ''Change in the Structure of Wages in the 1980s: An Evaluation of Alternative Explanations'', *American Economic Review,* June 1992, pp. 371-92.

Card, D., F. Kramarz and T. Lemieux, ''Changes in the Relative Structure of Wages and Employment: A Comparison of the United States, Canada, and France'', Université de Montréal, September 1995.

Card, D. and T. Lemieux, ''Multiple Modes of Adjustment: A Comparative Study of Youth in the US and Canada, 1970-1995'', Université de Montréal, May 1996.

Card, D. and W.C. Riddell, ''A Comparative Analysis of Unemployment in Canada and the United States'' in D. Card and R.B. Freeman (eds.), *Small Differences that Matter: Labor Markets and Income Maintenance in Canada and the United States,* Chicago: University of Chicago Press and NBER, 1993.

Corak, M., ''Unemployment Insurance Once Again: The Incidence of Repeat Participation in the Canadian UI Program'', *Canadian Public Policy,* June 1993, pp. 162-76.

Crompton, S., ''Employer-Supported Training – It Varies by Occupation'', *Perspectives on Labour and Income,* Statistics Canada, Spring 1994, pp. 9-17.

Davies, J. B., ''Training and Skill Development'' in *Adapting to Change: Labour Market Adjustment in Canada,* W.C. Riddell, ed., Toronto: University of Toronto Press, 1986, pp. 163-220.

Donald, S., D.A. Green and H.J. Paarsch, ''A Flexible Estimator of Distribution Functions and Lorenz Curves in the Presence of Covariates, with an Application to Differences in Wage Distributions Between Canada and the United States'', Department of Economics, University of British Columbia, 1997.

Freeman, R.B. and K. Needels, ''Skill Differentials in Canada in an Era of Rising Labour Market Inequality'' in *Small Differences that Matter: Labour Markets and Income Maintenance in Canada and the United States,* D. Card and R.B. Freeman (eds.), Chicago: NBER and University of Chicago Press, 1993.

Gaston, N. and D. Trefler, ''The Labour Market Consequences of the Canada-US Free Trade Agreement'', *Canadian Journal of Economics,* February 1997.

Gera, S. and P. Masse, ''Employment Performance in the Knowledge Based Economy'', Industry Canada, Working Paper No. 14, 1996.

Green, D.A. and W.C. Riddell, ''Job Durations in Canada: Is Long Term Employment Declining?'' in *Transition and Structural Change in the North Amercian Labour Market,* M.G. Abbott, C.M. Beach and R.P. Chaykowski, eds., Kingston: IRC Press, 1997, pp. 8-40.

Grenon, L. and B. Chun, ''Non-permanent Paid Work,'' *Perspectives on Labour and Income,* Statistics Canada, Autumn 1997, pp. 21-31.

Heisz, A., ''Changes in Job Tenure in Canada'', *Canadian Economic Observer,* 1996, pp. 3.1-3.9.

Human Resources Development Canada, ''Evaluation of the Employability Improvement Program'', 1998.

Juhn, C., K. Murphy, and B. Pierce, ''Wage Inequality and the Rise in Returns to Skill'', *Journal of Political Economy,* 101, pp. 410-42.

Kapsalis, C., ''An Explanation of the Increasing Age Premium'', Research Paper Series No. 112, Analytical Studies Branch, Human Resources Development Canada, June 1998.

Katz, L.F. and K.M. Murphy, ''Changes in Relative Wages, 1963-1987: Supply and Demand Factors'', *Quarterly Journal of Economics,* 107, February 1992, pp. 35-78.

Krahn, H., ''NonStandard Work on the Rise'', *Perspectives on Labour and Income,* Statistics Canada, Winter 1995.

Kuhn, P., ''Canada and the OECD Hypothesis: Does Labour Market Inflexibility Explain Canada's High Level of Unemployment?'', paper presented at the IRPP Conference on Adapting Public Policy to a Labour Market in Transition, Montreal, 1997.

Lavoie, M. and R. Roy, ‘‘Employment in the Knowledge-Based Economy: A Growth Accounting Exercise for Canada’’, Research paper R.98.9E, Human Resources Canada, June 1998.

Lowe, G.S., ‘‘Computers in the Workplace, *Perspectives on Labour and Income,* Statistics Canada, Summer, 1997, pp. 29-36.

Morissette, R., ‘‘The Declining Labour Market Status of Young Men’’, Statistics Canada, 1997.

Morissette, R., J. Myles and G. Picot, ‘‘What is Happening to Earnings Inequality in Canada?’’, Statistics Canada, Analytical Studies Branch, Research Paper No. 60, 1993.

Murphy, K., W.C. Riddell and P.M. Romer, ‘‘Wages, Skills and Technology in the United States and Canada’’, Western Research Network on Education and Training Working Paper 97.01, University of British Columbia, 1997.

Nickell, S., ‘‘Structural Changes and the British Labour Market’’ in H. Siebert (ed.), *Structural Change and Labour Market Flexibility: Experience in Selected OECD Economies,* Tubingen: Mohr, 1997.

OECD, *Education and Employment,* Paris, OECD, 1995*a.*

OECD, *Income distribution in OECD countries,* Paris, OECD, 1995*b.*

OECD, *Technology, Productivity and Job Creation,* Paris, OECD, 1996.

OECD, Education at a Glance: OECD Indicators, 1997, Paris, OECD, 1997.

OECD and Human Resource Development Canada. Literacy Skills for the Knowledge Society, Paris, OECD, 1997.

Packer, A., ‘‘Skill Deficiencies: Problems, Policies and Prospects’’, Journal of Labor Research. Summer 1993, pp. 227-245.

Richardson, D.H., ‘‘Changes in the Distribution of Wages in Canada, 1981-1992’’, Canadian Journal of Economics, August 1997, pp. 622-643.

Riddell, W.C., ‘‘Human Capital Formation in Canada: Recent Developments and Policy Responses’’ in K.G. Banting and C.M. Beach (eds.), *Labour Market Polarization and Social Policy Reform,* Kingston, School of Policy Studies, Queen’s University, 1995.

Riddell, W.C., ‘‘Human Capital Formation in Canada: Recent Developments and Policy Responses’’ in *Labour Market Polarization and Social Policy,* Kingston, School of Policy Studies, Queen’s University, 1995.

Riddell, W.C. and A. Sweetman, ‘‘Human Capital Formation in a Period of Rapid Change’’, paper presented at the IRPP Conference on Adapting Public Policy to a Labour Market in Transition, Montreal, 1997.

Riddell, W.C. and A. Sharpe, ‘‘The Canada-US Unemployment Rate Gap: An Introduction and Overview’’, *Canadian Public Policy: Special Edition on the CERF/CSLR Conference on the Canada-US Unemployment Rate Gap,* February 1998, pp. s1-s37.

Annex I

Indicators of monetary policy

Monetary conditions index

Over the last several years, the Bank of Canada has used the concept of monetary conditions – the combination of the movement of a short-term interest rate and the effective exchange rate – as the operational guide of policy. The main reason for that is to account for the fact that monetary policy actions have their effect through both interest rates and the exchange rate, especially in a small open economy. In determining the overall direction of monetary policy in the coming period, the Bank of Canada sets out a desired path for the MCI that will result in the rate of inflation being within the inflation-control target band (currently set at 1 to 3 per cent), taking account of the fact that changes in the MCI will have their main effect six to eight quarters ahead. The Bank, however, emphasises that the desired path of the MCI is not precise, but rather a zone, which needs to be constantly reassessed in the light of new information and economic shocks (such as the impact of the Asia crisis).

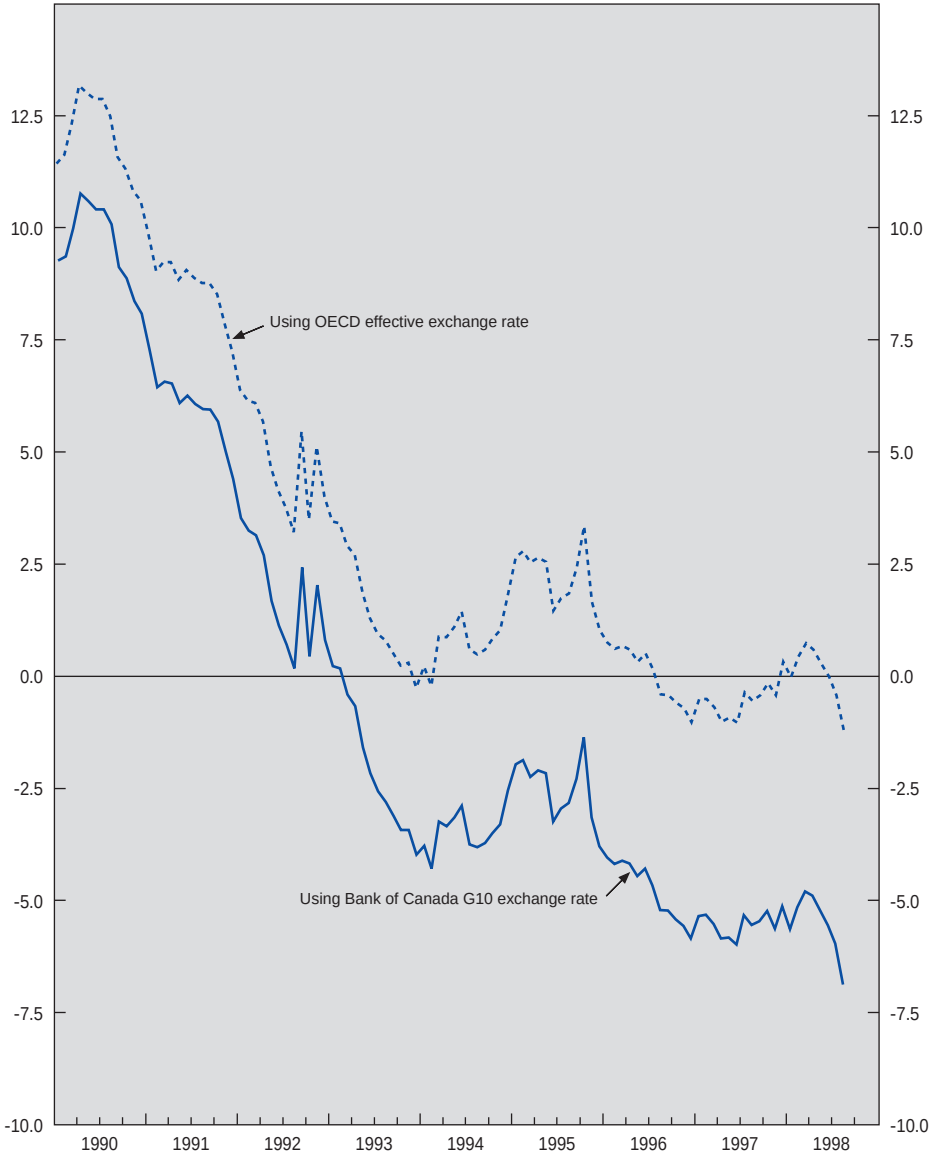
In technical terms, the MCI is defined as the combination of the changes in the 90-day commercial paper rate and the Group of 10 exchange rate from an arbitrary base period (which in the present case is January 1987, see Figure A1), with the latter given a weight of 1/3. These weights are based on empirical work that indicates that a 1 percentage point change in interest rates has about the same effect on aggregate demand as a 3 per cent change in the effective exchange rate (after six to eight quarters). Ideally, a real index based on the real interest rate and the real exchange rate should be used but relevant data are only available with some delay. However, as can be seen from Figure 2 in the text, differences between the nominal and real measure are relatively small. Likewise, as Figure A1 shows, the MCI is relatively insensitive to the choice of effective exchange rate lending support to the Bank's choice of a relatively small basket of currencies to calculate the measure, although exceptional developments – such as the recent turmoil in Asian financial markets – can have an impact.

Measures of core inflation[1]

Most central banks with inflation targets use a measure of core inflation similar to the Bank of Canada (which focuses on the consumer price index, excluding food, energy,

Figure A1. **MONETARY CONDITIONS INDEX**

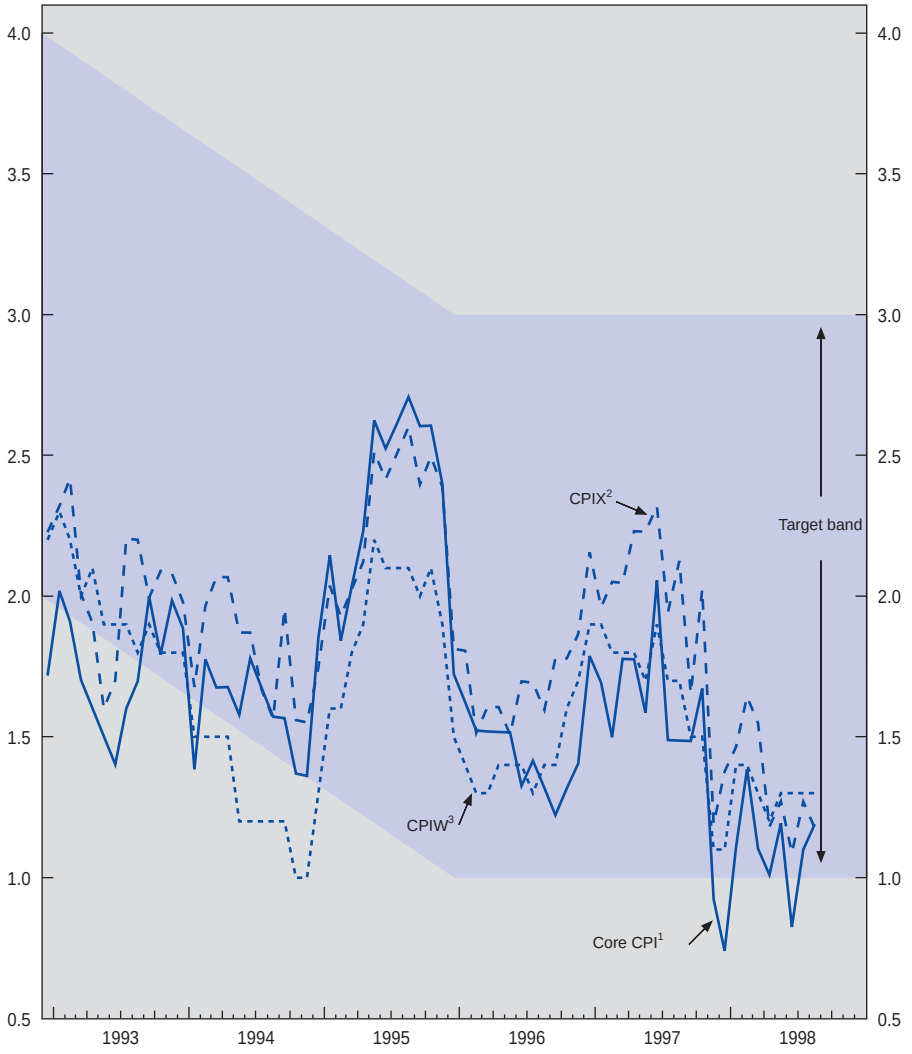
January 1987 = 0



Source : Bank of Canada; OECD.

Figure A2. **MEASURES OF UNDERLYING INFLATION**

Percentage change over 12 months



1. CPI excluding food, energy and the effect of indirect taxes.
2. CPI excluding the eight most volatile components as well as the effect of indirect taxes on the remaining components.
3. Each component of the total CPI is multiplied by an additional weight that is inversely proportional to the component's volatility.

Source: CANSIM-Statistics Canada; Bank of Canada.

and the effect of indirect taxes) as a guide in conducting monetary policy.[2] Some central banks (for example, the Reserve Bank of New Zealand and the Bank of England) also regularly publish other statistical measures of the trend inflation rate. These measures are based on the assumption that extreme fluctuations in relative prices generally translate into temporary shocks to the inflation rate without altering its underlying trend. The trend of inflation can be estimated using statistical measures but it cannot be measured precisely. The Bank, therefore, compares several measures deeming that when these convey the same message, its decisions are based on reliable guides. If, however, the estimates diverge significantly, the reasons for this divergence must be closely examined to help monetary policy keep the trend rate of inflation within the target range.

In this regard, the Bank of Canada monitors two other measures of trend inflation that seem to contain the most information about future movements in the total CPI. The first of these measures (CPIX) excludes the eight most volatile components from the CPI (fruit, vegetables, gasoline, fuel oil, natural gas, mortgage interest, inter-city transportation, and tobacco products) as well as the effect of indirect taxes on the remaining components (see Figure A2). The total weighting of these eight components in the CPI (16 per cent) is less than the weighting of total food and energy (26 per cent). The second measure (CPIW) retains all the CPI components, but multiplies each CPI basket weight by an additional weight that is inversely proportional to the component's variability, so that the most volatile components have less influence in the overall index.

As shown in Figure A2, both these measures have followed a path similar to that of the core CPI, although their levels differ slightly. They essentially tell the same story as the official measure, providing further checks that underlying inflation is within the bands.

Annex II

Federal and provincial fiscal policy

1998 Federal Budget

The Federal Budget was tabled in February. Its key feature is the zero balance (on a public accounts basis) planned for the 1998/99 and 1999/00 fiscal years (starting in April). The Budget continues to be based on three elements: 1) the two-year fiscal plan, premised on ''prudent'' economic assumptions that are more cautious than private sector forecasts; 2) the inclusion of a Contingency Reserve of C$ 3 billion in each year; and 3) the use of the Contingency Reserve, if not needed, for paying down debt. Other important elements of the Budget include the following:

- *Economic assumptions:* nominal GDP growth of 4.1 per cent in 1998 and 3.9 per cent in 1999 (compared with a private sector average of 4.7 and 4.9 per cent, respectively); short-term interest rates of 5.4 and 5.6 per cent over the next two years and long- term interest rates of 6.4 and 7.0 per cent (short-term interest rates are assumed to be 80 basis points higher than the private sector average in 1998 and 100 basis points higher in 1999, while for long-term rates the prudence factor is 50 basis points in 1998 and 100 in 1999).
- *Spending initiatives:* C$ 10.9 billion of new spending over the next four years, primarily for the proposed Canada Opportunities Strategy (which is intended to help expand access to education, lifelong learning and information technology, see Chapter III) and an increase in the Canada Health and Social Transfer to the provinces. Despite increased spending, programme expenditures as a share of GDP are projected to decline from the current level of nearly 13 per cent, to 11.5 in fiscal year 1999/2000.
- *Tax initiatives:* General tax relief for low- and middle-income earners, through an increase in the amount of income that can be earned tax free; the reduction or elimination (depending on income level) of the general surtax; and various measures related to the Canada Opportunities Strategy (primarily for students and upgrading skills/education). In sum, these actions will amount to over C$ 7 billion of tax relief over the next four years.
- *Debt repayment:* The Contingency reserve, if not required, will be used to pay down the public debt. Over the next two years, up to C$ 6 billion could thus be

used for this purpose. The net debt-to-GDP ratio is projected to drop to around 63 per cent in 1999/2000 from around 72 per cent in 1995/96.

Provincial budget themes

As noted in Chapter I, both the federal and provincial governments have contributed to a similar extent to the rapid improvement in public finances and are now both in a surplus position, on a *national accounts basis.* However, on a *public accounts basis,* while the federal budget was already in balance in 1997/98 the provinces, as a group, still showed a small deficit of around 0.4 per cent of total GDP and the aggregate net debt to-GDP-ratio was above that at the start of the decade (Table A1).[3] At present, five provinces are in budgetary balance or surplus (Nova Scotia, New Brunswick, Manitoba, Saskatchewan and Alberta), and eight (except Quebec and Ontario) have seen an improvement in their net debt-to-provincial GDP ratios since the mid-1990s.

Thus, while all provinces seem to be on the road to fiscal recovery, some are further along it than others, with the two largest (Quebec and Ontario) being the main laggards.

Table A1. **Provincial deficits and debt**

Public accounts basis, fiscal years, percentage

	Ratio of deficit/surplus to provincial GDP			Ratio of net debt to provincial GDP		
	1990/91	1994/95	1997/98	1990/91	1994/95	1997/98
Newfoundland	–3.8	–1.2	–0.2	39.0	48.3	40.0
Prince Edward Island[1]	–0.9	–0.1	–0.3	10.3	38.9	34.2
Nova Scotia	–1.5	–1.3	0.2	27.7	45.4	41.2
New Brunswick[2]	–2.4	–0.4	0.0	23.7	35.7	31.9
Quebec[3]	–2.1	–3.4	–1.1	24.0	33.4	35.7
Ontario	–1.1	–3.3	–1.5	15.1	29.0	30.3
Manitoba	–1.2	–0.8	0.1	21.8	28.4	23.0
Saskatchewan	–1.7	0.5	0.1	17.3	31.5	25.5
Alberta[4]	–2.5	1.1	2.6	7.7	15.0	5.9
British Columbia	–1.0	–0.4	–0.2	8.0	11.8	11.6
Total[5]	–1.5	–2.1	–0.6	16.5	26.9	26.1
Memorandum item:						
Federal government	–4.7	–4.9	0.0	57.6	71.6	68.2

1. Net debt as of March 1994 includes unfunded pension liabilities of C$ 337 million.
2. Net debt as of March 1993 includes unfunded pension liabilities of C$ 1.6 billion.
3. Including the accounting reform, the deficit is –1.2 per cent in 1997/98.
4. Accounting changes implemented in 1992-93 increased net debt by C$ 5.9 billion.
5. Total provincial deficit/debt as a share of national GDP, including the territories.

Source: Department of Finance, Canada.

Like at the federal level, this fiscal improvement can be attributed to expenditure discipline, and more recently to legislated restraints on deficits, debt accumulation and tax increases. Eight jurisdictions currently have, legislated fiscal rules ranging in coverage from New Brunswick's' four-year cumulative balanced-budget requirement to Manitoba's annual balanced budget obligation that also includes non-compliance penalties through ministerial salary cuts (see Table A2). Two jurisdictions have legislated debt reduction plans (Manitoba, and Alberta) and a third, the Yukon, has passed legislation which forbids the government from falling into a net debt position. Newfoundland, Prince Edward Island, Ontario and British Columbia remain the only provinces without any legislated fiscal rules.

In 1998, provincial budgets' continued, in general, to focus on reducing personal income taxes particularly for low-income families, while beginning to address in a more limited way tax changes on the corporate side. There have also been some increases in programme spending (primarily for health and education) for the first time in many years. Specific tax measures taken by provinces and territories, include the following:

- *Western Canada:* In all four western provinces, personal income tax rates were cut with British Columbia also reducing surtaxes and Manitoba lowering its payroll tax. In the corporate area, both British Columbia and Manitoba increased the exemption under the corporate capital tax and Saskatchewan announced a new corporate tax credit for research and development. (The latter province also introduced a new Child Benefit programme.)
- *Ontario:* The bulk of Ontario tax measures have been focused on reducing personal income taxes. The government announced the final phase of its personal income tax reduction, effective July 1998, six months ahead of schedule. As well, a new Child Care Supplement for Working Families was introduced. On the corporate side, the province announced that the small business tax rate would be reduced over the next eight years and several new business tax credits were introduced. With regard to payroll levies, the province increased the exemption threshold and eliminated the payroll tax on self-employed individuals starting in 1999. It also moved to lower the education tax on commercial and industrial property in municipalities with higher-than-average tax rates.
- *Quebec:* Quebec announced a major reform of their personal income tax system in their 1997 budget. In the 1998 budget speech, the focus was on initiatives for youth, stimulating private sector investment, and reducing the corporate tax burden. (One of the means by which the reduction in the corporate tax burden will be financed is through the elimination of the provincial small business deduction and the refundable tax credit for losses.)
- *Atlantic Canada:* There were few tax measures announced in the Atlantic provinces, given the focus in the region on attaining balanced budgets. New Brunswick, however, did announce a new low-income senior's benefit and Newfoundland increased the exemption threshold on its payroll tax. Nova Scotia introduced

Table A2. **Provincial-territorial legislated fiscal rules**

Province[1]	Fiscal rule (year enacted)	Highlights of the legislative requirements
Nova Scotia	Financial Measures Act (1996)	• Annual balanced budget from 1996-97 onwards. • Surplus must be used for tax cuts or debt reduction.
New Brunswick	Balanced Budget Act (1993; amended in 1995)	• Balanced budget over a four-year period beginning in 1996-97. • Netdebt must either remain at the level reached at the end of 1995-96 or be declining.
Quebec	Act Respecting the Elimination of the Deficit and a Balanced Budget (1996)	• Annual balanced budget by 1999-2000 and every year thereafter.
Manitoba	Balanced Budget, Debt Repayment and Taxpayer Protection Act (1995)	• An annual balanced budget from 1995-96 onward. • Debt Retirement Fund, with payments every five years after 1996-97.
Saskatchewan	Balanced Budget Act (1995)	• Balanced budget over a four-year period. • Debt Management Plan, requiring no specific debt repayment amount.
Alberta	Deficit Elimination Act (1993)	• Balanced budget by 1996-97.
	Balanced Budget and Debt Retirement Act (1995)	• Annual balanced budget every year starting in 1996-97. • Elimination of net debt excluding unfunded pension liabilities – they are dealt with in a separate plan – before March 31, 2010, with yearly minimum payments (note that better-than-expected surpluses could lead to the elimination of net det as early as March 31, 2000). • The first C$ 1 billion of any projected surplus will be used for debt repayment and inflation proofing of the Heritage Fund. The next C$ 250 million available for one-time initiatives. Any remaining surplus will be applied to debt reduction.
Yukon	Taxpayer Protection Act (1996)	• Prohibition of any debt accumulation.
Northwest Territories[2]	Deficit Elimination Act (1995; amended in 1996)	• Deficit targets leading to a balanced budget in 1998-99.

1. The Taxpayer Act introduced in British Columbia in 1991 was repealed following the election of a new provincial government in 1992. The current Financial Management Plan is not a legislated fiscal rule, but is subject to post-review and reporting by the provincial Auditor General.
2. Unexpected decreases in federal funding and increases in expenditure needs prevented the Northwest Territories from meeting the requirements set out in their 1995 legislation. The legislation was amended to be more flexible to unexpected changes in revenues and expenditures while still achieving the objectives of a balanced budget and a reduction in net debt by 1998-99.

Source: Department of Finance, Canada.

a new child Benefit and announced a sales tax rebate for residential electricity consumers.

– Territories: The Northwest Territories announced two new programmes in its 1998 budget, a Child Benefit and an Investment Tax Credit. The Yukon increased funding for health, training and assistance to low-income families.

In 1998/99, the aggregate provincial deficit (including the territories) is projected at some C$ 5.1 billion. While revenue growth should be somewhat limited by the tax reductions and slower economic growth, programme spending and debt charges are expected to increase slightly. Nevertheless, the 1998/99 outcomes might be better than expected, since several jurisdictions included both prudent economic assumptions and contingency reserves (totalling C$ 1.3 billion) in their projections.

Annex III

Classification of industries on the basis of technology

The classification of manufacturing industries used in Chapter II, is based on OECD Science Technology and Industry, *Scoreboard of Indicators,* details of which (SITC Revision 2) are shown in Table A3 below.

Table A3. **Classification of industries based on technology**

	ISIC Revision 2
High-technology industries	
Aircraft	3845
Office and computing equipment	3825
Drugs and medicines	3522
Radio, TV and communication equipment	3832
Medium-high-technology industries	
Professional goods	385
Motor vehicles	3843
Electrical machines excluding communication equipment	383 – 3832
Chemicals excluding drugs	351 + 352 – 3522
Other transport	3842 + 3844 + 3849
Non-electrical machinery	382 – 3825
Medium-low-technology industries	
Rubber and plastic products	355 + 356
Shipbuilding and repairing	3841
Other manufacturing	39
Non-ferrous metals	372
Non-metallic mineral products	36
Metal products	381
Petroleum refineries and products	353 + 354
Ferrous metals	371
Low-technology industries	
Paper, products and printing	34
Textiles, apparel and leather	32
Food, beverages and tobacco	31
Wood products and furniture	33

Source: OECD Science, Technology and Industry, Scoreboard of Indicators.

Annex IV

Estimating potential output

The OECD Secretariat estimates potential output for Canada (as for all Member countries) using a structural, production-function approach.[4] This involves a number steps. First, a two-factor Cobb-Douglas production function is estimated for the business sector for given sample average labour shares. The estimated residuals from this equation are then smoothed to give measures of trend total factor productivity (TFP). Second, potential output for the business sector is then calculated by combining this measure of TFP with the capital stock and estimates of ''potential'' employment, using the same estimated production function. The labour input is defined by the NAWRU (non-accelerating wage rate of unemployment); which in effect amounts to adjusting the actual labour input used in the estimated production function for the gap between actual unemployment and the estimated NAWRU level (see below). Third, potential output for the whole economy is then obtained by adding actual value-added in the government sector to the constructed business-sector potential (where actual value-added in the government sector is taken to be equal to potential output in that sector).

As a starting point, the identification of the NAWRU is derived using the method described by Elmeskov (1993). This method essentially assumes that the change in wage inflation is proportional to the gap between actual unemployment and the NAWRU. Assuming also that the NAWRU changes only gradually over time, successive observations on the changes in inflation and actual unemployment can then be used to calculate a time series corresponding to the implicit value of the NAWRU.[5]

The OECD Secretariat currently projects potential output growth for Canada to average around 2¾ per cent over the next several years based on the following assumptions: a pickup in total factor productivity growth to 0.2 per cent (from an average of –0.4 per cent over the first half of the 1990s); a decline in the structural unemployment rate from around 8½ per cent to below 8 per cent; growth in the capital stock of about 5¾ per cent, leading to a slight rise in the capital-output ratio; and growth in labour inputs of around ¾ per cent. Both the Department of Finance and the Bank of Canada currently estimate potential output growth of around 2½ to 3 per cent.

References

Elmeskov, Jorgen, ‘‘High and Persistent Unemployment: Assessment of the Problem and its Causes’’, *OECD Economics Department Working Paper*, No. 132, 1993.

Annex V

Some illustrative scenarios for economic growth

Better productivity performance and increases in the supply of capital, and in particular, labour inputs are important for maintaining Canada's high level of real income per capita, relative to the OECD average (based on purchasing power parities.[6]) As noted in Chapter II (Figure 9), Canada currently enjoys a relative income advantage of around 10 per cent compared to that average, but is significantly below the US level (by around 30 percentage points).

Under the ''status quo'', meaning no change in economic performance, Canada's per capita income is projected to decline slightly against the OECD average over the next two decades.[7] The scenario is largely constructed in two parts: a medium-term profile (1998-2003) which runs off the short-term outlook, and a longer-term profile (2003-2020) in which certain assumptions are employed. Over the medium-term, the OECD Secretariat projects potential output growth for Canada to average around 2¾ per cent based on (Table A4): a pickup in total factor productivity growth to 0.2 per cent (from an average of –0.4 per cent over the first half of the 1990s); a decline in the structural unemployment rate from around 8½ per cent to below 8 per cent; growth in the capital stock of about 5¾ per cent, leading to a slight rise in the capital-output ratio; and growth in labour inputs of around ¾ per cent. Over the longer term, potential output growth is driven solely by productivity improvement (which is assumed to reach recent US rates of 0.6 per cent per annum) and changes in the supply of labour (including a decline in the natural rate of unemployment to 7 per cent), since the capital stock is allowed to rise in line with potential output growth under the assumption of a constant cost of capital. Demographic projections are provided by Statistics Canada and are based on a medium fertility and mortality rate and net immigration of 250 000 per annum.

In the high performance scenario, total factor productivity growth of 1 per cent per annum is assumed; the structural rate of unemployment is lowered to 6 per cent; and there is an increase in the working age population both through increased labour force participation and, in particular immigration. On the other hand, the low performance scenario supposes total factor productivity growth continues on a path of around 0.4 per cent per annum; that the structural unemployment rate falls only slowly toward 7½ per cent; and that the working-age population declines, for the most part, through immigration.

Table A4. **Economic growth scenarios**

Average annual growth, per cent

	1998-2003	2003-2010	2010-2020
Canada[1]			
Real GDP, status quo	2.8	2.1	1.4
Real GDP, high performance	3.2	2.9	2.1
Real GDP, low performance	2.9	1.6	0.7
Population (status quo)	1.4	1.1	0.5
United States[2]			
Real GDP	2.4	2.4	2.1
Population	0.7	0.7	0.7
Japan[2]			
Real GDP	1.9	2.5	1.8
Population	0.2	0.0	–0.3
Total OECD[2]			
Real GDP	2.6	2.8	2.3
Population	0.5	0.4	0.4
	1996-2001	2001-2011	2011-2021
Memorandum item:			
Bank of Canada, long-term projections of potential output[3]			
Canada			
High-growth scenario	3.7	3.7	3.0
Medium-growth scenario	3.0	2.4	1.7
Low-growth scenario	1.4	1.5	0.8

1. The ''status quo'' assumes that over the long term the capital stock grows at the same pace as potential output (given unchanged cost of capital), so that growth is driven solely by productivity and labour input. In this case, total factor productivity growth rises to around 0.6 per cent (the same rate as the US over the 1990s); the natural rate of unemployment falls to 7 per cent, and population growth is based on a medium fertility and mortality rate and net immigration of 250 000 per annum. The high performance scenario implies total factor productivity growth of 1 per cent; a lowering of the structural unemployment rate to 6 per cent and an increase in the working age population both through increased labour force participation, and in particular immigration. The low performance scenario, on the other hand, assumes total factor productivity growth of around 0.4 per cent per annum; a structural rate of unemployment of 7½ per cent and a decline in the working-age population mainly through immigration.
2. Growth estimates are based on the average of ''high performance'' and ''business-as-usual'' scenarios outlined in *OECD Economic Outlook 62*, 1997 and *63*, 1998, population data come from UN, *World Population Prospects*.
3. Kuszcak, John and Richard Dion, ''Potential output growth: some long-term projections'', Bank of Canada Review, winter 1997-1998, pp. 43-60.

Source: UN, *World Population Prospects, 1950-2050* (revised), New York, 1996; Statistics Canada; OECD, *Economic Outlook 62* and *63*, and OECD Secretariat estimates.

In all three scenarios, Canadian per capita income declines relative to the OECD average. Even under the ‘‘status quo’’, the relative income advantage Canada now enjoys falls from the current level of around 10 per cent above that average, to around 15 per cent below some twenty or so years on. These scenarios highlight the importance for Canada of achieving high rates of economic growth, given its high population growth, compared to the average OECD country. In this regard, there are a number of structural policy priorities which need to be addressed, such as: improving the mobility of factors of production and of goods and service across provinces; augmenting the supply of capital, and in particular labour; gaining wider and more secure access to international markets; ensuring that the regulatory framework is able to deal with the changing nature of competition; and to roll back the relatively high tax burden faced by Canadians (see Chapter II for details).

Annex VI

The labour market in a knowledge based economy: empirical evidence

Employment, unemployment and participation rates

Canada's labour market experience over the last twenty years has been intermediate between those of the United States and European Countries. This is particularly apparent in the unemployment trends presented in Figure A3. Up to the early 1980s, the Canadian and US unemployment rates were at the same level and followed similar temporal patterns. Since then, the Canadian rate has still followed a broadly similar pattern to that in the United States but at levels more like those in Europe. Interestingly, the separation in the US and Canadian unemployment rates through much of the 1980s is not matched by a divergence in the employment rates (Figure A3, second panel). In the 1990s, however, though not falling as sharply as in Europe, Canada's employment rate has been consistently lower than that in the United States, with the differential emerging during the economic downturn early in that period and persisting since.

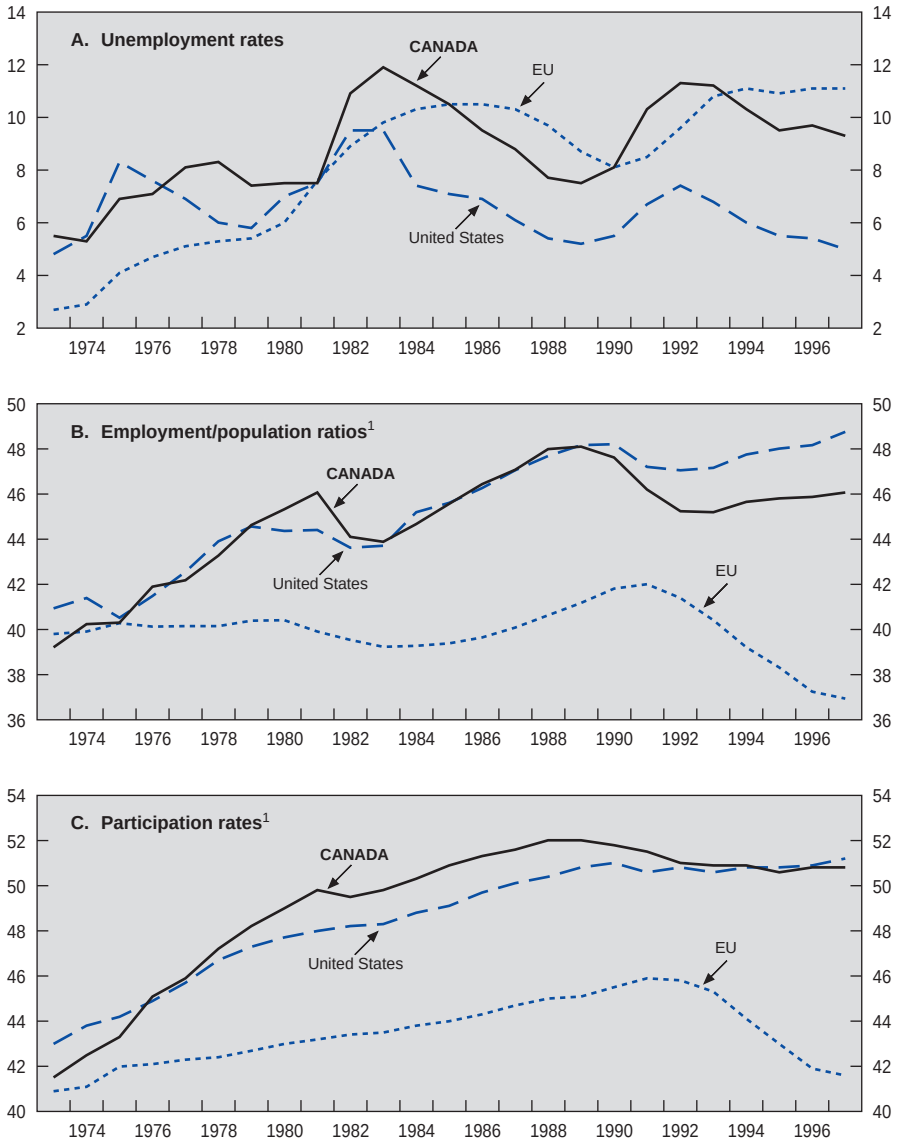
The labour market participation rate in Canada grew faster than that in either the United States or Europe up to the end of the 1980s (Figure A3, third panel), fuelled notably by increases for married women. In the 1990s, the Canadian participation rate has fallen, largely because of increased school attendance among youth in combination with a cessation of the upward trend in participation for older women. It is now at a level similar to that in the United States, like twenty years ago. Thus, in the long run, movements in the participation rate cannot help explain the deterioration in the unemployment performance relative to the United States.

Of key importance in investigating impacts of a potential shift to a ''Knowledge-based economy'' (KBE) is how these overall trends vary across age and education groups. Over the period from 1980 to 1993, the employment level of individuals with some or completed high school education in Canada fell 14 per cent while that of individuals with a university degree rose 77 per cent (Riddell, 1995). This reflects supply effects, as older less educated generations are replaced with younger and more educated ones in the labour force, as well as potential demand shifts favouring the more educated.

Figures A4 and A5 show employment to population rates[8] for different age groupings of high school and university educated individuals. They display trends by cohorts of labour market entrants rather than age groups. Cohorts here are defined as groups of

Figure A3. **LABOUR MARKET TRENDS BY INTERNATIONAL COMPARISON**

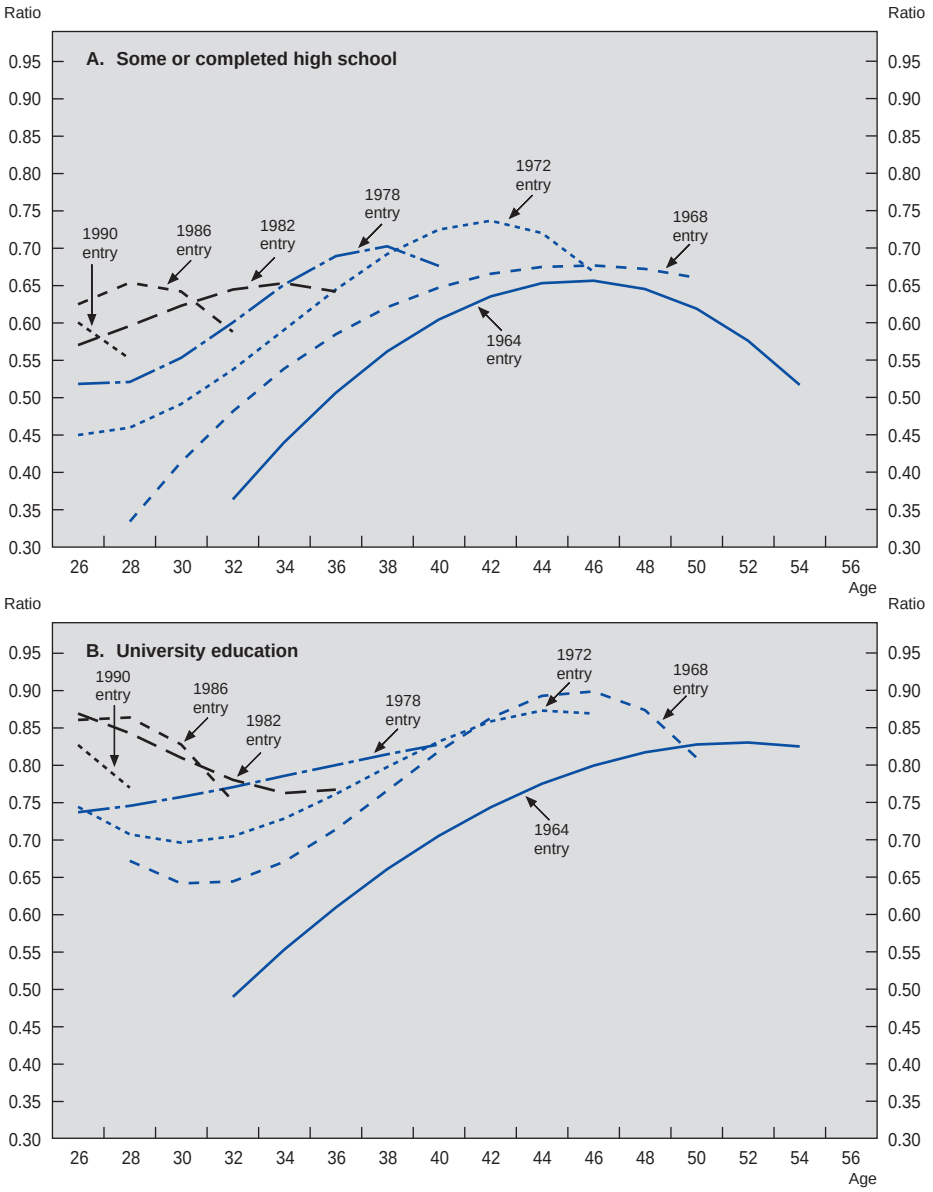
Per cent



1. Per cent of total population.

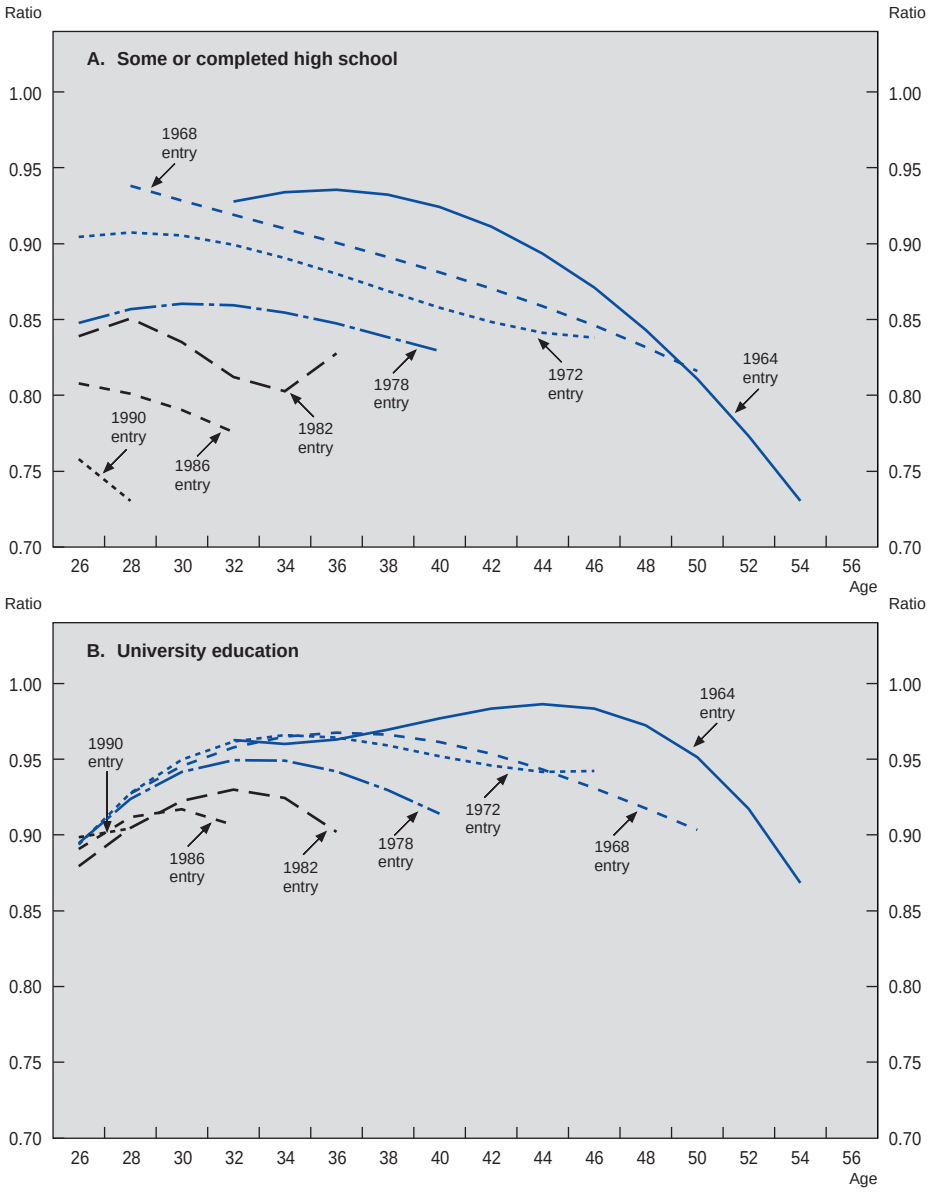
Source: Statistics Canada; OECD, *Labour Force Statistics.*

Figure A4. **SMOOTHED[1] EMPLOYMENT/POPULATION RATES BY COHORT: FEMALES**



1. Cycle effects removed.
Source: Beaudry and Green (1997*a*).

Figure A5. **SMOOTHED[1] EMPLOYMENT/POPULATION RATES BY COHORT: MALES**



1. Cycle effects removed.
Source: Beaudry and Green (1997*a*).

individuals who turn 25 or 26 in an even numbered year. The cohort line[9] then indicates the progress of the employment rate of this group of individuals as they age. Comparing employment rates at a given age across successive cohorts shows that those for females in both education groups have increased for each successive group of labour market entrants up until at least the early 1980s. For cohorts entering after 1982 the results are less clear cut and appear to point out a slowing down of the cross-generation employment rate growth.

In contrast, there have been continuous and dramatic declines in the employment rate for high school educated males both for given cohorts over time and across successive cohorts. Recent cohorts of university educated males have also experienced declining employment rates relative to their forbears at the same ages. The declines for the university educated are not nearly as large as those experienced by their high school educated counterparts. Thus, there has been a large shift toward the university educated in terms of employment levels, suggesting an increase in relative demand for the more educated. However, employment rates for more educated males and females are either flat or falling, indicating that the demand increases have been offset by increased supplies of university educated females and more than offset by increased supplies of university educated males.

The main conclusions from this evidence are ones which will be seen repeated in other contexts. *First,* the group suffering the worst trends, both in terms of levels and rates of employment, are men with high school or less education. Since this group made up over 45 per cent of males in the labour force in 1996, this has substantial impacts. *Second,* there is a clear advantage to having a university education but, at least for males, that does not mean that all trends are positive for the university educated. *Third,* there are substantial improvements for females of all age and education groups, reflecting the massive move into the labour force on the part of females over time. Indeed, the employment rate has grown faster for high school educated women, leading to a decline in the educational employment differential for women in the last twenty years. These trends seem to have slowed in the 1990s, however, and it is possible that women will experience the impacts of structural change in the labour market in a manner more similar to men in the ensuing decades. *Finally,* while trends for women have been better than those for men, males are still ahead of females with the same education level in terms of employment rates.

Participation rate trends are also very different for men and women (Table A5). For women, the period up to 1990 is defined by rising rates for all education and age groups, with the strongest growth for those over age 25. In the 1990s, the participation rate for those over age 25 stops changing while that for females aged 15-24 falls sharply. This withdrawal from the labour force is offset by increased school attendance. Men aged 15-24 display similar participation patterns to women in the same age group. Gradual increases up to 1990 are replaced with rapid declines in participation as younger workers stay in school longer. Older groups of men show declining participation rates over time.

Table A5. **Labour force participation rates**

	Males			Females		
	Age group					
	15-24	25-34	Over 25	15-24	25-34	Over 25
1976	0.68	0.95	0.81	0.57	0.54	0.41
1977	0.69	0.95	0.81	0.58	0.56	0.42
1978	0.70	0.95	0.81	0.59	0.60	0.44
1979	0.71	0.95	0.81	0.61	0.61	0.45
1980	0.72	0.95	0.81	0.63	0.63	0.46
1981	0.73	0.95	0.81	0.63	0.66	0.48
1982	0.70	0.94	0.80	0.63	0.66	0.49
1983	0.70	0.93	0.79	0.63	0.68	0.49
1984	0.70	0.94	0.79	0.64	0.70	0.51
1985	0.71	0.94	0.79	0.65	0.71	0.52
1986	0.72	0.94	0.79	0.66	0.74	0.52
1987	0.72	0.94	0.78	0.66	0.74	0.54
1988	0.73	0.94	0.78	0.67	0.75	0.55
1989	0.73	0.94	0.78	0.67	0.76	0.56
1990	0.71	0.94	0.77	0.66	0.77	0.56
1991	0.69	0.93	0.76	0.65	0.77	0.57
1992	0.67	0.92	0.75	0.63	0.76	0.56
1993	0.66	0.92	0.75	0.61	0.76	0.57
1994	0.65	0.91	0.75	0.60	0.76	0.56
1995	0.64	0.91	0.74	0.60	0.76	0.56
1996	0.62	0.91	0.74	0.58	0.77	0.57

Source: Statistics Canada, *Labour Force Survey.*

This is partly due to earlier retirement but the fact that participation rates for men aged 25 to 34 declined markedly in the 1990s indicates that this does not provide a complete explanation.

Movements in participation and employment for different education groups have resulted in relatively little change in the ratio of unemployment rates of the high school to the university educated, which fell from 2.77 to 2.44 from 1980 to 1993 (Riddell, 1995). This overall stability masks, however, quite different movements for education/age/gender subgroups. The ratio for males aged 31 to 40 rose from 1.65 to 2.5 between 1971 and 1991 while that for females of the same age group decreased from 2.7 to 1.25. In contrast, the unemployment rate ratio for different education groups aged 41 to 50 fell for males and increased for females over the same period (Riddell and Sweetman, 1997). Thus, it is difficult to find a pattern of increasing education differentials in employment related outcomes for all gender and age groups.

Employment outcomes in other economies also lead to somewhat mixed conclusions (OECD 1995*a* and 1997, Nickell, 1997). While there is a strong inverse correlation

between levels of education and unemployment rates, developments of education-related differentials over time show no clear trend. as in Canada, there is evidence of a general decline in employment outcomes for less educated males but even this trend is not universal. In the United States and the United Kingdom, for instance, the ratio of male unemployment rates of the low educated to the high educated declined in the 1990s following a substantial increase during the two preceding decades.

Sectoral employment patterns

While the overall employment and labour force patterns described in the previous section are useful in framing the discussion of a shift toward a KBE, additional information on sectoral differences is needed to evaluate the existence and nature of such a shift. Gera and Masse (1996) examine patterns in employment to see if there is a shift toward ''innovative industries'' which might be seen as at the centre of the second KBE definition discussed above. Further, they are interested in whether any such shift has been accelerating. One key finding from their study is that the pace of structural change across industries, however measured, has not increased over time. The main source of employment growth has been and continues to be the service sector, with employment gains observed in industries using both high and low skilled workers.[10]

The study attempts to assess the importance of shifts toward a KBE by investigating shifts in employment across high, medium and low knowledge industries. Its definition of the knowledge level of an industry is based on a combination of factors relating to R&D expenditures and associated employment and to the extent of highly educated workers generally and scientific and technical workers more specifically. Defined this way, the high-knowledge, ''innovative'' industries accounted for 10.7 per cent of employment in 1971, 13 per cent in 1981, and 15.4 per cent in 1991. Thus, they represent a relatively small share of the total and, while the sector is growing in importance, it is doing so only gradually.

Table A6 shows that more rapid employment growth has actually occurred in industries with a variety of knowledge intensity levels. Indeed, if one excludes education, health and social services on the grounds that they are public sector industries where employment may be driven by policy choice rather than exogenous technical change, only two of the top ten (number one and number ten) are high knowledge intensity industries. The office, store and business machines category, often pointed to as a key industry in the IT sector, does not make the top ten in growth rates, while industries classified as low knowledge intensity ones fill spots three through five. Moreover, while employment growth in industries which could be categorised as ''science-based'' has been strong, these are industries that were small in size at beginning of the observation period. thus, while employment is increasingly related to knowledge activities, the latter are far from dominating labour market trends. Overall, the sectoral evidence appears to fit

Table A6. **Annual average employment growth by level of knowledge intensity**
1971-1991

Industry	Knowledge intensity	Employment Growth rate (per cent)
Business services	High	5.90
Educational services	High	5.43
Health and social services	High	5.41
Personal services	Low	5.28
Recreation services	Low	4.98
Accomodation and food services	Low	4.59
Plastic products	Medium	4.55
Fishing and Trapping	Low	3.95
Mineral fuels	Medium	3.86
Services incidental to mining	Medium	3.83
Finance, real estate and insurance	Medium	3.83
Pipeline transport	High	3.71
Wholesale trade	Medium	3.08
Office, store and business machines	High	3.00
Aircraft and Aircraft parts	High	2.96
Other services	Low	2.90
Retail trade	Low	2.39
Electric power systems	High	2.15
Pharmaceutical and medecine products	High	2.08
Printing and publishing	Medium	2.06

Source: Gera and Masse (1996).

best with the general KBE shift model discussed in Chapter III: one in which any such shift has been continuing for a long time and is not accelerating.

Another finding of the above-mentioned study is that employment in high knowledge intensity industries is less cyclically sensitive than that in other sectors. Further, Canada appears to lag other developed countries – and the United States and Japan in particular – in its share of high technology manufacturing industries. As shown in Table 7 in Chapter II, both in terms of value added and employment, such industries account for a much smaller share of the manufacturing sector than on average in the OECD area. Together these observations may imply that an expansion of this sector is desirable and that there is room for such expansion relative to other similar economies. It has been argued, however, that given the presence of US based companies in Canada, the extent of employment in high-technology industries does not provide a good measure of the extent of technology usage in the Canadian economy, which can draw on US expertise.

Employment instability measures

Full year/full time work

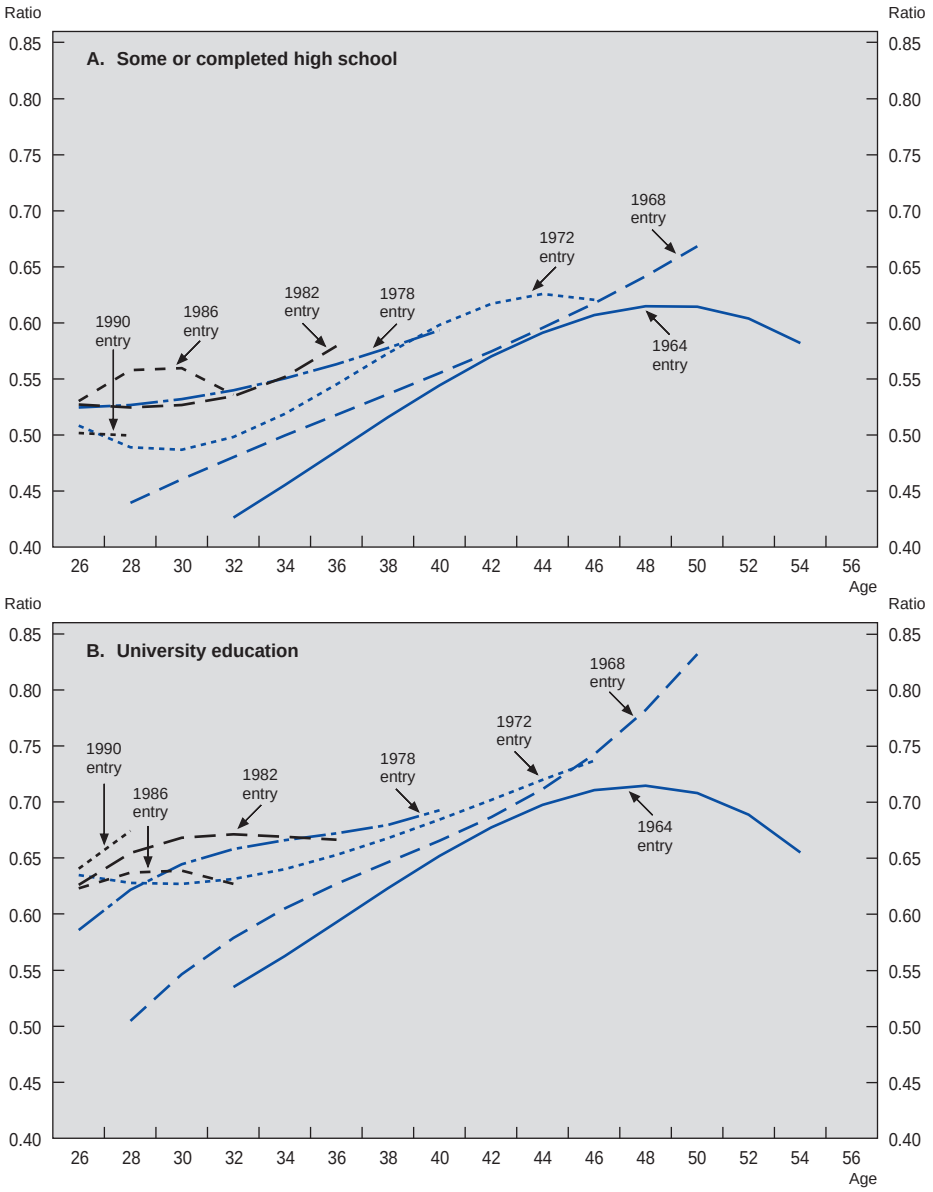
As discussed in Chapter III, the Accelerating KBE Shift model suggests an increase in employment instability. In particular, it implies a bifurcation in the labour force:

workers who know how to create and implement new technologies will obtain the ''core'' jobs of longer duration and longer hours as firms try to both hold onto this vital part of their operations and to get the most out of them; in contrast, workers in the remainder of the labour market will face shorter job durations and more part time work. Work patterns other than full year/full time jobs, often termed ''non-standard'' work, can arise through several routes (*e.g.* part time but full year jobs, full time but part year jobs, multiple job holding, etc.). Because there is typically substantial overlap among different types of non-standard work, it is preferable to characterise it by examining patterns in the opposite type of work: full-year/full-time work.

Figures A6 and A7 present the proportion of individuals who are recorded as full-year/full-time workers.[11] As in the analysis of employment rates, separate plots are provided by education group and gender, and the data are organised to follow cohorts through time. The results for females of both education groups show substantial increases in the full-year/full-time rate for cohorts entering the labour market before the mid-1970s. Thereafter, the cohort profiles are quite similar and generally flat. As with employment rates, the results actually indicate a decline in the university/high school differential over time, with high school educated females having witnessed a more substantial increase in full-year/full-time rates. For high school educated males, there are very large declines across cohorts observed at a given age. At age 32, for example, the 1986 cohort has a full-year/full-time rate of 0.75 compared to 0.90 for the 1968 cohort at the same age. However, unlike the employment rates, the profiles for given cohorts do not tend to slope downward at the end. Thus, while declining access to employment seems to affect high school educated males of all cohorts, access to full-year/full-time jobs among the employed is declining only for the new cohorts. Once a cohort obtains a certain level of access to these jobs, it appears to hold on to that access. This fits with a situation in which older workers are able to use seniority to keep a disproportionate number of stable jobs. In contrast, there is no obvious difference in full-year/full-time patterns across cohorts of university educated males. Thus, recent cohorts of university educated males may have part year or part time job rates higher than one might expect, but this is no different from cohorts entering the labour market in the 1960s and 1970s. University educated males, as a group, appear to wait until their early 30s to fully settle into stable work patterns. Overall, there is no evidence of increased instability in employment across the labour market according to this measure but one group, high school educated males, again fares poorly. As noted, an accelerating KBE shift should be expected to affect almost all workers, as the information revolution permits less skilled workers to be replaced and more skilled workers to work more irregular patterns.

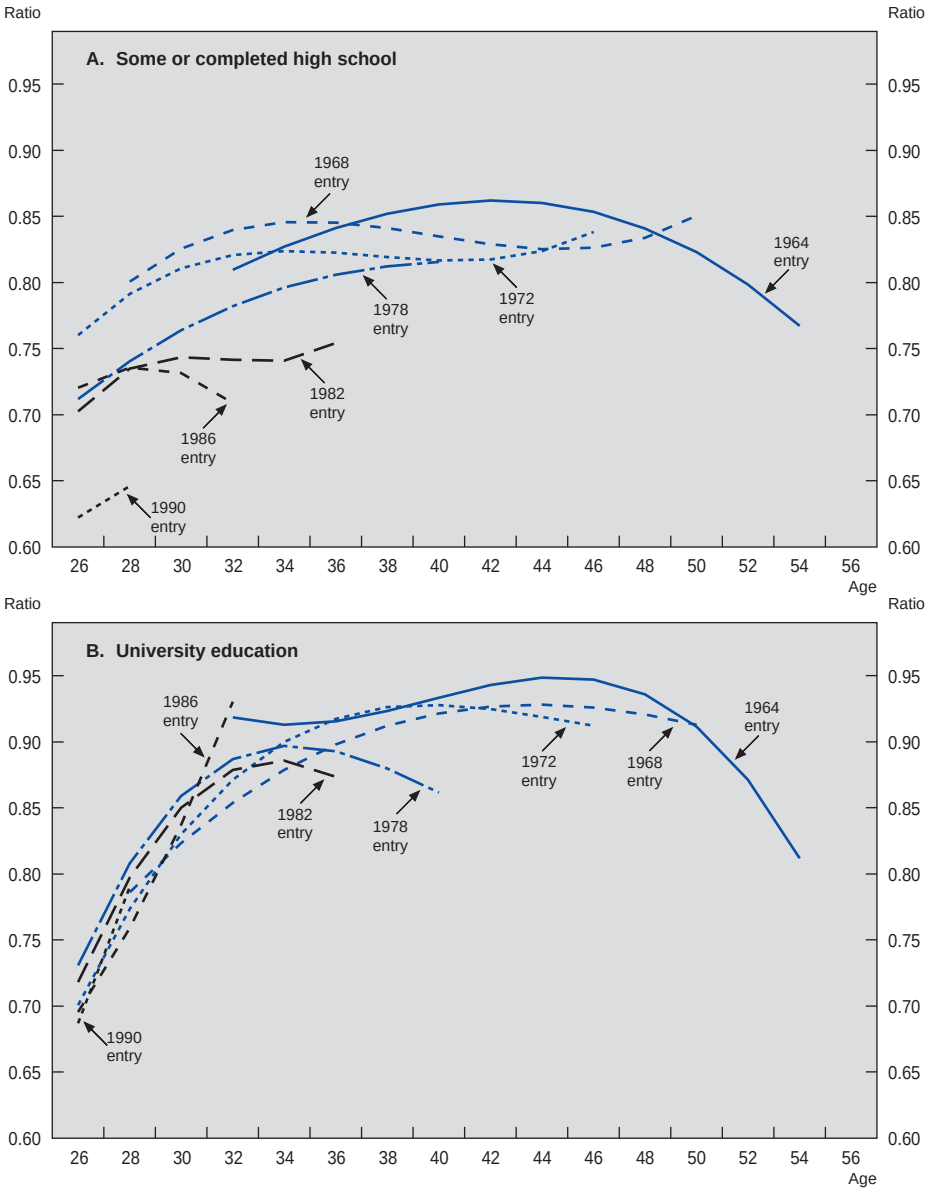
The result of overall stability for all groups other than high school educated males seems to contrast with a recorded increase in part-time work from 12½ per cent to 19 per cent of workers from the mid-1970s to the mid-1990s. In fact this rise is mainly due to increases in part time work for those aged 15 to 24. The focus of the above analysis is on individuals over age 25, and for those part time rates have remained relatively stable at

Figure A6. **SMOOTHED[1] FULL YEAR/FULL TIME RATES BY COHORT: FEMALES**



1. Cycle effects removed.
Source: Beaudry and Green (1997*a*).

Figure A7. **SMOOTHED[1] FULL YEAR/FULL TIME RATES BY COHORT: MALES**



1. Cycle effects removed.
Source: Beaudry and Green (1997*a*).

about 3 per cent for males and 20 per cent for females over the last twenty years (Krahn, 1995). The increase in part-time work for youth may be a natural accompaniment to the surge in school enrolment for this age group in this same period. While involuntary part-time work has increased in the 1990s, the proportion of those aged 25 to 34 who list themselves as involuntarily working part-time and actually search for a job with longer hours is still only in the order of 2 per cent of all workers, indicating that this form of instability affects only a small part of the workforce. Further, a measure of non-standard work combining part-time, short tenure and independent contract work (Betcherman *et al.*, 1994) rose by only 1 per cent between 1987 and 1993 (to 29 per cent of employment), an increase that can probably be accounted for by the cyclical downturn in the labour market. Thus, the level of non-standard employment may be relatively large in Canada, but claims of an accelerating upward trend are difficult to support.

Job tenure

The patterns in full-year/full-time work just discussed may mask increasing employment instability to the extent that workers need to combine more part-year jobs to achieve a full year of work than in the past. If this is true, one should expect to see a decline in job durations. In fact, average job tenure in Canada has displayed an upward trend over the last twenty years. However, movements in this summary statistic hide more complex patterns among sub-groups in the workforce. In particular, after controlling for cyclical factors, Green and Riddel (1997) find that young men, less educated men and young, lower educated women all experienced shortened job tenures over the last two decades. Offsetting these declines are increases in job tenure for older women and for some groups of prime age, higher educated males.

In part, these trends could be seen as reflecting increasing instability for workers with non-core jobs associated with an accelerating KBE shift. Indeed, the young and less educated workers who have experienced decreasing average job tenures may be the very group one would most expect to hold those ‘‘peripheral’’ jobs. The fact that some older, less educated workers have not had declining average job tenures may reflect a labour market in which union seniority rules protect older workers who would otherwise also face increased instability. This fits with results showing that declines in full-year/full-time rates for less educated males occur entirely across cohorts rather than within cohorts. However, an accelerating shift towards a KBE also implies increased instability for more educated workers, though to a lesser extent. This does not appear to have occurred.

In spite of the limited direct evidence of increased instability for most workers, survey responses indicate a widespread perception of insecurity. Thus, the percentage of Canadian workers indicating that they were satisfied with their job security fell from 60 per cent in 1992 to 56 per cent in 1996. A similar trend in perceptions is evident in other OECD countries, where, on average, this measure fell from 61 per cent to 56 per cent over this period. Across countries, however, trends in job tenure are mixed. For Australia, the United Kingdom, the United States, France and Germany average job

duration has either been flat or slightly increasing since 1979. At the same time, Australia, France and the United Kingdom have experienced increases in the percentage of workers with less than one year of job tenure while Canada, the United States and Germany all have witnessed a fall in this percentage. It is worth noting that job tenures could increase because workers are afraid to quit bad jobs in a deteriorating labour market. Thus, it is not impossible that increased job tenures could signal increased perceived uncertainty on the part of workers. More work is needed on exactly why average job tenures have been changing when they have.

A related measure of job instability is the percentage of jobs with fixed end dates. These are jobs that are either seasonal in nature or might be seen as part of any trend toward contracting out of services by firms. There has been only a modest change in the proportion of jobs of this kind, rising from 8 per cent of all jobs in 1989 to 9 per cent in 1994 (Krahn, 1995). As Grenon and Chun (1997) indicate, the industrial pattern of changes in this proportion may fit more with the small increase being due to the cyclical downturn in the early 1990s than with a structural change. As with job tenure, aggregate movements hide quite different patterns for demographic subgroups. Thus, men and women under age 24 have seen the largest increase in the proportion with fixed end date jobs, which may fit with younger workers having trouble getting access to more stable employment or with an increase in summer jobs as a much larger proportion of young people stay in school. All age groups of men under age 55 experienced increases in the proportion with jobs with fixed end dates while some age groups of women experienced declines in this proportion. Thus, as with job tenure, it is not possible to state that there has been an across the board increase in job instability by this measure. Moreover, if anything, there are distinctions by gender rather than by skill level, with men experiencing more instability of this type and women less.

Self-employment rate

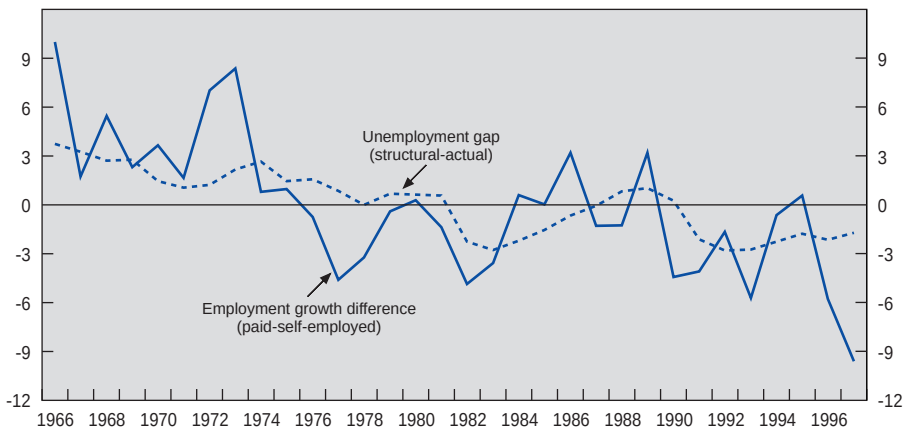
Another potential manifestation of a switch toward a KBE is an increase in the proportion of workers who are self-employed. Contracting out by firms and increased requirements in terms of worker flexibility might be expected to lead to more workers being self-employed. In the last 25 years or so, the overall self-employment rate has risen with the trend containing three main components: an increase from 12½ per cent to 14½ per cent in the early 1980s; little change in the remainder of the decade; and a period of strong growth in the 1990s, with the proportion of self-employed approaching 18 per cent in 1997. It is not immediately obvious whether the recent surge in self-employment is a feature of the prolonged downturn in the labour market since 1990 or part of a longer term trend that will persist past any revival. Approximately 90 per cent of the growth in self-employment in the 1990s was in the form of ''own account'' entrepreneurs who worked alone. This contrasts sharply with the 1980s when 66 per cent of new self-employed individuals set up businesses that employed others as well.

The fact that many own account workers list themselves as self-employed because they have no other job opportunities[12] suggests that a substantial part of the rise in the 1990s may be a form of ''hidden'' unemployment. If this is true, one would expect at least a partial reversal of the upward trend in self-employment as the labour market recovers and the surge in the 1990s should not be taken as evidence of a fundamental shift in the nature of work. Two pieces of evidence support this claim. *First*, as shown in Figure A8, over the last three decades, the difference between paid and self-employment growth has been associated with the unemployment gap (that is, the difference between the observed and structural unemployment rate). *Second*, more recently, increases in overall employment growth have indeed been accompanied by declining levels of self-employment.

To some extent, structural factors – such as shifts in the composition of demand by industry and in the age structure of the population – seem to have contributed to the rise in the share of self-employment in the 1990s. On the other hand, the trend toward an increased level of education in the workforce has had no significant impact. Available evidence suggests greater incidence of self-employment among workers with both low and high education.

Figure A8. **SELF EMPLOYMENT AND LABOUR MARKET SLACK**

Percentage points



Source: Statistics Canada; OECD.

There is no consistent pattern in the growth of self-employment across OECD countries. The self-employment rate in the United States has been flat at between 8 per cent and 8½ per cent for the last twenty years. The rate has declined steadily in France (from 18 per cent in 1976 to 11½ per cent), and it has also fallen over time in Germany. The United Kingdom experienced an increase in self-employment in the 1980s but the rate has stayed constant at about 13 per cent in the 1990s. Thus, although data are not strictly comparable,[13] there is no evidence of a world wide shift toward greater self-employment in response to trade or technology trends.

Earnings inequality

A key piece of evidence in the claim that Canada is facing a shift in technology toward a KBE is the rise in male earnings inequality over the last twenty years (Morissette *et al.*, 1993). Between 1981 and 1992, the male real weekly earnings distribution in Canada showed a 25 per cent decline for the 10th percentile while the 90th percentile maintained its real income (Richardson, 1997). A prominent explanation for these trends is that Canada is facing a skill biased demand shift, leading to higher wages for skilled relative to unskilled workers and thus an increase in inequality.

In fact, there has been virtually no change in the average wage differential between university and high school educated males since the early 1980s, but it has been argued that this is because demand shifts favouring university educated workers have been offset by large increases in the supply of this category (Freeman and Needels, 1993). There is, however, direct evidence of an increased experience differential in Canada: between 1981 and 1993, the difference between the average earnings of 45-54 year old males and 25-34 year old males increased by 18 per cent in real terms. Moreover, there has been a rise in relative inequality within education/experience groups (Beaudry and Green, 1997). While it thus appears that a skill biased demand shift is occurring in Canada, there are several problems with this hypothesis.

First, men and women under age 30 do experience an increase in the earnings difference between the university and high school educated over the 1980s. For older age groups this difference is either unchanging or declining (Bar-Or *et al.*, 1995). Yet, the large increase in the supply of university educated workers occurs among younger individuals. Thus, one would expect the educational differential to grow at a slower rate for the young than the old – the opposite of the pattern actually observed.

Second, the evidence on the increase in returns to experience also needs to be qualified. A given difference between the average wages of, say, 25 and 45 year olds in a cross-sectional dataset could represent the returns to experience in the economy, in which case those aged 25 could expect to earn the observed real average wages of the 45 year olds in 20 years time. Alternatively, they could in part reflect differences in the labour market outcomes of successive generations, often called ‘‘cohort effects’’, in which case the 25 year old would not expect to earn the average wage of the 45 year olds in 20 years.

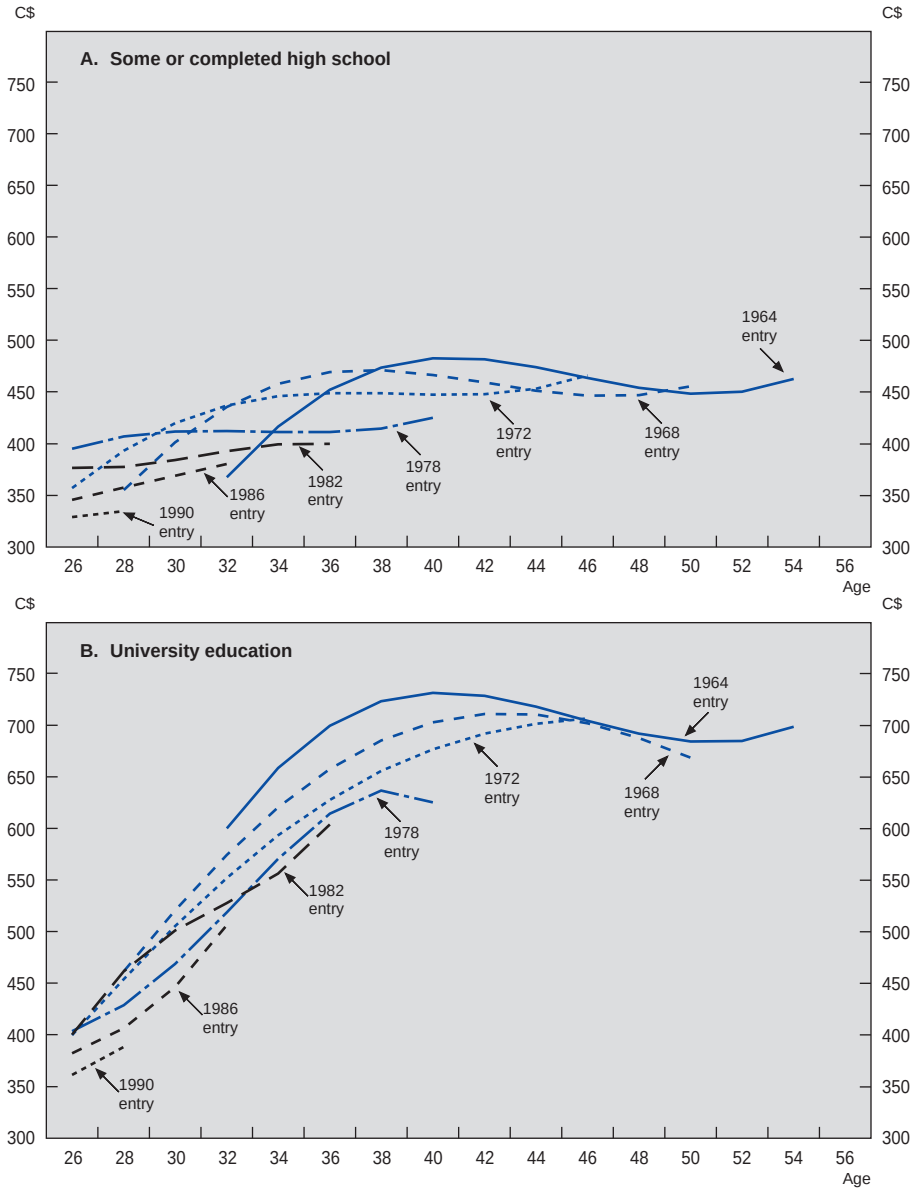
Similarly, an increase in the experience differential across successive cross-sections might represent an increase in the returns to experience, but it could also reflect declining earnings outcomes for successive cohorts of labour market entrants. Figures A9 and A10 depict the average earnings of cohorts of labour market entrants using a sequence of Surveys of Consumer Finance.[14] If returns to experience have been increasing over time, the age-earnings profile should be becoming steeper for successive cohorts. That is, one should observe that the earnings gain associated with moving from, say, age 25 to age 30 should be greater for more recent cohorts than it was for earlier ones.

For high school educated males, the analysis shows that for successive cohorts entering the labour market before 1978 average earnings observed at a given age rose. However, cohorts entering the labour market in the 1980s and 1990s experienced progressively lower wages at a given age. Thus, the 1992 entry cohort earned approximately 15 per cent less than its 1978 counterpart at age 26 in real terms. This adds to the evidence cumulated so far that high school educated males are facing worsening labour market conditions. In terms of earnings, though, substantial declines also occurred for university educated males. The deterioration across successive cohorts of university males has been continuous from those entering the labour market in the 1960s through to those entering in the 1990s: the 1986 entry cohort earned an average real weekly wage at age 32 that was 20 per cent less than that earned by the 1964 entry cohort at the same age. Neither educational group showed an increase in the slope of the age-earnings profiles for successive cohorts. For females as well, who have not experienced the pattern of declining earnings across cohorts observed for men, there is no evidence of increased slopes on cohort specific age-earnings profiles. By contrast, a relative shift in demand favouring more skilled workers of all types should have led to an increase in the rate at which an individual's earnings increase with age.

Third, regarding the above-mentioned evidence of an increase in inequality among individuals with the same experience and education levels, Beaudry and Green (1997*b*) point out that what appears to have happened in the last twenty years is that the entire distribution of real weekly wages for high school educated males has declined by a constant amount. That is, inequality has increased in relative but not absolute terms. There is no evidence that, among a group with the same education and experience, individuals at the top of the earnings distribution lost less or gained more than those at the bottom.

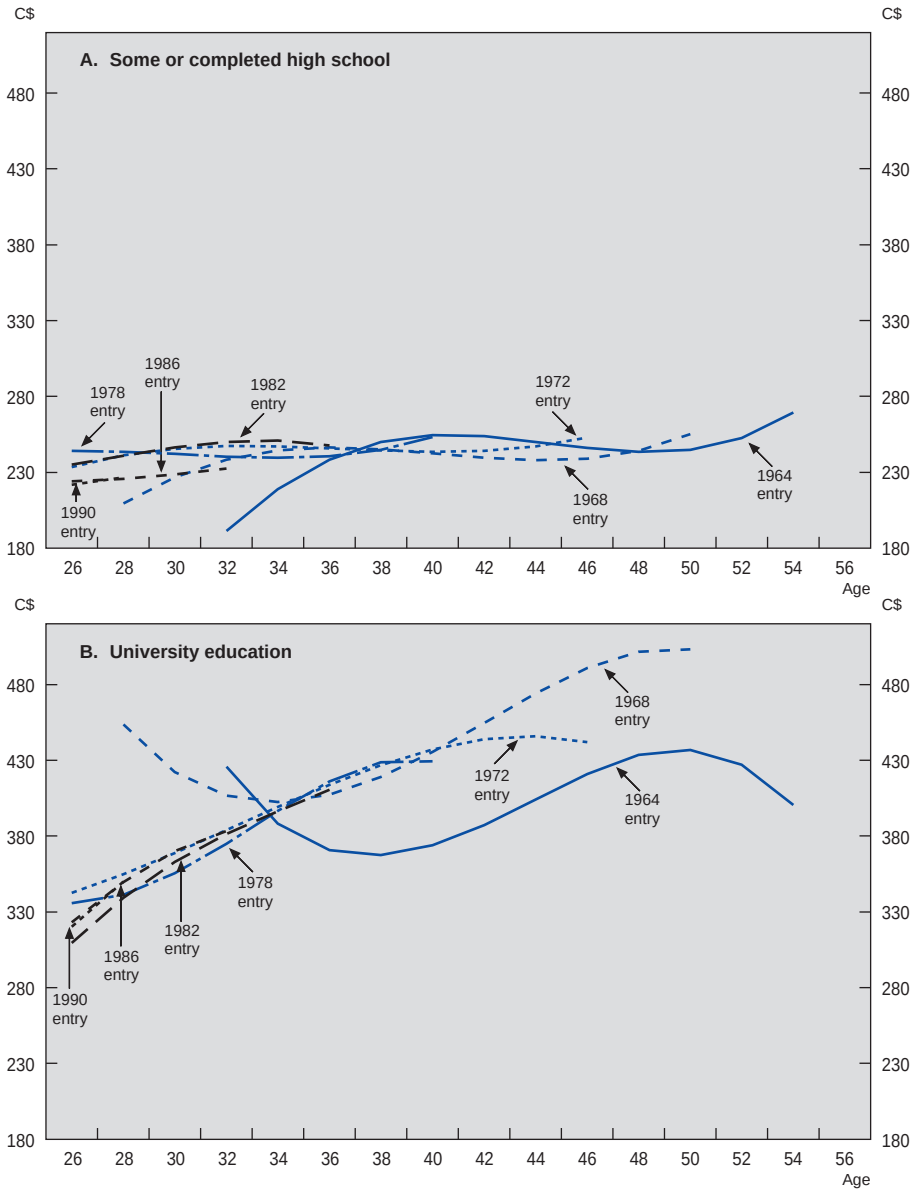
Perhaps more important is the fact that females do not exhibit any increase in earnings inequality over time. Thus, for a substantial portion of the workforce a main trend often quoted as an indication of a skill biased demand shift in the labour market is not present. Women also do not exhibit the pattern of declining average weekly earnings across cohorts observed for men. To some extent this is due to increases in hours worked per week since full year/full time working women exhibit similar, though smaller, declines across generations to men. Although differences by gender are still large and in

Figure A9. **REAL WEEKLY EARNINGS BY COHORT: MALES**[1]



1. Cycle effects removed.
Source: Beaudry and Green (1997*a*).

Figure A10. **REAL WEEKLY EARNINGS BY COHORT: FEMALES**[1]



1. Cycle effects removed.
Source: Beaudry and Green (1997*b*).

favour of the males, divergent trends over time between men and women are a key element of ongoing changes in the Canadian labour market.

A theory explaining changes in the Canadian wage structure must reconcile several key facts: declines in real earnings across successive cohorts of males, regardless of their education level, beginning around the late 1970s for less educated males but much earlier for the university educated; increased education differentials for workers under age 30 but not for older ones since 1980; no change in the level of inequality among workers with a common amount of experience and education since 1980; and no change across cohorts for women, regardless of their education level. These patterns do not easily fit with any version of a KBE shift model operating on its own, even if account is taken of offsetting supply responses to an exogenous demand shift. This leaves the door open for numerous possibilities. One is the declining access to union jobs among young people. Morrissette (1997) shows that the unionisation rate of those under age 25 has declined substantially. If this has occurred for males but not females then this may help in understanding the trends just discussed. In fact DiNardo, Fortin and Lemieux (1997) find that declining unionisation and minimum wages can explain a substantial component of increasing wage inequality in the United States. Kapsalis (1998) draws attention to the fact that young people have now to compete with older workers who not only have more experience but also better education than in the past.

Whether these trends are driving forces in themselves or mitigating factors reflecting larger shifts is not clear to this point. Alternatively, the large increase in supply of female labour could play a substantial role, particularly if changing attitudes toward the relative value of female labour (perhaps induced in part by equity policies) is involved. Thus, the increased female labour supply could lead to declining employment levels and wages for males for a given level of demand. This trend may not be evident for females because employers are willing (or forced by policy) to shift toward the cheaper female labour and pay equity policies prevent them from cutting female wages as much. However, what really is driving the wage structure patterns is not at all clear. If skill biased demand shifts are not the primary driving force, then other models in which institutional change and supply shifts are primary rather than mitigating forces become possible. More research needs to be done before a definitive alternative can be advanced.

The United States have experienced increasing male earnings inequality similar to Canada's, with the 10th percentile of the real earnings distribution falling by 9 per cent between 1981 and 1992 and the 90th percentile increasing by 6 per cent over this period. Juhn, Murphy and Pierce (1993) have explained this by an increase in returns to skills of all types which can be traced back at least to the early 1960s. MacCurdy and Mroz (1995) rearrange the US data into a cohort format and find that high school educated males have faced declining real median earnings across successive cohorts while university educated males have undergone little change in this regard. Also, they note that the spread of earnings within education-experience groups does not increase over time when it is measured in levels rather than relative terms. Both of these results are similar to those just

described in that they do not support claims of increasing returns to experience or unobserved skills over time. The main difference between US and Canadian male earnings patterns in the 1980s is that average earnings declined across successive generations of university males in Canada but not in the United States. Thus, in contrast to Canada, increased male earnings inequality has been associated with increased educational earnings differentials in the United States.

The pattern of earnings inequality and wage structure in other OECD countries is mixed (OECD, 1995*a*, and Gottschalk and Smeeding, 1997). Only the United Kingdom has experienced increases in inequality of the same magnitude and wage structure changes of the same pattern as the United States. Australia's experience appears to be quite similar to Canada's, with increases in male earnings inequality primarily due to declines at the bottom rather than increases at the top of the distribution. A range of other countries, including France, Germany, Italy, Sweden and Japan, have seen either much smaller or no increase in male earnings inequality over the 1980s. Similarly, very few OECD countries share the US experience of increases in educational and cross-sectional experience differentials along with increased earnings dispersion within skill groups. In fact, only the United Kingdom has experienced increases in educational differentials that compare with those in the United States. The cross-sectional experience differential has increased in a range of countries but, as for both Canada and the United States, this reflects differences in the labour market outcomes of successive generations. Also, one must take account of both offsetting supply effects and institutional effects in attempting to understand differences in cross-country patterns. Overall, though, the evidence seems to fit best with the kind of conclusion reached with respect to the Canadian data: a skill biased demand shift may have some role in changes in labour markets in developed economies but leaves many trends unexplained.

Annex VII

Chronology of economic events

1997

January

The Ontario proposals for restructuring provincial/municipal responsibilities were amended on 1 May 1997. Funding for education will be shared, with 50 per cent coming from property taxes and 50 per cent from provincial grants. Funding for social assistance will still be shared on an 80/20 basis between the province and the municipalities. Administrative costs, however, will be shared on a 50/50 basis. Social housing will be a municipal responsibility, but homes for special care and long-term care will be a provincial responsibility.

Employment Insurance (EI) moves to an hours-based system. To qualify for EI benefits claimants must work between 420 and 700 hours, depending on the region and 910 hours for new and re-entrants to the labour market.

February

The 1997 budget released by the federal government contained no new programme cuts and no new taxes. Other highlights include: increased spending of C$ 137 million for higher education and skills training; C$ 300 million over three years for investigating best practices related to health care, better access to medical information and a community action programme; C$ 600 million a year for child-tax benefits for low-income families; the Canada Foundation for Innovation with up-front funding of C$ 800 million; and incentives in the tax system to encourage charitable donations.

Draft legislation to reform the Canada Pension Plan was tabled.

March

The legislation for the harmonised federal and provincial sales tax in three Atlantic provinces was amended to not include the blended tax-in prices.

Quebec tabled its budget which included a 15 per cent reduction in personal income taxes for lower and middle-income earners, and a 1 percentage point rise in the sales tax to 7.5 per cent.

British Columbia released its budget which contained a 2 percentage point cut in personal income taxes announced in the 1996 Budget and continuing with the freeze on car insurance, hydro and tuition fees.

The Manitoba government announced its 1997/98 fiscal-year budget which contained no new taxes or tax increases. The government anticipates a C$ 27 million surplus – its third straight – achievable due to a C$ 100 million transfer from the province's fiscal Stabilisation Fund. (The Manitoba surplus includes a C$ 75 million instalment into the debt retirement fund.)

Saskatchewan tabled it fourth balanced budget in a row, with plans to reduce the public debt by almost C$ 4 billion over seven years while maintaining key social programmes. Retail sales taxes were cut from 9 per cent to 7 per cent.

The Newfoundland government released its budget which contained a three-year programme to eliminate 1 100 civil servant positions and balance the books by the end of the century. The budget contained no new tax increase, but many user fees were increased.

The federal government introduces changes to the EI programme, allowing for ''small weeks'' to be used in the calculation of EI benefits in 29 high unemployment regions.

April

Alberta tabled its budget which contained an expected C$ 154 million surplus following a C$ 2.3 billion surplus last year. It calls for a modest increase in spending and no new tax increases.

Nova Scotia tabled its budget with a forecast surplus of C$ 4 million over the next two years.

Prince Edward Island released its budget with a projected C$ 17 million deficit. It contained a rise in the corporate income tax rate and spending cuts. Gasoline taxes will increase by one cent a litre, license and registration fees were raised, and the personal income tax base was expanded.

May

Ontario tabled its budget which included a deficit target of C$ 6.6 billion; personal income tax cuts; increased spending for health care and job creation; C$ 500 million for a ten-year research and development fund; and tax credits for small businesses.

June

The Liberal government was re-elected for a second mandate but with a reduced majority. The Liberals retained 155 of the 301 seats of the House of Commons, down from 174 in their first term in office. The Reform Party replaced the Bloc Quebecois as the official opposition party.

Changes to Financial Institutions Legislation comes into force, including measures to strengthen consumer protection, ease the regulatory burden on financial institutions and ensure the financial institutions statutes remain current with evolving trends.

The Bank of Canada raised its bank rate ¼ percentage point to 3.5 per cent on 26 June. The increase, the first since February 1995, did not trigger any increase in administered rates such as the prime.

Quebec announced an increase in its minimum wage of 10 cents an hour to C$ 6.80 effective 1 October.

August

Ontario Hydro announced the shutdown of seven of its nuclear reactors in response to an assessment of the performance of the nuclear division. To make up for the loss in production, Hydro will increase electricity production from its fossil fuel plants and restore production at others that had been closed.

September

The federal government introduced legislation to reform the Canada Pension Plan. Under the reforms, premium rates will climb from 5.85 per cent of insurable earnings this year to 9.9 per cent by 2003 and then be capped. (Premiums are split 50-50 between employers and employees and are paid on earnings of up to a maximum this year of C$ 35 800.) The CPP fund will no longer be lent to the provinces at below-market rates and the fund will be managed professionally by a board that will be appointed after consultation with the provinces.

Five-year mortgage rates fell to a 32-year low of 6.85 per cent.

Effective 1 October, Old Age Security benefit rates will increase by 0.3 per month (over the previous three months) for the three-month period from October to December 1997. The basic OAS pension, paid to persons 65 years of age and over, will be C$ 406.34.

October

Ontario teachers began a strike on 27 October, cancelling classes for 2.1 million children. The walkout, affecting up to 126 000 teachers, occurred after talks between the government and teachers' union over changes to Bill 160 failed. Bill 160 will give the

provincial government, not local school boards, control over class size, teachers' preparation time, the length of school days, and the academic year.

The Bank of canada raised the Bank Rate and its operating band for the overnight rate by ¼ percentage point on 1 October.

November

Ontario's 126 000 striking teachers returned to work on 10 November, ending a ten-day walkout over changes to Bill 160. About 45 000 members of the Canadian Union of Postal Workers sent on strike on 20 November, shutting down mail service across the country.

The federal government announced that it will cut employment insurance premiums by C$ 1.4 billion in 1998, double the reduction that had previously been announced. As of 1 January, EI premiums paid by employees will drop to C$ 2.70 for every C$ 100 of insurable income from C$ 2.90. For employers, EI will be C$ 3.78 for every C$ 100 of payroll, down from C$ 4.06.

The Ontario government released plans for changes to Ontario Hydro by 2000. The corporation will be split into three new Crown corporations: the Ontario Electricity Generation Corporation will take the ownership of Ontario Hydro's generation assets; the Ontario Electric Services Corporation will be a holding company for other electricity business including transmission, retail and operation contracts; the independent Market Operator will be a non-profit corporation to run an electricity exchange, dispatch power based on leased cost bids and arrange financial settlements between buyers and sellers.

The Bank of Canada raised the Bank Rate ¼ percentage point to 4 per cent on 24 November, following an increase on 1 October. Commercial banks followed suit, raising the prime lending rate to 5.5 per cent from 5.25 per cent.

December

A two-week postal strike ended on 3 December, as 45 000 workers were legislated back to work. The bill also imposed a wage settlement of about 5.15 per cent over three years.

The Canadian dollar slipped below 70 cents (US) for the first time in nearly twelve years. In support of the dollar, the Bank of Canada raised its key rate by 50 basis points to 4.5 per cent. This prompted a similar hike in the prime rate to 6 per cent.

New Brunswick tabled its 1998-99 budget on 4 December, forecasting a budgetary surplus of C$ 18.5 million despite spending increases on health care and education. The province will continue with its plans to a 10 per cent reduction in the personal income tax rate by 1999.

1998

January

Severe ice storms paralysed Eastern Ontario and Western Quebec early in the month, knocking out power in most regions and creating a state of emergency. Canadian troops were deployed to assist hydro workers and the federal and provincial governments announced substantial aid packages. More than two million people were without power for varying amounts of time and business was shut down in Ottawa and Montreal for several days.

The Royal Bank of Canada and the Bank of Montreal announced a C$ 40 billion merger plan on 23 January. Combined, the two banks have 17 million customers, C$ 435 billion in assets and a stock market value of over C$ 40 billion.

The Canadian dollar slid to a record low of 68.70 cents US at month-end, the lowest value in its 140-year history, while the Bank Rate was hiked half a point to 5 per cent.

February

The federal government released its 1998-99 fiscal year budget on 24 February announcing that the budget will be balanced in 1997-98 and also in 1998-99 and 1999-2000. Highlights include a C$ 2.5 billion multi-year Canada Millennium Scholarship Foundation which is to provide 100 000 scholarships a year to low and middle-income students; as of 1 July income that can be earned tax-free rises up to C$ 500 for individuals and up to C$ 1 000 for families; the 3 per cent general surtax will be removed on taxable incomes up to C$ 50 000; increased limits on child-care expense deductions; government grants of up to C$ 400 a year to parents who set up registered education savings plans (RESPs) for their children; employers will not have to pay unemployment insurance premiums for young Canadians hired in 1990 and 2000. Overall, the budget has C$ 10.9 billion in new programme spending over four years and C$ 7.2 billion in tax cuts.

Alberta tabled its budget mid-month with a slight surplus forecast. Highlights include a 1.5 per cent cut in the personal income tax rate retroactive to 1 January; C$ 24 million in bonuses to civil servants worth about 2 per cent of salary; a planned 5.5 per cent increase in doctors' fees over the next three years and doctors received an 8 per cent salary increase over 3 years. Both the civil servants and doctors had accepted salary cuts four years ago.

Ontario Hydro reported a C$ 6.3 billion loss last year, the biggest annual loss in Canadian corporate history. The loss included a one-time charge of C$ 6.6 billion largely related to a restructuring programme for Hydro's nuclear operations.

The official 1 to 3 per cent inflation target is extended until the end of 2001.

March

British Columbia forecast a deficit of C$ 95 million, with C$ 1.25 billion for new schools, hospitals, roads and other infrastructure, an increase of C$ 275 million; increased spending on health care, education and child protection; a personal income tax cut of 1 percentage point to 49.5 per cent of the federal rate, effective next 1 January; a cut in the corporate capital tax by raising the asset level at which it kicks in from C$ 1.5 million to C$ 5 million by 1 January 2001; and a cut in the small business corporate income tax rate to 8 per cent from 9 per cent over two years.

Quebec projected a deficit of C$ 1.1 billion before the accounting reform. Highlights include no tax increases and no substantial new programme spending; cuts in payroll taxes for small and mid-sized businesses over two years; and C$ 232 million over two years in job creation for youths, internship and day care.

Saskatchewan tabled its fifth consecutive balanced budget. Highlights include a 2 percentage point cut to personal income taxes to 48 per cent of the federal rate, effective 1 July; over C$ 200 million in new funding for child welfare, health care, education, highways and policing; and a C$ 500 million reduction in the debt due to a surplus in general revenue and money from the Crown Investments Corp.

Newfoundland forecast a C$ 10 million deficit, with new spending for health care and education, and a 7 per cent increase in monthly social assistance payments over three years. Prince Edward Island projected a shortfall of C$ 3.4 million, with no tax changes and almost no changes in spending plans.

Manitoba's budget projected a fourth consecutive surplus in 1998-99. The personal income tax rate was reduced from 52 per cent of Basic Federal Tax to 51 per cent for the 1998 taxation year and to 50 per cent for 1999. There were new spending indicators on health, education, children and support to families, as well as a doubling of the payment towards debt to C$ 150 million.

April

The Canadian Imperial Bank of Commerce and Toronto-Dominion Bank, Canada's first and fifth-largest banks by assets, respectively, announced plans to merge in a C$ 47 billion deal. The proposed union follows less than three months after the proposed merger of the Royal Bank of Canada and the Bank of Montreal.

May

The Ontario government tabled its 1998-99 budget on 5 May with 36 targeted tax cuts and a forecast C$ 4.2 billion deficit. These include a reduction in the Ontario corporate income-tax rate for small business, while the final instalment of the 30 per cent personal income tax cut will go into effect 1 July, six months ahead of schedule. Other

highlights include increased spending for health, youth, employment, education and child care.

The federal government banned the fishing of coho salmon on the West Coast in an effort to preserve stocks. The fish has been an integral part of the C$ 750 million sports fish industry.

June

Nova Scotia's budget was tabled on 4 June. It projected the third consecutive balanced budget and included new initiatives in health, education and child support.

August

The Bank of Canada raised its Bank rate from 5 per cent to 6 per cent, the first increase since January.

September

The Bank of Canada lowered its Bank Rate by ¼ percentage point.

Notes

1. This section draws heavily from Technical Box 1, Measuring the Trend of Inflation, in Bank of Canada, *Monetary Policy Report*, November 1997.
2. Central banks pursuing inflation targets but which do not focus on a core measure of inflation include: Italy where the focus is the private consumption deflator; and both Spain and Sweden where the focus is the headline CPI. See Table 10 in *OECD Economic Surveys: Canada, 1997.*
3. The national accounts measure the deficit differently from the public accounts. The purpose of the public accounts measure is to provide information to Parliament on the government's financial activities as required by legislation. The budgetary balance of the public accounts measures the amount by which the governments' revenues exceed the obligation incurred during the course of the fiscal year. In the case of the federal government and certain provinces, the budgetary balance includes the total annual pension-related obligations (the government's contribution as an employer for current service cost's plus interest on its borrowing from the pension accounts). In contrast, the budget balance on a national accounts basis measures revenues and expenditures, including employee and employer premiums paid plus interest earned less the annual pension benefits paid out.
4. The detailed methodology is described in Giorno, Claude *et al.*, ''Estimating Potential Output, Output Gaps and Structural Budget balances'', *Economics Department Working Papers,* No. 152, Paris, 1995, and the following is drawn heavily from that reference.
5. In algebraic terms: it is assumed that the rate of change of wage inflation is proportional to the gap between actual unemployment and the NAWRU and thus,
   $D^2 \log W = -a(U - NAWRU)\ a > 0$
   where D is the first-difference operator and W and U are the levels of wages and unemployment; assuming further that NAWRU is constant between any two consecutive time periods, an estimate of a can be calculated as:
   $a = -D^3 \log W/DU$
   which, in turn can be used to give the estimated NARWU as:
   $NARWU = U - (DU/D^3 \log W)*D^2 \log W$
   The resulting series is then smoothed to eliminate erratic movements.
6. The purchasing power parities (PPPs) used are those calculated by the OECD and published in *Main Economic Indicators.* In the scenarios, PPPs are assumed to be unchanged from 1996.
7. Economic assumptions for the United States, Japan and the OECD area come from OECD, *Economic Outlook, 62,* December 1997, Tables 14 and 17, and are an average of the ''high performance'' and ''business-as-usual'' projections contained therein. Population projections

for the United States, Japan and the OECD come from the UN, while those for Canada are provided by Statistics Canada (see Table A4).

8. Note that these employment to population ratios differ from those in figure A3. There, numbers are calculated as the ratio of employment to the total population while those in figures A4 and A5 are ratios of employment to the non-institutional population age 16 and over. The group of high school educated consists of those with some or completed high school education.
9. The lines in the figures are constructed using ''synthetic'' cohort techniques following a given cohort across a series of separate cross-sectional datasets. The employment rate paths have been smoothed and had cycle effects removed using a detrended unemployment rate variable (see Beaudry and Green, 1997*a*).
10. More recent research done by Industry Canada suggests that increased demand for more-skilled workers takes place within rather than between industries. Moreover, evidence of such a shift can be found at the occupational rather than sectoral level (Lavoie and Roy, 1998).
11. Statistics Canada's defines full-year/full-time workers as individuals who work at least 49 weeks in a year and report themselves as ''mostly working full time'' (*i.e.* at least 30 hours per week).
12. The 1995 Survey of Work Arrangements indicated that own-account self-employed were more than twice as likely to be pushed into self-employment because no other work was available than were self-employed employers.
13. For instance, Canadian statistics include incorporated self-employed, which are not included in US statistics.
14. The definitions of cohorts and methodology are the same as those described earlier for cohort patterns in employment rates.

STATISTICAL ANNEX AND STRUCTURAL INDICATORS

Table A. **Selected background statistics**

	Average 1988-97	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997
A. Percentage changes											
Private consumption [1]	2.3	4.4	3.6	1.3	–1.4	1.8	1.9	3.1	1.7	2.4	4.1
Gross fixed capital formation [1]	2.3	9.8	5.9	–3.6	–3.5	–1.3	–2.9	7.1	–2.8	4.8	11.4
Public investment [1]	2.3	4.7	11.2	6.8	7.0	–0.3	–2.1	7.0	–1.2	–3.7	–5.4
Private investment [1]	2.3	10.5	5.3	–5.1	–5.1	–1.4	–3.0	7.1	–3.1	6.3	14.1
Residential [1]	–0.6	2.2	4.2	–10.2	–14.5	7.2	–3.8	3.7	–14.1	10.9	13.1
Non-residential [1]	4.2	16.8	6.0	–1.6	0.5	–5.9	–2.5	9.0	3.0	4.2	14.5
GDP [1]	2.0	4.9	2.5	0.3	–1.9	0.9	2.5	3.9	2.2	1.2	3.7
GDP price deflator	2.3	4.5	4.6	3.1	2.7	1.3	1.2	1.2	2.6	1.4	0.5
Industrial production	2.1	6.1	–0.4	–2.8	–3.8	1.1	5.1	6.0	4.1	1.5	5.0
Employment	1.2	3.2	2.1	0.6	–1.9	–0.6	1.4	2.1	1.6	1.3	1.9
Compensation of employees (current prices)	4.2	9.7	7.8	5.2	2.8	2.3	1.9	2.6	3.4	2.5	3.8
Productivity (real GDP/employment)	0.8	1.6	0.4	–0.3	0	1.5	1.1	1.8	0.5	0	1.8
Unit labour costs (compensation/real GDP)	2.1	4.6	5.2	4.9	4.7	1.4	–0.6	–1.3	1.2	1.3	0.1
B. Percentage ratios											
Gross fixed capital formation as per cent of GDP at constant prices	18.3	19.0	19.7	18.9	18.6	18.2	17.2	17.8	16.9	17.5	18.8
Stockbuilding as per cent of GDP at constant prices	0.1	0.4	0.6	–0.3	–0.8	–0.9	0.1	0.4	1.0	0.1	0.9
Foreign balance as per cent of GDP at constant prices	0.5	0.1	–1.0	–0.5	–0.8	–0.4	0.6	1.4	2.3	2.5	1.0
Compensation of employees as per cent of GDP at current prices	53.7	53.2	53.5	54.4	55.5	55.5	54.5	53.2	52.4	52.4	52.1
Direct taxes as per cent of household income	16.0	15.5	15.3	16.5	16.2	15.8	15.4	15.6	15.9	16.7	17.4
Household saving as per cent of disposable income	9.1	10.7	11.5	11.5	11.7	11.4	10.2	8.1	8.0	5.9	1.7
Unemployment rate	9.5	7.8	7.5	8.2	10.4	11.3	11.2	10.4	9.5	9.7	9.2
C. Other indicator											
Current balance (billion dollars)	–14.5	–14.9	–21.8	–19.8	–22.4	–21.0	–21.8	–13.0	–4.7	3.3	–9.2

1. At constant 1992 prices.

Table B. **Supply and use of resources**

Million Canadian dollars, current prices

	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997
Private consumption	339 728	366 851	386 913	399 932	412 940	429 994	445 879	460 033	477 927	505 896
Public consumption	128 408	139 000	151 977	162 765	169 262	170 760	169 599	171 072	168 965	168 459
Gross fixed investment	133 662	145 676	140 996	130 343	127 045	125 447	137 850	135 342	141 178	158 566
Final domestic demand	601 798	651 527	679 886	693 040	709 247	726 201	753 328	766 447	788 070	832 921
	(9.4)	(8.3)	(4.4)	(1.9)	(2.3)	(2.4)	(3.7)	(1.7)	(2.8)	(5.7)
Stockbuilding	3 467	4 063	–2 660	–5 882	–6 562	561	2 142	8 164	1 043	7 256
	(0.1)	(0.1)	(–1.0)	(–0.5)	(–0.1)	(1.0)	(0.2)	(0.8)	(–0.9)	(0.8)
Total domestic demand	605 265	655 590	677 226	687 158	702 685	726 762	755 470	774 611	789 113	840 177
	(9.5)	(8.3)	(3.3)	(1.5)	(2.3)	(3.4)	(4.0)	(2.5)	(1.9)	(6.5)
Exports	163 842	168 936	175 513	172 161	189 784	220 106	261 692	302 199	320 739	343 536
Imports	159 117	168 723	174 624	176 093	192 393	220 174	253 724	278 276	289 319	329 193
Foreign balance	4 725	213	889	–3 932	–2 609	–68	7 968	23 923	31 420	14 343
	(–0.3)	(–0.7)	(0.1)	(–0.7)	(0.2)	(0.4)	(1.1)	(2.1)	(0.9)	(–2.1)
Statistical discrepancy	1 795	387	20	13	–1 532	–1 774	–1 187	595	–210	583
GDP (market prices)	611 785	656 190	678 135	683 239	698 544	724 920	762 251	799 129	820 323	855 103
	(9.6)	(7.3)	(3.3)	(0.8)	(2.2)	(3.8)	(5.1)	(4.8)	(2.7)	(4.2)

Note: Figures in parentheses are annual growth rates; for stockbuilding and the foreign balance they are contributions to GDP growth.
Source: CANSIM – Statistics Canada.

Table B. **Supply and use of resources** *(cont.)*

Million Canadian dollars, 1992 prices

	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997
Private consumption	392 093	406 034	411 343	405 783	412 940	420 595	433 812	441 263	451 682	470 177
Public consumption	152 897	157 195	162 937	167 541	169 262	168 864	165 888	165 244	163 164	162 988
Gross fixed investment	130 658	138 425	133 392	128 682	127 045	123 420	132 127	128 401	134 592	149 895
Final domestic demand	675 648	701 654	707 672	702 006	709 247	712 879	731 827	734 908	749 438	783 060
	(5.4)	(3.8)	(0.9)	(–0.8)	(1.0)	(0.5)	(2.7)	(0.4)	(2.0)	(4.5)
Stockbuilding	2 452	4 388	–1 926	–5 806	–6 562	437	2 918	7 459	942	6 982
	(–0.1)	(0.3)	(–0.9)	(–0.5)	(–0.1)	(1.0)	(0.3)	(0.6)	(–0.9)	(0.8)
Total domestic demand	678 100	706 042	705 746	696 200	702 685	713 316	734 745	742 367	750 380	790 042
	(5.3)	(4.1)	(0)	(–1.4)	(0.9)	(1.5)	(3.0)	(1.0)	(1.1)	(5.3)
Exports	162 162	164 203	171 977	175 926	189 784	212 603	237 684	259 695	274 456	296 534
Imports	161 382	171 580	175 482	181 120	192 393	208 046	227 054	242 306	254 908	288 933
Foreign balance	780	–7 377	–3 505	–5 194	–2 609	4 557	10 630	17 389	19 548	7 601
	(–0.8)	(–1.2)	(0.6)	(–0.2)	(0.4)	(1.0)	(0.8)	(0.9)	(0.3)	(–1.6)
Statistical discrepancy	7 296	4 912	3 223	1 241	–1 532	–1 750	–1 155	553	–198	540
GDP (market prices)	686 176	703 577	705 464	692 247	698 544	716 123	744 220	760 309	769 730	798 183
	(4.9)	(2.5)	(0.3)	(–1.9)	(0.9)	(2.5)	(3.9)	(2.2)	(1.2)	(3.7)

Note: Figures in parentheses are annual growth rates; for stockbuilding and the foreign balance they are contributions to GDP growth.
Source: CANSIM – Statistics Canada.

Table C. **Industrial production, employment and other business indicators**

Seasonally adjusted

	1993	1994	1995	1996	1997	1997 II	1997 III	1997 IV	1998 I	1998 II
Indices of industrial production (1990 = 100)										
Total	102.1	108.3	112.7	114.4	120.2	119.3	121.5	122.3	121.9	122.7
Durable manufactures	100.6	109.2	117.4	119.0	128.3	126.9	130.8	131.5	133.0	132.6
Non-durable manufactures	98.1	102.1	104.2	104.9	109.5	108.8	110.3	111.1	111.5	112.5
New Residential construction (annual rates)										
Building permits (million C$)	16 433	17 590	13 242	15 718	18 304	17 841	18 570	18 069	18 396	17 590
Starts (thousand units)	155.4	153.9	112.6	123.4	148.3	145.0	148.2	148.3	151.1	137.9
Employment and unemployment (thousands, monthly averages)										
Civilian labour force	14 665	14 827	14 926	15 149	15 347	15 326	15 392	15 439	15 517	15 574
Non-agricultural employment	12 565	12 866	13 076	13 224	13 513	13 472	13 590	13 655	13 739	13 822
Employment										
Mining	125	137	134	131	143	144	151	145	144	146
Manufacturing	1 617	1 655	1 697	1 738	1 800	1 810	1 840	1 825	1 814	1 888
Durables	850	893	931	963	1 004	1 007	1 021	1 026	1 023	1 060
Non-durables	767	762	766	775	796	803	818	799	791	828
Transportation, communication and other utilities [1]	851	858	857	839	851	858	863	849	857	876
Unemployment (thousands)	1 649	1 536	1 419	1 472	1 410	1 433	1 384	1 369	1 342	1 305
Unemployment (percentage of civilian labour force)	11.2	10.4	9.5	9.7	9.2	9.3	9.0	8.9	8.7	8.4
Average weekly hours worked in manufacturing	38.6	38.9	38.5	38.4	39.3	39.6	39.2	39.1	38.5	38.7
Retail sales ($ million, monthly averages)	16 194	17 320	17 814	18 369	19 773	19 600	19 918	20 264	20 326	20 640
Orders and inventories in manufacturing ($ million)										
New orders (monthly averages) [2]	26 107	29 868	33 027	33 879	36 892	36 464	37 304	37 864	37 224	36 959
Unfilled orders (end of period)	29 172	34 761	34 256	34 265	42 394	38 164	39 954	42 394	43 518	44 645
Total inventories (end of period)	36 638	40 976	45 569	45 767	47 815	46 760	47 231	47 815	48 189	49 301

1. Includes storage, electric power gas and water utilities.
2. 3-month averages for quarters.

Source: OECD, *Main Economic Indicators;* CANSIM – Statistics Canada.

Table D. **Prices, wages and finance**

	1993	1994	1995	1996	1997	1997 II	1997 III	1997 IV	1998 I	1998 II
Prices (1990 = 100)										
Consumer prices, all items	109.2	109.4	111.7	113.5	115.3	115.3	115.6	115.5	116.1	116.4
of which:										
Food	106.1	106.6	109.1	110.6	112.3	112.2	112.4	112.4	114.1	114.3
Non-food	109.9	110.0	112.3	114.2	116.0	116.0	116.3	116.2	116.6	117.0
Producer prices, manufactured goods	103.0	109.3	117.4	117.9	118.8	119.1	118.9	119.0	118.3	118.4
Wages and profits										
Hourly earnings in manufacturing (1990 = 100)	110.7	112.5	114.1	117.7	118.7	118.9	117.1	119.8	121.4	122.3
Corporate profits before tax ($ million, annual rates)	40 354	59 467	70 355	67 988	78 988	74 892	80 180	82 992	76 096	74 312
Banking ($ million, end of period)										
Chartered banks:										
Canadian dollar deposits	377 231	406 349	433 558	449 541	490 148	457 926	474 580	490 148	482 887	488 194
of which:										
Personal savings deposits	263 766	280 291	297 559	292 444	289 697	286 411	292 147	289 697	286 390	285 099
Liquid assets	82 181	87 238	91 788	87 099	78 257	82 942	72 054	78 257	74 378	71 667
Holdings of Government of Canada direct and guaranteed securities	32 829	42 291	49 284	55 361	51 073	53 973	47 085	51 073	44 367	49 286
Total loans	382 459	403 234	439 761	505 826	547 468	538 428	542 603	547 468	542 603	549 552
Currency outside banks	23 672	25 429	26 121	27 141	28 638	27 871	28 079	28 638	29 247	29 651
Interest rates (per cent, end of period)										
Prime corporate paper (3 month)	3.9	7.2	5.7	3.1	4.8	3.3	3.6	4.8	4.8	5.1
Yield of long-term Govt. bonds	7.3	9.2	7.5	6.8	5.9	6.6	6.2	5.9	5.6	5.5
Miscellaneous										
Share prices Toronto stock exchange (1990 = 100)	114.1	125.2	129.6	154.0	188.8	183.1	200.0	195.4	208.0	220.4

Source: OECD, *Main Economic Indicators;* CANSIM – Statistics Canada.

Table E. **Balance of payments**

Million US dollars

	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997
Merchandise exports	116 582	124 115	130 289	128 840	135 219	147 561	166 843	193 035	205 727	217 762
Merchandise imports	107 794	117 572	120 817	122 722	127 747	137 648	152 712	168 457	175 671	200 893
Trade balance	8 787	6 543	9 472	6 118	7 471	9 913	14 131	24 577	30 056	16 869
Services, credits	15 649	17 547	19 178	20 350	20 780	22 092	23 858	26 163	28 513	29 878
Services, debits	21 007	24 427	28 292	30 314	30 809	32 451	32 510	33 751	35 767	36 250
Services, net	–5 357	–6 880	–9 114	–9 964	–10 028	–10 359	–8 651	–7 588	–7 254	–6 372
Investment income, credits	13 208	13 431	15 064	12 930	11 391	10 690	14 981	18 894	19 098	20 283
Investment income, debits	30 674	33 918	34 451	30 329	28 873	31 504	34 903	41 279	39 412	43 279
Investment income, net	–17 465	–20 487	–19 386	–17 399	–17 482	–20 814	–19 922	–22 385	–20 314	–22 996
Official transfers, net	–726	–769	–755	–1 111	–997	–711	–549	–460	–57	35
Private transfers, net	–126	–204	–41	–6	55	115	186	312	249	199
Current account	–14 887	–21 797	–19 824	–22 362	–20 981	–21 856	–14 806	–5 544	2 680	–12 265
Capital account	3 914	4 628	5 315	5 593	7 092	8 296	7 498	4 946	5 921	5 488
Direct investment, assets	–6 222	–5 266	–5 235	–5 833	–3 589	–5 867	–9 118	–11 145	–8 501	–12 942
Portfolio investment, assets	–3 640	–4 620	–2 224	–10 178	–9 719	–13 859	–6 470	–5 241	–13 529	–8 231
Other investment, assets	3 830	–6 100	–8 351	976	–3 369	–478	–20 577	–8 211	–22 582	–19 773
Total assets, net flows	–6 032	–15 986	–15 810	–15 035	–16 677	–20 204	–36 165	–24 597	–44 612	–40 946
Direct investment, liabilities	6 123	6 010	7 580	2 879	4 722	4 747	8 457	10 761	6 398	8 246
Portfolio investment, liabilities	18 728	19 603	15 924	27 483	20 433	41 124	17 052	17 170	14 689	15 183
Other investment, liabilities	2	9 758	9 624	–255	–2 218	–6 101	16 421	–3 101	18 023	24 480
Total liabilities, net flows	24 853	35 371	33 128	30 107	22 937	39 770	41 930	24 830	39 110	47 909
Change in reserves	–8 264	–690	–1 068	1 834	4 756	–935	358	–2 752	–5 498	2 446
Financial account	10 556	18 695	16 249	16 911	11 015	18 631	6 124	–2 521	–11 001	9 409
Errors and omissions	415	–1 525	–1 741	–141	2 871	–5 072	1 184	3 116	2 399	–2 632

Source: Statistics Canada; OECD Secretariat.

Table F. **Public sector**

	A. Budget indicators: general government accounts (per cent of GDP)				
	1961	1970	1980	1996	1997
Current receipts[1]	28.7	36.8	38.9	45.7	46.3
Non-interest expenditure[1]	23.1	28.4	33.2	35.8	34.4
Primary budget balance	–0.5	1.9	–1.3	3.2	5.8
Net interest payments	1.6	1.4	1.8	5.2	4.9
General government budget balance	–2.1	0.5	–3.1	–2.0	0.9
of which:					
Federal	–1.0	0.3	–3.4	–1.6	0.8
Provincial, local, hospitals	–1.0	–1.1	–0.6	–0.1	0.4
Pension plans[2]	0.0	1.3	1.0	–0.3	–0.3
General government debt					
Gross debt	68.9	54.2	45.6	100.5	96.1
Net debt	27.1	12.4	13.5	70.0	66.4
of which: Federal	19.1	6.1	14.6	54.3	51.4

	B. The structure of outlay and capital accounts (per cent of GDP)				
Outlay					
Total general government expenditure	26.0	32.0	38.5	45.0	43.1
Current consumption[1]	15.9	20.7	21.7	21.6	20.6
Current transfers	7.2	7.7	11.5	14.2	13.8
Persons	6.1	6.5	8.4	12.9	12.5
Businesses (Subsidies)	0.9	0.9	2.7	0.9	0.9
Non-residents	0.2	0.3	0.3	0.4	0.3
Total current program expenditure	23.1	28.4	33.2	35.8	34.4
Gross interest payments	2.9	3.7	5.4	9.2	8.8
Capital					
Net capital transfers[3]	–0.1	–0.2	–0.4	–0.4	–0.2
Non-financial capital acquisition[4]	4.7	4.0	3.0	2.2	2.2

	C. General government expenditure by function, financial management system basis (per cent of GDP, fiscal year)					
	1965-66	1970-71	1980-81	1990-91	1994-95	1996-97
Social services	4.8	6.0	7.4	10.0	10.7	n.a.
Education	4.6	6.2	5.0	5.7	5.5	n.a.
Health	2.6	4.3	4.4	6.1	5.9	n.a.
Transport and Communication	3.3	2.8	2.6	2.2	2.0	n.a.
National Defence	2.7	1.9	1.6	1.8	1.5	1.5
General Services	1.5	2.3	2.4	2.7	2.3	n.a.

1. Exclude capital consumption allowances.
2. Canada and Quebec Pension Plans.
3. Include net capital formation assistance to persons, businesses and non-residents, and other net capital transfers.
4. Includes fixed capital formation, inventories and existing assets.

Source: Department of Finance.

Table G. **Financial markets**

	1970	1980	1992	1997
Size of the financial sector (percentages)[1]				
Sector employment/total employment	4.8	5.7	6.2	5.7
Net financial assets/GDP	6.3	4.3	7.5	n.a.
Structure of financial assets and liabilities				
Financial institutions' share in domestic financial assets (per cent)	33.1	38.6	41.5	45.0
Government securities in NFB[2] total financial assets (per cent)	0.3	0.1	2.2	1.7
Structure of structure of NFB liabilities				
Debt to equity ratio[3]	1.17	1.42	1.48	1.44
Short-term:				
Securities and mortgages ($ billion)	8.7	33.4	118.9	130.0
Trade payables ($ billion)	12.4	54.9	101.6	140.9
Long-term:				
Bonds ($ billion)	13.7	28.7	97.5	133.9
Loans and corporate claims ($ billion)	27.0	105.5	252.5	316.7
Internationalisation of markets				
Share of foreign currency assets and liabilities in the banking sector[4]				
Assets	28.9	39.1	30.8	41.5
Liabilities	28.6	40.2	33.3	43.5
Foreign purchases of Canadian securities[5]	12.4	18.2	75.4	. .
Canadian purchases of foreign securities[5]	–1.2	0.6	20.8	. .
Debt (per cent of GDP)				
Private non-financial sector				
NFB[2, 6]	77.3	81.6	87.8	94.3
Households[7]	45.3	52.5	63.2	63.7

1. Public and private financial institutions and insurance.
2. NFB = non-financial corporate business, excluding farms.
3. (Liabilities-shares)/shares.
4. Per cent of consolidated balance sheet of chartered banks; excludes other deposit-taking institutions.
5. Per cent of net issues on domestic securities markets. Data include new issues as well as secondary market transaction.
6. Liabilities less shares.
7. Persons and unincorporated business liabilities less trade payables, other loans and other Canadian bonds.

Source: *Bank of Canada Review*; Labour Force Survey; National Authorities (National Balance Sheet; Security transactions with non-residents).

Table H. **Labour market indicators**

	A. Labour market performance			
	1976	1980	1990	1997
Standardised unemployment rate	7.2	7.5	8.1	9.2
Unemployment rate: Total	7.2	7.5	8.1	9.2
Male	6.4	6.9	8.1	9.2
Woman	8.4	8.4	8.1	9.1
Youth[1]	12.6	13.1	12.7	16.7
Share of long-term unemployment in total unemployment[2]	2.5	3.7	5.7	12.5
Dispersion of regional unemployment rates[3]	3.0	3.0	3.5	4.2
	B. Structural or institutional characteristics			
	1976	1980	1990	1997
Participation rate:[4] Total	67.9	71.8	76.8	74.9
Male	84.5	85.9	84.9	81.8
Women	51.2	57.8	68.6	68.0
Employment/population (15-64 years)	63.0	66.4	70.5	67.9
Average hours worked (manufacturing)	38.7	38.5	38.2	39.3
Part-time work (as per cent of dependent employment)	14.2	16.5	19.8	23.1
Non-wage labour costs[5] (as percentage of total compensation)	8.0	8.5	10.1	13.0
Government unemployment insurance replacement ratio[6]	74.2	64.6	55.7	45.6
Unionisation rate	29.2	28.6	27.2	. .
	Average percentage changes (annual rates)			
	1980/1976	1990/1980	1996/1990	
Labour force	3.3	1.8	0.9	
Employment: Total	3.2	1.7	0.6	
Goods-producing	2.0	0.1	–0.6	
Services	3.8	2.5	1.1	

1. People between 15 and 24 years as a percentage of the labour force of the same age group.
2. People looking for a job since one year or more.
3. Measured by standard deviation for 10 provinces.
4. Labour force as a percentage of relevant population group, aged between 15 and 64 years.
5. Supplementary labour income (including contributions to social security, pension funds and other forms of SLI).
6. Average weekly benefits (regular)/average weekly earnings (including overtime).

Source: Canadian authorities; OECD, *National Accounts.*

Table I. **Production structure and performance indicators**

	A. Production structure					
	Per cent share of GDP at factor cost (constant prices)			Per cent share of total employment		
	1976	1985	1996	1976	1985	1996
Agriculture	2.9	2.1	2.2	4.9	4.1	3.3
Mining and quarrying	5.1	4.3	4.4	1.5	1.7	1.2
Manufacturing	20.6	19.7	18.8	20.3	17.6	15.2
of which:						
Food and beverages	3.1	2.7	2.4	2.6	2.2	1.8
Paper and paper products	2.1	1.6	1.4	1.5	1.1	0.9
Primary metal industries	1.6	1.4	1.5	1.3	1.1	0.8
Fabricated metal products, machinery and equipment	4.7	4.2	5.0	4.6	3.9	3.6
Chemicals and chemical products	1.2	1.4	1.5	1.0	0.8	0.7
Construction	6.8	6.1	4.8	6.7	5.2	5.3
Market services	57.1	59.7	63.4	64.5	69.5	73.1
of which:						
Transport, storage and communication	7.1	7.9	9.1	8.7	7.8	7.5
Wholesale and retail trade	11.3	11.2	12.2	17.3	17.8	17.3
Finance, insurance and real estate	13.5	15.1	16.3	5.3	5.6	5.8
Community, business social and personal services	22.0	22.0	22.5	27.1	32.3	27.1
Government services	7.9	7.1	5.7	7.2	7.1	7.2

	B. Manufacturing sector performance *(constant prices)*		
	Productivity growth by sector (GDP/employment, annual rate)		
	1980/1976	1990/1980	1996/1990
Manufacturing	–0.7	2.5	2.3
of which:			
Food and beverages	0.1	1.3	1.3
Paper and paper products	1.3	0.3	1.2
Primary metal industries	–3.2	2.6	4.1
Chemicals and chemical products	1.5	4.4	3.8

Source: Canadian authorities; OECD, *National Accounts.*

BASIC STATISTICS:

INTERNATIONAL COMPARISONS

BASIC STATISTICS: INTERNATIONAL COMPARISONS

	Units	Reference period[1]	Australia	Austria	Belgium	Canada	Czech Republic	Denmark	Finland	France	Germany	Greece
Population												
Total	Thousands	1996	18 289	8 060	10 157	29 964	10 316	5 262	5 125	58 380	81 877	10 465
Inhabitants per sq. km	Number	1996	2	96	333	3	131	122	15	106	229	79
Net average annual increase over previous 10 years	%	1996	1.3	0.6	0.3	1.3	0	0.3	0.4	0.5	3	0.5
Employment												
Total civilian employment (TCE)[2]	Thousands	1996	8 344	3 737 (94)	3 675 (95)	13 676	4 918	2 593	2 087	21 951	35 360	3 824 (95)
of which: Agriculture	% of TCE	1996	5.1	7.2 (94)	2.5 (95)	4.1	6.3	4	7.1	4.6	3.3	20.4 (95)
Industry	% of TCE	1996	22.5	33.2 (94)	26.7 (95)	22.8	42	27	27.6	25.9	37.5	23.2 (95)
Services	% of TCE	1996	72.4	59.6 (94)	71.4 (95)	73.1	51.7	69	65.3	69.5	59.1	56.4 (95)
Gross domestic product (GDP)												
At current prices and current exchange rates	Bill. US$	1996	398.9	228.7	268.2	579.2	56.2	174.9	125.1	1 536.6	2 353.5	91.2 (95)
Per capita	US$	1996	21 812	28 384	26 409	19 330	5 445	33 230	24 420	26 323	28 738	8 722 (95)
At current prices using current PPPs[3]	Bill. US$	1996	372.7	172.4	222	645.1	..	118	96.7	1 198.6	1 736.1	133.5
Per capita	US$	1996	20 376	21 395	21 856	21 529	..	22 418	18 871	20 533	21 200	12 743
Average annual volume growth over previous 5 years	%	1996	3.9	1.6	1.2	2.2	2	2.2	1.6	1.2	1.4	1.3 (95)
Gross fixed capital formation (GFCF)	% of GDP	1996	20.3	23.8	17.3	17.7	33	16.7	16.1	17.4	20.6	17 (95)
of which: Machinery and equipment	% of GDP	1996	10.2 (95)	8.8 (95)	7.5 (95)	6.6	..	7.9 (95)	6.4 (95)	7.8	7.6	7.7 (95)
Residential construction	% of GDP	1996	4.6 (95)	5.9 (95)	4.6 (95)	5.4	..	3.2 (95)	3.5 (95)	4.4	7.3	3.3 (95)
Average annual volume growth over previous 5 years	%	1996	5.6	2.1	0.3	2.2	9.4	2	–4.1	–1.5	0.2	0.5 (95)
Gross saving ratio[4]	% of GDP	1996	18	21.9	22.2	17.8	..	17.6	19.6	18.7	20	16 (95)
General government												
Current expenditure on goods and services	% of GDP	1996	17	19.8	14.5	18.7	21.5	25.2	21.9	19.4	19.8	20.8 (95)
Current disbursements[5]	% of GDP	1995	35.6	48.6	52.2	45.8	..	59.6	55.9	50.9	46.6	52.1
Current receipts	% of GDP	1995	34.9	47.4	49.9	42.7	..	58.1	52.8	46.9	45.9	45
Net official development assistance	% of GNP	1995	0.36	0.33	0.38	0.38	..	0.96	0.32	0.55	0.31	0.13
Indicators of living standards												
Private consumption per capita using current PPPs[3]	US$	1996	12 596	12 152	13 793	12 959	..	12 027	10 282	12 506	12 244	9 473
Passenger cars, per 1 000 inhabitants	Number	1994	460	433	416	466	282	312	368	430	488	199
Telephones, per 1 000 inhabitants	Number	1994	496	466	449	576	209	604	551	547	483[8]	478
Television sets, per 1 000 inhabitants	Number	1993	489	479	453	618	476	538	504	412	559	202
Doctors, per 1 000 inhabitants	Number	1995	2.2 (91)	2.7	3.7 (94)	2.2	2.9	2.9 (94)	2.8	2.9	3.4	3.9 (94)
Infant mortality per 1 000 live births	Number	1995	5.7	5.4	7.6 (94)	6.3 (94)	7.7	5.5	4	5.8 (94)	5.3	8.1
Wages and prices (average annual increase over previous 5 years)												
Wages (earnings or rates according to availability)	%	1996	1.7	5.2	2.7	2.4	..	3.2	3.8	2.6	4.2	11.8
Consumer prices	%	1996	2.4	2.9	2.2	1.4	11.9	1.9	1.5	2	3.1	11.6
Foreign trade												
Exports of goods, fob*	Mill. US$	1996	60 288	57 870	170 223[7]	202 320	21 910	51 030	40 576	288 450	521 263	11 501
As % of GDP	%	1996	15.1	25.3	63.5	34.9	39	29.2	32.4	18.8	22.1	12.9 (95)
Average annual increase over previous 5 years	%	1996	7.5	7.1	7.6	9.7	..	6.2	12.1	6.3	5.4	5.8
Imports of goods, cif*	Mill. US$	1996	61 374	67 376	160 917[7]	170 931	27 721	44 987	30 911	271 348	455 741	27 402
As % of GDP	%	1996	15.4	29.5	60	29.5	49.3	25.7	24.7	17.7	19.4	30.4 (95)
Average annual increase over previous 5 years	%	1996	9.7	5.9	5.9	7.7	..	5.6	7.3	3.9	3.3	6.6
Total official reserves[6]	Mill. SDRs	1996	10 107	15 901	11 789[7]	14 202	8 590	9 834	4 810	18 635	57 844	12 171
As ratio of average monthly imports of goods	Ratio	1996	2	2.8	0.9	1	..	2.6	1.9	0.8	1.5	5.3

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Including non-residential construction.
10. Refers to the public sector including public enterprises.

Sources: Population and Employment: OECD, *Labour Force Statistics.* GDP, GFCF and General Government: OECD, *National Accounts,* Vol. I and *OECD Economic Outlook,* Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators.* Foreign trade: OECD, *Monthly Foreign Trade Statistics,* Series A. Total official reserves: IMF, *International Financial Statistics.*

BASIC STATISTICS: INTERNATIONAL COMPARISONS *(cont'd)*

	Units	Reference period[1]	Hungary	Iceland	Ireland	Italy	Japan	Korea	Luxembourg	Mexico	Netherlands	New Zealand
Population												
Total	Thousands	1996	10 193	270	3 621	57 473	125 864	45 545	418	96 582	15 494	3 640
Inhabitants per sq. km	Number	1996	77	3	52	191	333	458	161	48	380	14
Net average annual increase over previous 10 years	%	1996	–0.3	1.1	0.2	0	0.4	1	1.3	2	0.6	1.1
Employment												
Total civilian employment (TCE)[2]	Thousands	1996	3 605	142	1 307	20 036	64 860	20 764	212 (95)	32 385 (95)	6 983	1 688
of which: Agriculture	% of TCE	1996	8.4	9.2	10.7	7	5.5	11.6	2.8 (95)	23.5 (95)	3.9	9.5
Industry	% of TCE	1996	33	23.9	27.2	32.1	33.3	32.5	30.7 (90)	21.7 (95)	22.4	24.6
Services	% of TCE	1996	58.6	66.2	62.3	60.9	61.2	55.9	66.1 (90)	54.8 (95)	73.8	65.9
Gross domestic product (GDP)												
At current prices and current exchange rates	Bill. US$	1996	43.7 (95)	7.3	70.7	1 214.2	4 595.2	484.8	17	329.4	396	65.9
Per capita	US$	1996	4 273 (95)	27 076	19 525	21 127	36 509	10 644	40 791	3 411	25 511	18 093
At current prices using current PPPs[3]	Bill. US$	1996	..	6.3	68.8	1 148	2 924.5	618.5	13.5	751.1	324.5	63.6
Per capita	US$	1996	..	23 242	18 988	19 974	23 235	13 580	32 416	7 776	20 905	17 473
Average annual volume growth over previous 5 years	%	1996	–2.4 (95)	1.5	7.1	1	1.5	7.1	4.8	1.7	2.3	3.7
Gross fixed capital formation (GFCF)	% of GDP	1996	19.3 (95)	17.5	17.2	17	29.7	36.8	20.8	18	19.7	20.9
of which: Machinery and equipment	% of GDP	1996	..	6.7	5.5 (95)	8.8	10.1 (95)	13	..	8.8	9.4	10
Residential construction	% of GDP	1996	. .	3.9	4.9 (95)	4.5	5.3 (95)	7.6	..	4.7	5	5.6
Average annual volume growth over previous 5 years	%	1996	–0.9 (95)	–1.4	6	–1.4	1.3	6.9	0.2	–0.7	2.2	9.6
Gross saving ratio[4]	% of GDP	1996	..	15.6	21.7	20.5	31.4	34.2	37.5	22.7	25.7	16
General government												
Current expenditure on goods and services	% of GDP	1996	24.9 (95)	20.8	14.1	16.4	9.7	10.6	13.6	9.7[10]	14	14.4
Current disbursements[5]	% of GDP	1995	..	35.1	39.2 (94)	49.5	28.5	15.1	..	..	51.8	..
Current receipts	% of GDP	1995	..	36	39.3 (94)	44.5	32	25.1	..	..	50	..
Net official development assistance	% of GNP	1995	..	..	0.29	0.15	0.28	0.03	0.36	..	0.81	0.23
Indicators of living standards												
Private consumption per capita using current PPPs[3]	US$	1996	..	14 244	10 020	12 224	13 912	7 354	17 811	5 045	12 477	10 895
Passenger cars, per 1 000 inhabitants	Number	1994	212	434	264	517	342	115	544	91	383	457
Telephones, per 1 000 inhabitants	Number	1994	170	557	350	429	480	397	564	93	509	470
Television sets, per 1 000 inhabitants	Number	1993	427	335	301	429	618	215	261	150	491	
Doctors, per 1 000 inhabitants	Number	1995	3.4	3.9 (94)	3.4	3.0 (94)	1.7	1.6 (92)	1.8 (94)	1.1	2.2 (93)	1.6
Infant mortality per 1 000 live births	Number	1995	11	6.1	6.3	6.6 (94)	4.3	9	5.3 (94)	17 (94)	5.5	7.2 (94)
Wages and prices (average annual increase over previous 5 years)												
Wages (earnings or rates according to availability)	%	1996	..	..	3.7	3.5	1.8	..	..	–1.6	2.4	1.5
Consumer prices	%	1996	23.2	2.6	2.2	4.5	0.7	5.3	2.4	19.7	2.5	2
Foreign trade												
Exports of goods, fob*	Mill. US$	1996	15 674	1 891	48 416	250 842	411 067	129 715	..	96 000	203 953	14 316
As % of GDP	%	1996	35.9	26	68.5	20.7	8.9	26.8	..	29.1	51.5	21.7
Average annual increase over previous 5 years	%	1996	8.9	4	14.8	8.2	5.5	12.5	..	17.6	8.9	8.2
Imports of goods, cif*	Mill. US$	1996	18 105	2 032	35 763	206 904	349 149	150 340	..	89 469	184 389	14 682
As % of GDP	%	1996	41.4	27.9	50.6	17	7.6	31	..	27.2	46.6	22.3
Average annual increase over previous 5 years	%	1996	9.6	3.4	11.5	2.5	8	13.9	..	12.4	7.8	11.8
Total official reserves[6]	Mill. SDRs	1996	6 812	316	5 706	31 954	150 663	23 670	..	13 514	18 615	4 140
As ratio of average monthly imports of goods	Ratio	1996	..	1.9	1.9	1.9	5.2	..	..	1.8	1.2	3.4

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Including non-residential construction.
10. Refers to the public sector including public enterprises.

Sources: Population and Employment: OECD, *Labour Force Statistics.* GDP, GFCF and General Government: OECD, *National Accounts,* Vol. I and *OECD Economic Outlook,* Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators.* Foreign trade: OECD, *Monthly Foreign Trade Statistics,* Series A. Total official reserves: IMF, *International Financial Statistics.*

BASIC STATISTICS: INTERNATIONAL COMPARISONS *(cont'd)*

	Units	Reference period[1]	Norway	Poland	Portugal	Spain	Sweden	Switzerland	Turkey	United Kingdom	United States
Population											
Total	Thousands	1996	4 370	38 618	9 935	39 270	8 901	7 085	62 695	58 782	265 557
Inhabitants per sq. km	Number	1996	13	123	108	78	20	172	80	240	28
Net average annual increase over previous 10 years	%	1996	0.5	0.3	–0.1	0.2	0.6	0.8	2	0.3	1
Employment											
Total civilian employment (TCE)[2]	Thousands	1996	2 110	14 977	4 475	12 394	3 963	3 803	20 895	26 088	126 708
of which: Agriculture	% of TCE	1996	5.2	22.1	12.2	8.7	2.9	4.5	44.9	2	2.8
Industry	% of TCE	1996	23.4 (95)	31.7	31.4	29.7	26.1	27.7	22	27.4	23.8
Services	% of TCE	1996	71.5 (95)	46.2	56.4	61.6	71	67.4	33.1	71	73.3
Gross domestic product (GDP)											
At current prices and current exchange rates	Bill. US$	1996	157.8	117.9 (95)	103.6	584.9	251.7	294.3	181.5	1 153.4	7 388.1
Per capita	US$	1996	36 020	3 057 (95)	10 425	14 894	28 283	41 411	2 894	19 621	27 821
At current prices using current PPPs[3]	Bill. US$	1996	106.7	..	130.1	587.2	171.4	180.6	383.3	1 095.5	7 388.1
Per capita	US$	1996	24 364	..	13 100	14 954	19 258	25 402	6 114	18 636	27 821
Average annual volume growth over previous 5 years	%	1996	4.1	2.2 (95)	1.5	1.3	1	0.1	4.4	2.2	2.8
Gross fixed capital formation (GFCF)	% of GDP	1996	20.5	17.1 (95)	24.1	20.1	14.8	20.2	25	15.5	17.6
of which: Machinery and equipment	% of GDP	1996	8.4	..	11.7 (93)	6.1 (95)	7.9	9.3	11.9	7.6	8.3 (95)
Residential construction	% of GDP	1996	2.6 (94)	..	5.2 (93)	4.3 (95)	1.9	11[9]	8.4 (95)	3	4.1 (95)
Average annual volume growth over previous 5 years	%	1996	2.8	5.4 (95)	2.2	–1	–2.6	–0.8	6.9	1.3	6.9
Gross saving ratio[4]	% of GDP	1996	29.9	..	21.6	20.7	16	27.1	22.3	14.6	16.6
General government											
Current expenditure on goods and services	% of GDP	1996	20.5	16.9 (95)	18.5	16.3	26.2	14.3	11.6	21.1	15.6
Current disbursements[5]	% of GDP	1995	45.8	..	42.5 (93)	41.2	63.8	47.7	..	42.3 (94)	34.3
Current receipts	% of GDP	1995	50.9	..	39.8 (93)	37.9	57.5	53.8	..	37.2 (94)	32.1
Net official development assistance	% of GNP	1995	0.87	..	0.27	0.24	0.77	0.34	0.07	0.28	0.1
Indicators of living standards											
Private consumption per capita using current PPPs[3]	US$	1996	11 593	..	8 522	9 339	10 096	15 632	4 130	11 865	18 908
Passenger cars, per 1 000 inhabitants	Number	1994	381	186	357	351	406 (93)	451	47	372	565
Telephones, per 1 000 inhabitants	Number	1994	554	131	350	371	683	597	201	489	602
Television sets, per 1 000 inhabitants	Number	1993	427	298	190	400	470	400	176	435	816
Doctors, per 1 000 inhabitants	Number	1995	2.8	2.3	3	4.1 (93)	3.1	3.1 (94)	1.2	1.6 (94)	2.6 (94)
Infant mortality per 1 000 live births	Number	1995	4	13.6	7.4	6 (94)	4	5	46.8 (94)	6.2 (94)	8 (94)
Wages and prices (average annual increase over previous 5 years)											
Wages (earnings or rates according to availability)	%	1996	3.2	..	..	5.8	4.8	..	..	4.9	2.7
Consumer prices	%	1996	1.9	..	5.6	4.7	2.7	2.2	81.6	2.7	2.9
Foreign trade											
Exports of goods, fob*	Mill. US$	1996	49 576	24 417	24 614	102 067	84 836	79 581	23 301	259 941	625 075
As % of GDP	%	1996	31.4	20.7	23.8	17.5	33.7	27	12.8	22.5	8.5
Average annual increase over previous 5 years	%	1996	7.8	..	8.6	11.2	9	5.3	11.1	7	8.2
Imports of goods, cif*	Mill. US$	1996	35 575	37 185	35 192	121 838	66 825	78 052	43 094	287 033	795 289
As % of GDP	%	1996	22.5	31.5	34	20.8	26.5	26.5	23.7	24.9	10.8
Average annual increase over previous 5 years	%	1996	6.9	..	6.1	5.5	6	3.2	15.1	6.5	10.3
Total official reserves[6]	Mill. SDRs	1996	18 441	12 409	11 070	40 284	13 288	26 727	11 430	27 745	44 536
As ratio of average monthly imports of goods	Ratio	1996	6.2	..	3.8	4	2.4	4.1	3.2	1.2	0.7

* At current prices and exchange rates.
1. Unless otherwise stated.
2. According to the definitions used in OECD *Labour Force Statistics*.
3. PPPs = Purchasing Power Parities.
4. Gross saving = Gross national disposable income minus private and government consumption.
5. Current disbursements = Current expenditure on goods and services plus current transfers and payments of property income.
6. End of year.
7. Data refer to the Belgo-Luxembourg Economic Union.
8. Data refer to western Germany.
9. Including non-residential construction.
10. Refers to the public sector including public enterprises.

Sources: Population and Employment: OECD, *Labour Force Statistics.* GDP, GFCF and General Government: OECD, *National Accounts,* Vol. I and *OECD Economic Outlook,* Historical Statistics. Indicators of living standards: Miscellaneous national publications. Wages and Prices: OECD, *Main Economic Indicators.* Foreign trade: OECD, *Monthly Foreign Trade Statistics,* Series A. Total official reserves: IMF, *International Financial Statistics.*

OECD PUBLICATIONS, 2, rue André-Pascal, 75775 PARIS CEDEX 16
PRINTED IN FRANCE
(10 98 01 1 P) ISBN 92-64-16008-6 – No. 50309 1998
ISSN 0376-6438